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MODELLING OF WORK HARDENING AND STRESS SATURATION IN FCC METALS

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Abstract—A work hardening theory has been developed based on a microstructural concept comprising three elements; the cell/subgrain size, δ , the dislocation density inside the cells, ρ_i , and the cell boundary dislocation density or the sub boundary misorientation, ρ_b or ϕ . The theory is based on a statistical approach to the storage of dislocations. This approach predicts that the slip length, L , scales with the inverse square root of the stored dislocation density, $\rho^{-1/2}$, and also, predicts a substructure evolution which is consistent with the concept of microstructural scaling (similitude) at zero degree Kelvin, at stress $\tau < \tau_{III}$. The model provides a solution to the basic ‘dislocation–book-keeping-problem’ by defining a differential equation which regulates the storage of dislocations into (i) a cell interior dislocation network, (ii) increases in boundary misorientation and (iii) the creation of new cell boundaries. By combining such a solution for the dislocation storage problem with models for the dynamic recovery of network dislocations and sub-boundary structures, the result becomes a general internal state variable solution which has the potential of giving the flow stress as a function of strain for any combination of strain-rate and temperature. The theory predicts that the dislocation density ρ_i inside the cells saturates at the end of the Stage II–III transition, a saturation effect which regulates the subsequent stress–strain behaviour which is then controlled by the continuous refinement of the cell/subgrain structure. The characteristic features of Stages III and IV are well accounted for by the model. The present work hardening model also provides the necessary basis for the construction of constitutive laws for the saturation stresses; τ_{III} , $\tau_{III,s}$ and τ_s and steady-state creep. Beyond Stage II breakdown only two rate controlling dynamic recovery mechanisms are involved, relating to dislocation network growth and subgrain growth. The activation energy of both growth reactions is that of self diffusion. Steady-state creep laws are presented covering the entire range from low temperature creep to Harper–Dorn creep. © 1998 Elsevier Science Ltd. All rights reserved

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A_1, A_2	stacking-fault energy parameters associated with thermally activated cross-slip
b	magnitude of Burgers vector
B_1, B_2, B_1^{cs}	constants
$B_\rho, B_\rho^{cs}, B_\delta, B_I$	pre-exponential constants associated with thermal activation, respectively, dislocation climb, cross-slip, subgrain growth and short range interactions
C	proportionality constant; $C = L\sqrt{\rho}$
f	volume fraction of cell boundaries
f_{ob}	fraction of dislocations stored in old boundaries
F	force per unit length of dislocation
g	dislocation dipole density
G	elastic shear modulus
h	cell boundary thickness
k	Boltzmann constant
$K's$	constants
l	characteristic length
l_g	dislocation dipole separation
l_{sa}	spacing of solute atoms along dislocation
l_ρ	spacing of short range obstacles to dislocation glide in a pure metal
L	average slip distance of dislocations
m	strain-rate sensitivity
M	Taylor factor
n	stress exponent
N	density of stored dislocation loops
N_m	density of mobile dislocation loops
N_{nb}, N_{ob}	density of dislocation loops stored in new and old boundaries, respectively
p, p_{nb}	statistical dislocation storage parameters
P	driving pressure for subgrain growth
q_c, q_b, q_n, q_{IV}	microstructure scaling parameters; $q_c = \delta\sqrt{\rho l}$, $q_b = \sqrt{\rho_b/\rho_l}$, $q_n = h/\delta$ and $q_{IV} = \delta_{III}/\delta_{IV}$
r	radius
S	dislocation storage parameter
S_{scs}, S_{IV}	respectively, values of S when microstructural scaling (similitude) applies and during Stage IV
t	time
T	temperature
U	activation energy
U_{SD}, U_{cs}, U_I	respectively, activation energies for self diffusion, cross-slip and short range interactions
v	dislocation velocity
v_c	dislocation climb velocity

V_a	activation volume
w	number of dislocations in pileup
y	slip line separation
Z	Zener–Hollomon parameter
α, α_1	dislocation interaction constants relating dislocation spacings and stress components
α_2	dislocation interaction constant relating sub-boundary spacings and stress components
α_3	dislocation–stress interaction constant; $\alpha_3 = \alpha_1 + \alpha_2/q_c$
α_4	dislocation–stress interaction constant; $\alpha_4 = \alpha_1 + q_{IV}\alpha_2/q_c$
β_1, β_2	constants
δ	average cell or subgrain size
$\delta_{III}, \delta_{IV}, \delta_s$	value of δ at the initiation of Stage III and IV, and at stress saturation, respectively
$\epsilon, \dot{\epsilon}$	respectively, uniaxial strain and strain rate
$\gamma, \dot{\gamma}$	respectively, shear strain and strain rate
$\bar{\gamma}$	equivalent shear strain
$\bar{\Gamma}$	stacking-fault energy
φ	sub-boundary misorientation
φ_{IV}	Stage IV sub-boundary misorientation
κ	geometric constant
ν_D	Debye frequency
ν_g	collapse frequency of dislocation dipoles
ν_r	dislocation annihilation rate in a one-parameter description
θ	work hardening rate
θ_h	athermal work hardening rate in a one-parameter description
θ_{II}	athermal work hardening rate in Stages II
$\theta_{III}, \theta_{IV}$	makrosopic work hardening rates in Stages III and IV, respectively
ρ	total density of stored dislocations
ρ_m	mobile dislocation density
ρ_i	density of free dislocations inside subgrains
ρ_b	density of cell wall/subgrain boundary dislocations
ρ_{III}	saturation density of free dislocations inside subgrains
σ	uniaxial normal stress
τ	resolved shear stress
τ_i	friction stress
τ_{III}	shear stress at deviation from linear hardening
τ_{III}^*	shear stress at the end of Stage II–III transition, corresponds to saturation in ρ_i ($\rho_i = \rho_{III}$)
τ_{IIIs}	Stage III saturation stress
ω, ω_{sa}	geometric constants
ξ_ρ, ξ_δ	stress intensity factors
C–S	Cottrell–Stokes
fcc	Face-centered cubic
H–D	Harper–Dorn
K–M	Kocks–Mecking
K–W	Kuhlmann-Wilsdorf
RT	room temperature
SFE	stacking-fault energy
TEM	transmission electron microscopy

1. INTRODUCTION

In order to solve the problems of work hardening and stress saturation the following three equations need to be defined and solved in combination

$$\tau = f(\tau_i, \rho) \quad (1)$$

$$\tau_i = f(\rho_m, \rho, \dot{\gamma}, T) \quad (2)$$

and

$$\frac{d\rho}{d\gamma} = \frac{d\rho^+}{d\gamma} + \frac{d\rho^-}{d\gamma} \quad (3)$$

The first equation defines the flow stress, τ , in terms of a frictional stress, τ_i , and the stored dislocation density, ρ . This latter quantity, however, is unfortunately not a single, well-defined parameter, but is a relevant measure describing the complex deformation sub-structure, $\rho = f(\rho_i, \delta, \varphi \dots)$ where ρ_i represents an elementary dislocation density term, δ a cell size, φ a cell or sub-boundary misorientation, etc. The second equation defines the frictional stress in terms of the density of mobile dislocations, ρ_m , the stored dislocations, the strain-rate and the temperature. The term 'frictional stress' is given a rather broad physical content here as by using this term we understand the total contribution to the flow stress due to short range interactions experienced by the mobile dislocations, i.e. adding together the lattice contribution (the Peierls stress) and contributions due to the interactions with atoms in solid solution, cutting of trees and the associated dragging of jogs. The dislocation density, ρ , is obtained by solving the third equation which is in a differential form. This relation simply reflects that dislocations are stored at a rate determined by the balance between deformation induced dislocations $d\rho^+/d\gamma$ and dislocations lost due to dynamic recovery $d\rho^-/d\gamma$. The dislocation storage term, $d\rho^+/d\gamma$, is defined by a differential equation of the form:

$$\frac{d\rho^+}{d\gamma} = \frac{2}{bL} \quad (4)$$

where $L = f(\rho)$ is the average distance the mobile dislocations migrate before being stored. This equation is an important element in all theories of work hardening. However, except for the standard one-parameter case, this equation has so far not been developed to a level which makes it applicable in a more general multi-parameter case where the objective is to define the storage rates of different microstructural elements, i.e. discriminate between $d\rho_i^+/d\gamma$, $d\delta^-/d\gamma$, $d\varphi/d\gamma$ etc.

The objective of the present work is to introduce a statistical approach to the storage of dislocations, from which the relevant $d\rho^+/d\gamma$ -terms are obtained and subsequently balanced with the dislocation annihilation effect due to dynamic recovery, equation (3). The dynamic recovery treatment will be based on a further development of recent work on static recovery by Nes⁽¹⁾, Furu *et al.*⁽²⁾, and Nes and Sæter⁽³⁾. As will be demonstrated in the following, the result becomes, in principle, a general solution to the problem of work hardening as defined by equations (1) and (2) equation (3) above. It needs to be emphasized, however, that although this multi-parameter approach to work hardening is capable of producing complete stress-strain curves for any specified combination of strain rate and temperature, this modelling effort will, necessarily, have to incorporate certain elements of empiricism in order to cover phenomena which are poorly understood. Some preliminary results of this reconsideration of the problems of work hardening have recently been published in two short communications^(4,5), works pertaining to the special problem of deriving constitutive laws for the steady state saturation stress. A short version of this work has also been presented at a recent conference on the modelling of thermo-mechanical processing of metals⁽⁶⁾. The problems of work hardening are classic ones within material science. The area has been repeatedly reviewed, with an excellent recent coverage by Gil Sevillano⁽⁷⁾.

A school of particular relevance for the present treatment is that initiated by Kocks⁽⁸⁻¹⁰⁾ and further consolidated as a result of a close collaboration over many years between

Kocks and Mecking^(11–14). In more recent years Estrin^(15–17) has made significant contributions to this tradition. The Kocks–Mecking–Estrin models, which are one-parameter approaches in terms of an average dislocation density, ρ , provide phenomenological descriptions of work hardening and steady-state creep. These theories have received widespread attention, and are extensively used in the modelling of metal plasticity. A useful reference to the works of Kocks and Mecking is the recent review by Rollett and Kocks⁽¹⁸⁾. The present work differs from those of Kocks, Mecking and Estrin in two ways; firstly, in terms of the number of parameters involved and secondly in terms of the interpretation and handling of some of the underlying physical phenomena responsible for dislocation storage and annihilation. In order to compare and contrast the different models a special Section 3.5.3 is devoted to a brief discussion and re-examination of the one-parameter approach to work hardening.

2. BACKGROUND

2.1. The Stress–Strain Curve

This paper is concerned with work hardening in fcc-metals, deformed at a constant temperature and strain-rate. It is well established that the flow curve at relatively low tempera-

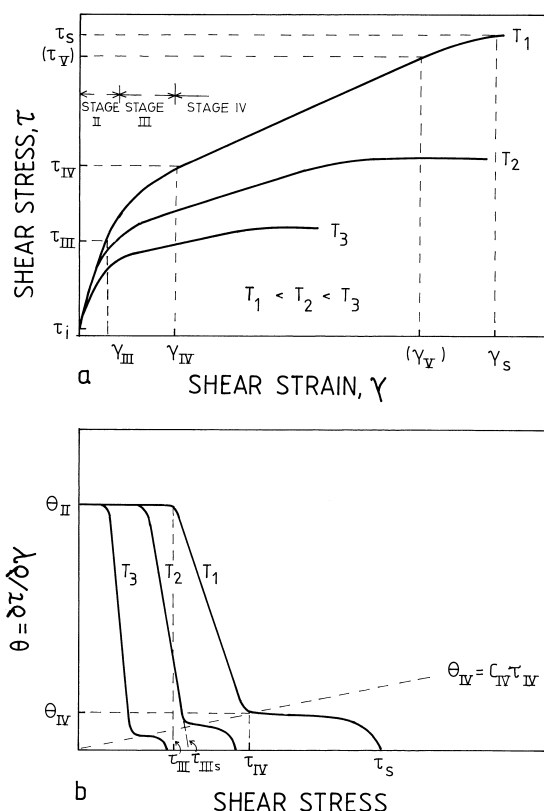


Fig. 1. (a) Schematic stress–strain curves and (b) strain hardening (θ) vs flow stress (τ) curves for fcc-metals

tures, $T < (0.3-0.4) T_m$, can be divided into four stages as indicated in Fig. 1(a). (The following treatment assumes that a multiple-slip situation applies for all strains, i.e. Stage I work hardening will be ignored, except for a brief reconsideration in relation to the modelling of Harper–Dorn creep, Section 3.6.1) The various stages of work hardening are most clearly distinguished on a diagram of θ vs τ where $\theta \equiv \delta\tau/\delta\gamma$, Fig. 1(b), a plot commonly referred to as a Kocks–Mecking diagram.

Stage II is characterized by a linear work hardening rate $\theta \simeq G/200$ (for multiple-slip) which is only weakly sensitive to temperature and strain rate (except for the temperature dependence of the shear modulus G).

Stage III begins when the flow curve deviates from linearity. This stress, τ_{III} , is strongly strain rate and temperature dependent, a dependence associated with a thermally activated process where the rate controlling reaction is generally believed to be cross-slip.

Stage IV is most easily distinguished from the θ - τ plot [Fig. 1(b)]. In the work hardening curve this stage often appears with a linear hardening rate (although deviation from linearity is frequently observed) with $\theta_{IV} \approx 2 \cdot 10^{-4} G$. Another characteristic aspect associated with the temperature dependence of Stage IV and indicated in Fig. 1(b) is that θ_{IV} is linearly dependent on τ_{IV} (i.e. $\theta_{IV} = C_{IV}\tau_{IV}$), C_{IV} typically having a value of 0.1^(7,16,19). Following the logic of this flow curve characterization Zehetbauer and Seumer⁽²⁰⁾ introduced a Stage V before reaching the saturation stress τ_s . It follows from the treatment below, however, that within the context of the present analysis such a Stage V seems to be redundant.

The temperature dependence of the stress–strain curve is indicated in a qualitative way in Fig. 1. Copper has been most extensively investigated and the temperature dependence of τ_{III} , $\tau_{III,s}$ [$\tau_{III,s}$: τ_{III} -saturation is extrapolated from the θ - τ plot, see Fig. 1(b)], τ_{IV} and τ_s in this metal is illustrated in Fig. 2. It can be seen that each of these quantities decreases ap-

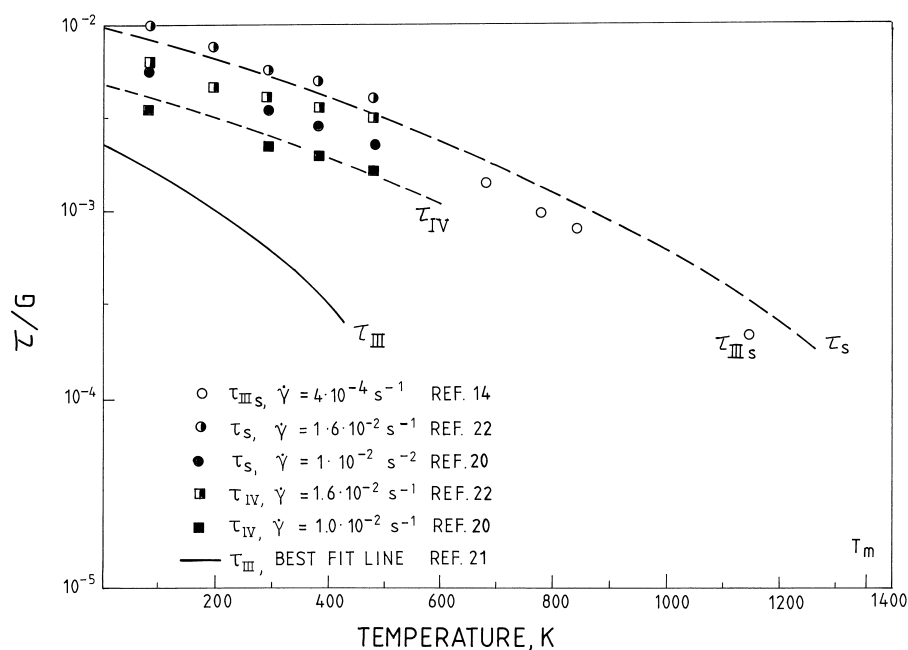


Fig. 2. Temperature dependence of the saturation stresses (τ_{III} , $\tau_{III,s}$, τ_{IV} and τ_s) in copper, from Refs. (14, 20–22)

proximately exponentially with temperature, the 0K values being $\tau_{III}^0 = 2.3 \cdot 10^{-3} G_o$, $\tau_{IV}^0 = 4.8 \cdot 10^{-3} G_o$ and $\tau_s^0 = 9 \cdot 10^{-3} G_o$. It is noted that while τ_{IV} , τ_{IIIs} and τ_s appear to decline with temperature at approximately the same rate, τ_{III} falls off more steeply. In analytical terms the temperature and strain-rate dependence of any of these stresses is usually expressed by (see Ref.⁽²¹⁾):

$$\ln \frac{\tau_{III,IV,s}}{G} = \ln \frac{\tau_{III,IV,s}^0}{G} + \frac{kT}{Gb^3} \ln \frac{\dot{\gamma}}{K_1 \nu_D} \quad (5)$$

where ν_D is the Debye frequency and K_1 is a constant depending on the metal. Mecking *et al.*⁽¹⁴⁾ have carefully examined the variation in the τ_{III} -saturation-stress (τ_{IIIs}) with tempera-

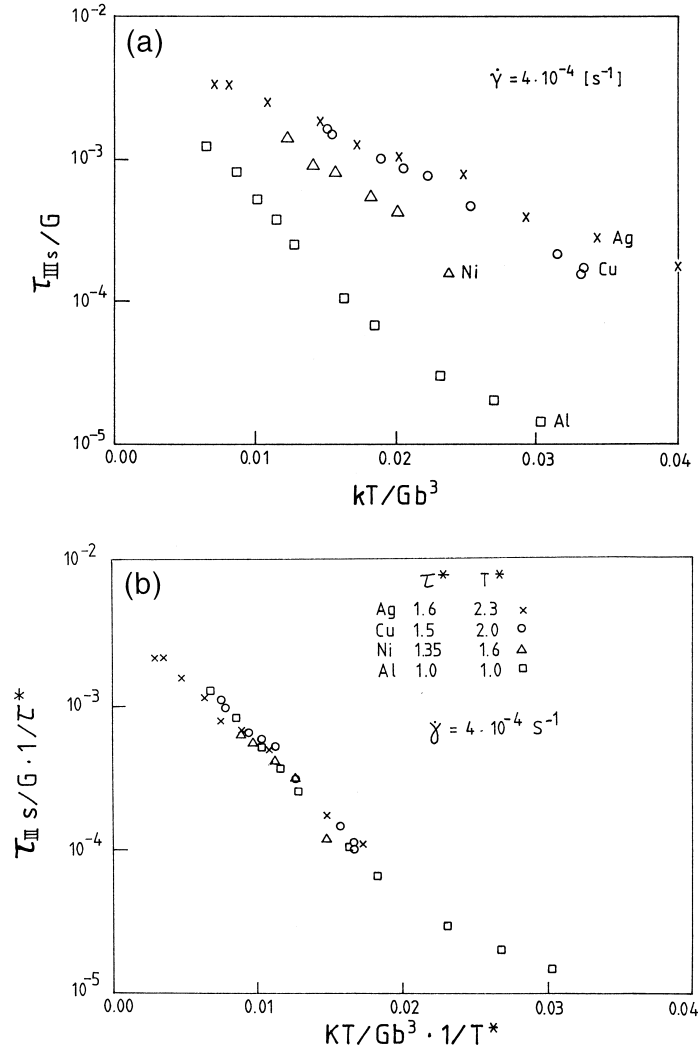


Fig. 3. (a) Stage III saturation stress ($\tau_{III s}$) as a function of normalized temperature (kT / Gb^3), from Ref.⁽¹⁴⁾. (b) The results in (a) with the saturation stress $\tau_{III s} / G$ scaled with τ^* and the normalized temperature kT / Gb^3 scaled with T^*

ture for four different fcc metals, Fig. 3(a), from which it can be seen that equation (5) applies in each case. They further demonstrated that by introducing appropriate scaling factors, τ^* and T^* , the different curves in Fig. 3(a) would all fall on the same master line, Fig. 3(b). Such a scaling, however, is of no relevance *per se*, the interesting aspect is that Mecking *et al.* demonstrated that these scaling factors apparently correlated with the stacking fault energy.

The theory on the initiation of Stage III dates back to the mid 1950s in work by Seeger and coworkers^(23–25), later commonly referred to as the Stuttgart school. This school regards Stage III as a breakdown of Stage II which arises because cross-slip occurs when the local pile-up stresses become high enough. The model predicts an exponential variation in τ_{III} with temperature, in accordance with the results presented in Fig. 2. This pile-up based theory, however, is still controversial, a dispute which dates back to Friedel's treatise of 1957⁽²⁶⁾. A current follow-up is the review by Nabarro⁽²⁷⁾ where he writes: "Although it leads to satisfactory estimates of stacking-fault energies, the theory of the onset of Stage III is open to *severe* criticism". The theory on τ_{III} developed by the French school^(26, 28, 29), in contrast to that of Seeger and coworkers, predicts τ_{III} to vary linearly with temperature. This Friedel–Escaig cross-slip theory does not require pile-ups but relies on thermal activation of cross-slip from constricted regions of screw dislocations. In a more recent treatment of the mechanics and energetics of cross-slip by Duesbery *et al.*⁽³⁰⁾ the 'French approach' is favoured although it is argued that significant discrepancies between the theoretical and experimental orientation dependence of the cross-slip stress cast doubt on prevailing mechanistic theories. It needs to be added that cross-slip is not only associated with Stage II breakdown, but is commonly believed to control dynamic recovery throughout Stage III⁽¹⁴⁾.

The opinion of present author is that the physics of cross-slip has not been developed to a level sufficient to be incorporated into models for crystal plasticity, a view which will be illustrated here by considering predictions derived from either the French or the Stuttgart schools. On the other hand, in order to understand the effect of dynamic recovery on work hardening in general (not only Stage II breakdown), cross-slip based mechanisms must, of course, be considered. It is, however, the present opinion that the importance of climb in this context, in general, has been overlooked. Therefore, the present work will focus on an alternative approach based on thermal activation of vacancies. The general view has been that a climb based theory, involving exchanges of vacancies in terms of local equilibria and diffusion⁽³¹⁾, cannot be applied in the ambient temperature regime. The present author supports that view, but that does not exclude climb based on stress enhanced thermal activation of vacancies⁽³¹⁾ involving no long-range diffusion. It is interesting to note that Nabarro⁽²⁷⁾, after reviewing the operation of five possible temperature-dependent mechanisms for dynamic recovery, suggests that it may be controlled by climb mediated by vacancies produced by plastic deformation.

Based on extensive investigations of the steady state conditions for creep and hot working Sellars and McG. Tegart⁽³²⁾ demonstrated that the saturation stress was well represented by the following relation:

$$\tau_s = K_2 \operatorname{Arcsinh} \left(\frac{Z}{K_3} \right)^{n'} \quad (6)$$

where K_2 , K_3 and n' are constants. $Z = \dot{\epsilon} \exp(U/kT)$ is the Zener–Hollomon parameter* where U is an activation energy which at temperatures $T > T_m/2$ is found to be approximately equal to that of self diffusion. equation (6) can be applied only at relatively high temperatures. That this equation fails at low temperatures is easily seen from the fact that it predicts an infinite saturation stress at 0K. Accordingly, the conclusion becomes that both equations (5) and (6) can only be taken as approximate empirical expressions for the temperature and strain-rate dependency of τ_{III} , $\tau_{III,s}$, τ_{IV} and τ_s . No generally accepted theory has been developed which predicts the temperature and strain rate dependence of these strength parameters from first principles.

A general comment is needed here. It has been common to approach the problems of work hardening of metals in different ways depending on the temperature of deformation. When applying the phrase ‘work hardening’, one usually associates deformations at temperatures from 0K to room temperature, while deformation at temperatures higher than $0.4 T_m$ generally is referred to as hot deformation or creep. Although there may be practical reasons for such a distinction, it becomes artificial when studying the basic mechanisms of strain hardening of pure metals, provided the effects due to allotropic transformations and dynamic recrystallization can be excluded. In the analysis of the hot deformation properties of metals found in the literature, rate reactions controlled by diffusion are readily accepted, while in the low temperature region such reactions are frequently excluded. However, the results given in Figs 2 and 3 indicate no major shift in mechanism (controlling the saturation stress) in the temperature region from RT (or below) to $0.4 T_m$. The attitude taken in the following is that the mechanisms controlling the storage of dislocations are in principle temperature invariant. And it follows that the change in deformation behaviour with temperature reflects the rate changes due to mechanisms which are controlled by thermal activation. As more than one mechanism is involved, the balance between different mechanisms may of course change with temperature. A similar attitude was also expressed by Nix *et al.*⁽³³⁾ in their approach to time-dependent deformation of metals.

2.2. The Microstructural Evolution

The evolution of microstructure during deformation has been extensively studied over the last decades: for excellent reviews see Refs.^(34–36). With our modelling objective in mind we want to extract from this literature a general and yet simple picture, and we have selected a microstructural description which includes three global parameters: (i) The dislocation density inside the cells, ρ_i , (ii) the cell/subgrain size, δ , and (iii) the cell/sub-boundary misorientation ϕ (or at low strains the cell wall thickness and dislocation density in the wall).

2.2.1. The subgrain size

If metals are deformed at room temperature one finds that the variation in the average subgrain/cell size with strain follows a relationship which is of a similar type for a range of metals, and independent of the mode of deformation as shown in Fig. 4, taken from the work by Gil Sevillano *et al.*⁽³³⁾. In Fig. 4 the variation in the inverse cell size is also given (broken line), and it can be seen that this quantity varies linearly with strain for $\bar{\epsilon} > 1$. This is an interesting observation in view of the fact that the flow stress is commonly correlated

*The strain rate $\dot{\epsilon}$ and the shear strain rate $\{\dot{\gamma}\}$ are related through $\dot{\gamma} = M\dot{\epsilon}$ where M is the Taylor factor.

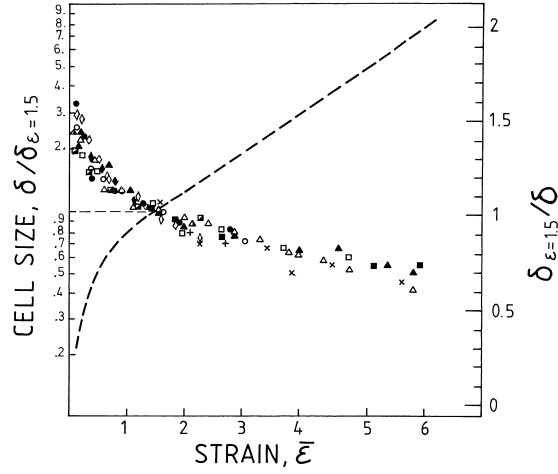


Fig. 4. Subgrain/cell size as a function of equivalent strain, from Ref.⁽³⁴⁾

with the inverse cell size. Accordingly, this result is consistent with a linear Stage IV work hardening behaviour. From Fig. 4 it follows that deformation at room temperature for strains $\varepsilon > \gamma/M \approx 1$ gives:

$$\frac{1}{\delta} = (0.7 + 0.09\gamma)\delta_{\varepsilon=1.5} \quad (7)$$

which, following the assumption that the flow stress scales with the inverse cell size (see equation (11) below), gives a Stage IV work hardening rate of $\theta_{IV} = 0.09\alpha_2 Gb/\delta_{\varepsilon=1.5}$. For Cu, $\delta_{\varepsilon=1.5} = 0.29 \mu\text{m}$, $\alpha_2 \approx 2.5$ and $b = 2.56 \cdot 10^{-10} \text{ m}$ (Gil Sevillano *et al.*⁽³⁴⁾) which gives $\theta_{IV} \approx 2 \cdot 10^{-4} \text{ G}$, a very reasonable RT-value indeed.

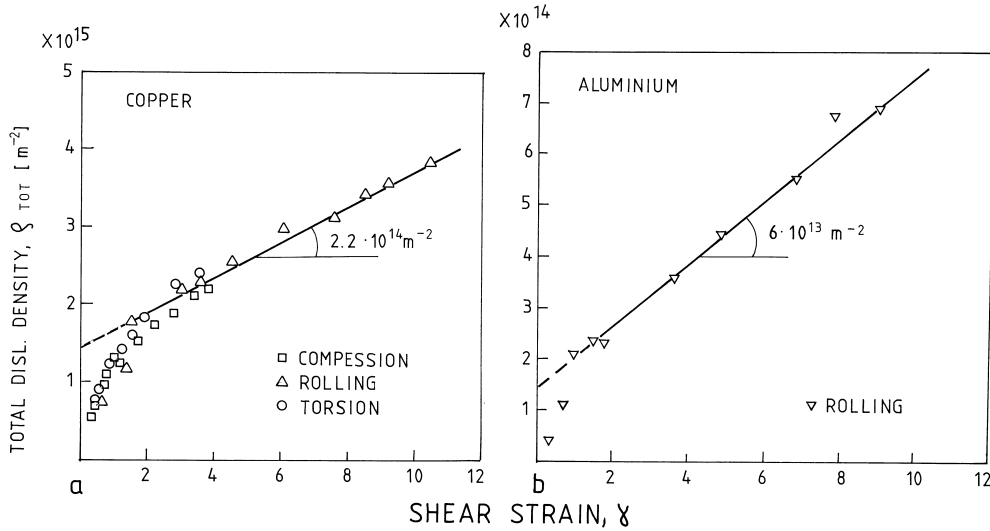


Fig. 5. Total dislocation density as a function of strain. (a) copper⁽²⁰⁾ and (b) aluminium⁽³⁸⁾

2.2.2. The dislocation density in the subgrain interior

Although the substructure evolution during cold deformation of metals has been the subject of extensive investigations in TEM over the last decades little quantitative information is available regarding the dislocation density inside the subgrains (ρ_i). However, the impression derived from the literature is that the variation in ρ_i with strain during large strain deformation is not on the same scale as that found for the subgrain size δ . The following quotation from the classic work on wire drawing (iron) by Langford and Cohen⁽³⁷⁾ seems to be representative: “Most of the dislocations present are confined to the cell walls; the average dislocation density within the cells is comparatively low”. Speculation about the extent to which this dislocation density saturates during Stage IV has been brought up by, for instance, Nix *et al.*⁽³³⁾.

While quantitative TEM observations on the variation in ρ_i are lacking, more indirect measurements of dislocation densities based on resistivity and calorimetry measurements have recently been reported by Zehetbauer and Seumer⁽²⁰⁾ and Zehetbauer⁽³⁸⁾. Figure 5(a) and (b) give their results as total dislocation density vs strain for copper and aluminium respectively. The results clearly indicate that the total dislocation density increases linearly with strain in Stage IV ($\gamma > 1.5$). It is interesting to compare these ρ_{tot} vs γ relationships to that which can be derived from the data compiled by Gil Sevillano *et al.*⁽³⁴⁾ (Fig. 4) and expressed as $\delta_{e=1.5}/\delta$ vs γ by equation (7). On the assumption that ρ_i prior to Stage IV has reached a saturation level ρ_i^s , an estimate for the total dislocation density during Stage IV deformation can be expressed by

$$\rho_{tot}^{IV} = \rho_i^s + \frac{\kappa\varphi}{b\delta} \quad (8)$$

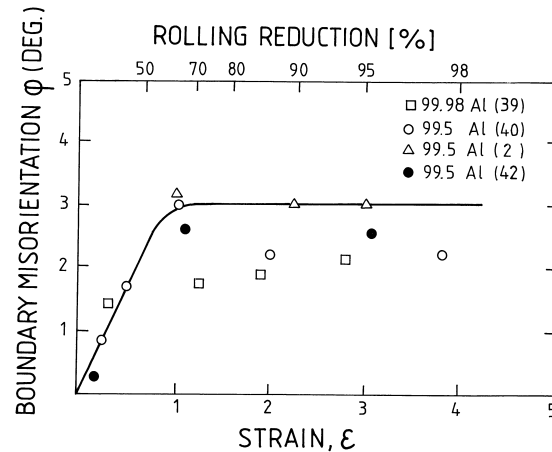
where φ is the sub-boundary misorientation which during Stage IV has reached a saturation level of about 3 degrees (see following subsection) and for definition of κ see equation (10). Combining equations (7) and (8) the total dislocation density vs strain becomes:

$$(\rho_{tot}^{IV})_{Cu} = (\rho_i^s + 1.5 \cdot 10^{15} + 2 \cdot 10^{14}\gamma)m^{-2}, \text{ and } (\rho_{tot}^{IV})_{Al} = (\rho_i^s + 4 \cdot 10^{14} + 5 \cdot 10^{13}\gamma)m^{-2} \quad (9)$$

Note that these relationships predict a linear variation in the total dislocation density in accordance with the experimental results presented in Fig. 5. In deriving these expressions the following additional observations reported by Gil Sevillano *et al.*⁽³⁴⁾ have been used: $\delta_{e=1.5,Cu} = 0.29 \mu\text{m}$ and $\delta_{e=1.5,Al} = 0.6 \mu\text{m}$. Both these expressions are in satisfactory agreement with the experimental results presented in Fig. 5(a) and (b). The conclusion is that the assumption that ρ_i is constant in Stage IV seems reasonable.

2.2.3. The average sub-boundary misorientation

The variation in the average sub-boundary misorientation, φ , with increasing strain has not been the subject of such an extensive investigation as that of the scale of the substructure, for obvious experimental reasons. With the new techniques, convergent beam diffraction in TEM and channelling and back scattering patterns in SEM, this is rapidly changing. But earlier work based on TEM/SAD and Kikuchi patterns and other methods did provide some valuable information on sub-boundary misorientation. By measuring inter-dislocation spacing in sub-boundaries in aluminium deformed to various strain levels, Schuh and von

Fig. 6. Sub-boundary misorientation vs strain⁽²⁾

Heimendahl⁽³⁹⁾ found that the boundary misorientation varied with strain as shown in Fig. 6. As can be seen from this figure, their results agree with later observations by Nes *et al.*⁽⁴⁰⁾, using Kikuchi patterns from individual subgrains. This diagram clearly shows that the average boundary misorientation increases rapidly with strain, reaching about 2° – 3° at a strain of about 1, after which it remains constant up to rolling strains as high as 4. A similar trend has also been reported in copper⁽⁴¹⁾. Later work based on convergent beam in TEM and EBSD in SEM confirms these results⁽²⁾. Figure 6 also includes observations from high temperature deformation of aluminium⁽⁴²⁾, and it is interesting to note that the same variation in boundary misorientation with strain is observed irrespective of the temperature of deformation. It needs to be pointed out that this saturation of the boundary misorientation refer to sub-boundaries within stable texture components. The effects on the average boundary misorientation due to 'old' grain boundaries or transition banding have purposely been excluded.

2.3. Theories of Work Hardening

No attempt will be made here to review the various theories, the exception being a discussion of some of the basic elements in the Kocks–Mecking model (Section 3.5.3). The problems of work hardening are defined in the introductory section above and a solution to these problems require a different approach than that tried so far. It is the opinion of the present author that the traditional approach to work hardening is 'over-detailed' in focusing on subtle dislocation interactions within the context of a one-parameter substructure description. On the other hand, observations which relate directly to the storing of dislocation during plastic deformation, as illustrated in Figs 4–6, are generally ignored in all existing theories. It is true that recent efforts do acknowledge that more than one single microstructural parameter is required in order to establish an improved theoretical description of work hardening^(33,43,44). However, none of these contributions have succeeded in reformulating the basic equations (3) and (4) to be consistent with a more suitable picture of the substructure evolution.

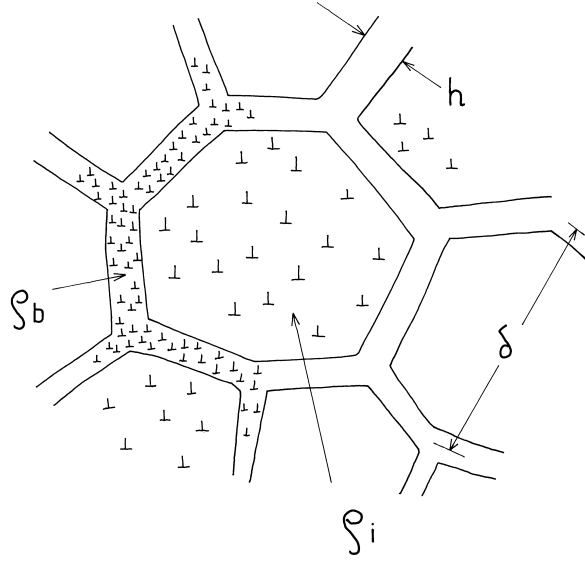


Fig. 7. A schematic representation of the microstructure; cell diameter, δ , cell wall thickness, h , cell wall dislocation density, ρ_b and dislocation density within the cells, ρ_i

3. THEORY

3.1. Identification of the Relevant Microstructural Elements and their Effect on the Flow Stress

In order to solve the work hardening problem as defined in Section 1 above the first objective becomes to define the content of the microstructural term labelled ρ (equations (1) and (3)). What is needed is a description of the microstructural evolution in terms of a limited number of parameters without unduly simplifying the problem. Based on the experience gained in modelling static recovery⁽¹⁻³⁾ the two-parameter internal-state variable approach used there will be further developed here.

The following theory rests on the assumption that at any stage during the deformation of a pure metal the substructure can be adequately described by a few microstructural elements, the two most important ones being the cell/subgrain size δ and the dislocation density in the cell interior, ρ_i . With reference to the sketch in Fig. 7, the following additional elements are required in the present 'composite' description of the deformation substructure. The cell structure consists of a volume fraction of boundaries $f \approx \kappa h / \delta$ having a 'local' dislocation density ρ_b (h is the cell-boundary thickness and κ is a geometric constant which for a regular cell structure is of the order 3; this expression for f applies only as long as $h \ll \delta$). The density of stored dislocations, ρ , then becomes:

$$\rho = (1 - f) \rho_i + f \rho_b \quad (10)$$

Based on this microstructural description several possible approaches can be taken in order to calculate the flow stress, as discussed in detail in Ref.⁽³⁾. Two interesting treatments are the link-length model by Kuhlmann-Wilsdorf^(45,46) and the composite theory of Mughrabi^(47,48) and Pedersen *et al.*⁽⁴⁹⁾. The last of these approaches predicts a relationship

of the form

$$\tau = \tau_i + \alpha_1 Gb\sqrt{\rho_i} + \alpha_2 Gb\frac{1}{\delta} \quad (11)$$

where α_1 and α_2 are constants, the appropriate values of which will be discussed in Section 3.3 below and τ_i is a frictional stress (see comments to equation (2)). The significance of the frictional stress contribution to the flow stress will be the topic of Section 5 below, a section which also includes some comments to the kinetic-relationship used in the present analysis. Prior to this treatment, the τ_i -contribution is assumed to be of second order and accordingly will be ignored.

Nes and Sæter⁽³⁾ have recently presented a critical evaluation of the general validity of using equation (11) in calculating the flow stress, both in the as-deformed state and during subsequent recovery annealing. The effect of the cell wall dislocation density or sub-boundary misorientation on the flow stress was considered in particular. Application of equation (11) requires either that the flow stress is a function of the cell/subgrain size *per se*, or the square root of the cell wall dislocation density scales with the inverse cell size (the latter assumption is implicit in the Kuhlmann-Wilsdorf's^(45,46) link-length theory). The mechanism of sub-boundary strengthening will not be discussed here, but experimental evidence presented by Nes and Sæter shows that as long as $h \ll \delta$ the magnitude of the dislocation density and their arrangement within the cell walls has no (first order) effect on the flow stress. Furthermore, in line with this observation, Nes and Sæter⁽³⁾ also provided evidence showing that the magnitude of the sub-boundary misorientation in well recovered substructures has no (first order) effect on the flow stress. *The conclusion becomes that the spacing of the walls/boundaries rather than their internal structure controls the flow stress.* This conclusion violates both the link-length theory and the composite model⁽⁴⁸⁾, or in other words no current flow stress theory seems capable of accounting for the general validity of equation (11). Although the evidence presented by Nes and Sæter⁽³⁾ in support of equation (11) seems clear, this evidence is after all limited and other interpretations have been offered. Accordingly, since this conclusion has not previously been spelled out equally clearly, it follows that more experimental and theoretical work is needed in order to understand the effect of the cell wall/sub-boundary structure on the flow stress. However, the following work hardening model will be based on this simple flow stress equation, and the work hardening problem is 'reduced' to the solving of equation (3) for a microstructural situation where the stored dislocations are split between cell-boundaries and a cell-interior forest sub-structure.

3.2. The Principles of Dislocation Storage

In addition to the general microstructural description given above, the present theory rests on the assumption that the storage of dislocations can be treated in a statistical way, the rules of which are independent of stress, strain, strain rate and temperature. The basis of such an approach is that during deformation a given mobile dislocation travels on average a distance L before being stored, which is long in comparison to the spatial separation of the microstructural elements with which it interacts, i.e. $L\sqrt{\rho}$ is typically in the range 50–100 and L/δ is of the order 10. But this is only as seen in terms of numbers of intercepts in the radial direction of migration. When an average dislocation loop has been stored, it has interacted with about 10^4 individual dislocations and cut through or encircled about 100 cells. A reasonable approach must be that such an average dislocation loop, in covering

the area (πL^2), will experience any part of it as rather similar. Accordingly, the reaction which eventually locks it must be of a statistical nature. Furthermore, the present statistical approach simply recognizes that the dislocations have a natural tendency to distribute themselves into a cell structure, which, for the purpose of the subsequent treatment is characterized in terms of the microstructural parameters introduced above (Fig. 7). In this respect, the present treatment differs from that of Kocks⁽¹⁰⁾ who used a statistical approach to rationalize the occurrence of cell formation and the associated scaling phenomenon which ties the cell size to the elementary obstacle spacing.

The statistical approach to the storage of dislocations proposed here postulates that the substructure evolution during plastic straining is regulated through three fundamental storage parameters labelled p , p_{nb} and f_{ob} . As will be demonstrated below, this postulate makes it possible to obtain a set of rules regulating dislocation storage, on the bases of which the problems of work hardening as defined in Section 1 can be solved.

The parameter p defines the statistical chance (per unit area of slip plane and per unit total dislocation density, ρ) for a moving dislocation to become stored in the substructure. It then follows that if an increment in strain, $\Delta\gamma$, causes the dislocation sources to induce an additional number, N_m , of mobile dislocations loops per unit volume, dN of these loops will be stored when the loops expand from r to $r + dr$, where:

$$dN = (N_m - N)2\pi r p \rho dr \quad (12)$$

where N is the number of loops stored between $r = 0$ and $r = r$. As the loops expand from the source towards infinity, this number as a function of r becomes:

$$N(r) = N_m(1 - \exp - \pi r^2 p \rho) \quad (13)$$

The statistical average radius $\bar{r} = L$ to which a dislocation loop expands before being stored is given by:

$$\begin{aligned} L &= \frac{1}{N_m} \int_0^{N_m} r dN = 2\pi \int_0^\infty r^2 p \rho (\exp - \pi r^2 p \rho) dr \\ &= \frac{C}{\sqrt{\rho}} \end{aligned} \quad (14)$$

where $C = 1/(2p^{1/2})$. A relationship of this form appears in all work hardening theories in order to account for a linear Stage II-behaviour. It is usually justified on the basis of the principle of similitude (see below) or empirically, based on slip-line measurements.

In terms of the present microstructural representation (Fig. 7), the N_m loops will be stored in one of three principal ways, i.e., within the cell interior, as new cell boundaries or in old cell boundaries. The parameter p_{nb} defines the statistical probability (per unit area and per unit dislocation density) for a dislocation moving through a forest of *uniform* dislocation density, $(1-f)\rho_i$, to become stored in a *new* boundary. Accordingly, when the loops expand from r to $r + dr$, dN_{nb} of the N_m loops will become stored in new boundaries, where:

$$dN_{nb} = (N_m - N)(1 - f_{ob})2\pi p_{nb}(1 - f)\rho_i r dr \quad (15)$$

Here f_{ob} is the fraction of loops which will be trapped in old boundaries, i.e. $N_{ob} \equiv f_{ob}N_m$. By combining equations (13)–(15) it follows that:

$$\begin{aligned}
N_{nb} &= (N_m - N_{ob}) \int_0^\infty 2\pi p_{nb}(1-f)\rho_i r \exp -\frac{\pi r^2}{L^2} dr \\
&= 4(N_m - N_{ob})p_{nb}(1-f)\rho_i L^2 = 4N_m(1-f_{ob})(1-f)p_{nb}\rho_i L^2
\end{aligned} \tag{16}$$

As the number of mobile dislocation loops, N_m , is the result of an incremental strain $\Delta\gamma$, it follows that the corresponding increase in dislocation density becomes $\Delta\rho = 2\pi L N_m$, and equation (16) can be written (in differential from):

$$d\rho = d\rho_{ob} + \frac{1}{4p_{nb}(1-f)\rho_i L^2} d\rho_{nb} = \frac{1}{S(1-f)\rho_i L^2} d\rho_{nb} \tag{17}$$

where the storage parameter $S = 4(1-f_{ob})p_{nb}$. The objective of this statistical approach is to establish a ‘book-keeping system’ capable of accounting for the dislocations stored at a given strain. Such a ‘system’ is required in order to solve the fundamental equation (3). Or, in other words, the objective is to describe in analytical terms how the stored dislocations are distributed, defining the cell structure. Equation (17) has the potential of taking care of this ‘book-keeping’ problem. Within the context of the present theory, this expression provides a solution to the problem of athermal storage of dislocations. In order to solve equation (17), however, the parameters p , p_{nb} and f_{ob} must be defined, which becomes an important objective in the continued treatment below. Another important aspect related to equation (17) is that it is consistent with *microstructural scaling (similitude)* during deformation (at zero degree Kelvin), at stresses less than τ_{III} , i.e. in the absence of dynamic recovery. Before demonstrating this, the principle of scaling in the present context needs to be defined.

The principle of scaling means that the substructure is ‘self-similar’ at any strain except for the scale. In terms of the microstructural picture drawn above (Fig. 7), similitude requires

$$\sqrt{\rho_i} = q_c \frac{1}{\delta} \tag{18}$$

$$h = q_h \delta \tag{19}$$

$$\sqrt{\rho_b} = q_b \sqrt{\rho_i} \tag{20}$$

where q_c , q_h and q_b are constants, the values of which are discussed below. Equation (20) simply expresses that the separation of dislocations in the cell boundaries is $1/q_b$ -th of that in the cell interior. With this definition of substructure scaling, the stored dislocation density can be expressed in various ways:

$$\rho = (1-f)\rho_i + \kappa q_h (q_c q_b)^2 \frac{1}{\delta^2} = (1+f(q_b^2 - 1))\rho_i = (1+f(q_b^2 - 1))\frac{q_c^2}{\delta^2} \tag{21}$$

As long as $h \ll \delta$, $f = \kappa q_h$ becomes a good approximation. A cell free material corresponds to $q_c \rightarrow \infty$ or $q_b = 1$. These two parameters, however, are not independent of each other. If q_c increases q_b will decrease, the boundary condition being that as q_c increases q_b will approach unity. On the other hand, a substructure consisting of cell boundaries only ($\rho_i = 0$) corresponds to $q_b \rightarrow \infty$.

The condition defined by equation (18) is an important new element compared to previous work hardening treatments. Kuhlmann-Wilsdorf⁽⁴⁶⁾ defined similitude on the bases of equation (19) and 20 only ($\rho_i=0$ in her theory). It is the opinion of the present author that without equation (18) the necessary physical condition for scaling has not been established. In the K-W treatment the cell structure scales with the dislocation density inside the cell boundaries, and it follows that new boundaries will have to be the result of chance-reactions between the mobile dislocations when crossing the ‘free space’ between cell boundaries. In the present approach such chance-reactions are seen as being of low probability. However, this probability will be considerably enhanced by the moderating effect of the dislocation forest (ρ_i) on the mobile dislocations (i.e. $dN_{nb} \propto p_{nb}\rho_i$, equation (15)). An important consequence of this is a *self-consistent substructure evolution model*, as will now be demonstrated.

It follows from equation (10) that an increase in strain by an infinitesimal amount $d\gamma$ will cause an increase in the dislocation density by

$$d\rho = (1-f)\frac{\partial\rho_i}{\partial\gamma}d\gamma + \frac{\kappa}{\delta}\frac{\partial(h\rho_b)}{\partial\gamma}d\gamma + \kappa h\rho_b\frac{\partial}{\partial\gamma}\left(\frac{1}{\delta}\right)d\gamma \quad (22)$$

These dislocations are stored in three principal ways:

1. In the forest network in the cell/subgrain interior:

$$d\rho_i = \frac{\partial\rho_i}{\partial\gamma}d\gamma \quad (23)$$

2. In ‘old’ cell boundaries:

$$d\rho_{ob} = \frac{\kappa}{\delta}\frac{\partial(h\rho_b)}{\partial\gamma}d\gamma \quad (24)$$

3. and in ‘new’ cell boundaries:

$$d\rho_{nb} = \kappa h\rho_b\left(\frac{\partial}{\partial\gamma}\frac{1}{\delta}\right)d\gamma \quad (25)$$

However, when the principle of scaling applies these differential terms become coupled to each other: $d\rho_{ob} = d\rho_{nb} = (fq_b^2/2)d\rho_i$. In such a case it follows from equation (22) that $d\rho$ can be expressed in terms of $d\rho_{ob}$ and/or $d\rho_{nb}$:

$$d\rho = d\rho_{ob} + (2(1-f)/fq_b^2 + 1)d\rho_{nb} = 2(1 + (1-f)/fq_b^2)d\rho_{nb} \quad (26)$$

The important point here is that this expression, like equation (17), relates $d\rho$ to $d\rho_{nb}$. At zero degree Kelvin, and for stresses less than τ_{III} , all N_m dislocations are stored, and by combining equations (14), (17) and (21) an expression which is identical to equation (26) is

obtained, provided the storage parameter $S = S_{sc}$ is defined as

$$S_{sc} \equiv \frac{fq_b^2}{2C^2(1-f)} \quad (27)$$

Since $S \equiv 4(1-f_{ob})p_{nb}$ and $d\rho_{nb} = d\rho_{ob} \equiv f_{ob}d\rho$ (equation (15)) it follows that the storage parameters f_{ob} and p_{nb} become

$$f_{ob} \equiv \frac{fq_b^2}{2[1+f(q_b^2-1)]} \quad (28a)$$

$$p_{nb} \equiv \frac{[1+f(q_b^2-1)]fq_b^2}{4C^2(1-f)[2[1+f(q_b^2-1)]-fq_b^2]} \quad (28b)$$

3.2.1. A general storage principle

With the expression for $d\rho$ given by equation (17) it follows that equation (4) can be written in the following general form:

$$\frac{d\rho_{nb}^+}{d\gamma} = \frac{2CS(1-f)\rho_i}{b\sqrt{\rho}} \quad (29)$$

where in the special case of microstructural scaling $S = S_{sc}$ is given by equation (27). It is reiterated that this statistical approach to dislocation storage is consistent with microstructural scaling at 0 K and predicts the average slip length to be $L = C/\sqrt{\rho}$, the latter applies whether we have scaling or not.

While the basic elements included in the present cell structure model are observed at all stages of deformation, independent of strain-rate and temperature, it is, of course, not difficult to point out characteristic differences in comparing the substructure generated during different stages of deformation. Still, however, as a first approach towards a general theory of work hardening, equation (29) is assumed to have general validity and the objective of the following treatment is to apply this differential equation in the modelling of all stages of deformation, at any temperature and strain rate.

3.3. The Flow Stress and the Constants α_1 and α_2

The constants α_1 and α_2 in equation (11) need to be more precisely defined. The way in which the resolved shear stress depends on the dislocation density has been investigated in numerous studies, see the review by Gil Sevillano⁽⁷⁾. Basinski and Basinski⁽⁵⁰⁾ have compiled data spanning more than six orders of magnitude of dislocation densities, the linear regression of which gives

$$\tau = \alpha Gb\sqrt{\rho} \quad (30)$$

where $\alpha = 0.14/\ln(b\sqrt{\rho})$. In practical terms the logarithmic dependency on ρ in α can be ignored and $\alpha \simeq 0.3$ is the value to be used in this treatment.

The data collected by Basinski and Basinski⁽⁵⁰⁾ are from metals which all develop cell structures. If we assume that during Stage II the principle of scaling is approximately fulfilled, then the combination of equations (11), (21) and (30) gives

$$\alpha = \frac{1}{\sqrt{1 + f(q_b^2 - 1)}} \left(\alpha_1 + \frac{\alpha_2}{q_c} \right) \quad (31a)$$

since $\alpha_1 \equiv \alpha$ this gives the following relationship between the scaling parameters f , q_c and q_b :

$$q_c = \frac{\alpha_2}{\alpha_1 (\sqrt{1 + f(q_b^2 - 1)} - 1)} \quad (31b)$$

This latter equation is consistent with the definition of microstructural scaling given above in the sense that for a cell free material $q_c \rightarrow \infty$ when $q_b \rightarrow 1$. Further, for a pure boundary structure ($\rho_i = 0$) $q_c \rightarrow 0$ when $q_b \rightarrow \infty$.

Equation (11) will be applied irrespective of whether the principle of scaling applies or not, which implies that the constant α_2 needs to be determined based on experimental observations^(3, 37, 50). Typical α_2 -values to be used in the following will be in the range 2–3. For a given α_2 -value, equation (31b) provides a consistent flow stress description between scaling and non-scaling situations.

3.4. Dynamic Recovery

Recovery during static annealing of deformed metals has recently been treated by Nes⁽¹⁾, Furu *et al.*⁽²⁾ and Nes and Sæter⁽³⁾. These works demonstrate that static recovery can be satisfactorily treated on the basis of an internal state variable approach comprising two variables, i.e. the cell/subgrain size and the dislocation density in the cell interior. A similar approach will be tried here in handling the dynamic case.

From the literature, dynamic recovery emerges as a rather less well-defined reaction. Kocks⁽¹⁰⁾ defines dynamic recovery in terms of the rate sensitivity involved: “Dynamic recovery is characterized by a weak rate sensitivity as compared to that of simultaneous *static* recovery”. In a work on time dependent deformation of metals by Nix *et al.*⁽³³⁾, dynamic recovery is distinguished from static recovery processes in terms of the nature of the driving forces involved, the applied stress in the former and the defect interaction in the latter. There is, however, a fundamental difference in the way dynamic recovery is conceived, comparing the views of the Kocks–Mecking school and that of Prinz and Argon⁽⁴³⁾, Nix *et al.*⁽³³⁾, the present author and others. According to Mecking⁽⁵¹⁾: “In dynamic recovery the dislocation annihilation is not due to the fact that time goes on (like in static recovery) but since strain goes on”. From this, they claim, it follows that the dynamic dislocation annihilation frequency, in contrast to that in static recovery, becomes independent of the instantaneous dislocation density and only a function of temperature at a constant strain-rate. The other view is that the time aspects cannot be ignored. Since both static and dynamic recovery are driven by forces derived from the free energy changes associated with local reductions in stored energy, it follows that the dislocation annihilation frequency necessarily must reflect the instantaneous separation of dislocations in both cases. This local separation, however, may be widely different comparing the dynamic and static cases. From this it follows that dynamic recovery will be associated with a dislocation annihilation frequency which at a constant strain-rate will be a function of both the temperature and the dislocation density. A comparison of these different views in terms of work hardening predictions will be further discussed in Section 3.7.

The view here is that the main difference between the static and the dynamic situations is believed to be that in the dynamic case the stored dislocations, i.e. the dislocation network inside the subgrain and the sub-boundaries, are continuously exposed to interactions with the mobile dislocations. The physical implication of such interactions for the stability of the stored dislocation networks and boundary structures is that the recovery rate becomes orders of magnitude larger in the dynamic case as demonstrated by the model presented in Section 3.5.1 below.

Thermally activated cross-slip is the mechanism commonly believed to be responsible for Stage II breakdown. As mentioned in Section 2.1, an objective of the present treatment is to explore whether or not a stress enhanced climb mechanism (based on thermal activation of vacancies without long range diffusion) will control dynamic recovery rates during plastic deformation beyond the Stage II–III transition. Accordingly in the subsequent development of a work hardening model, dynamic recovery will be incorporated by relying on the following two equations for the relevant dislocation dynamics involved:

1. A dislocation segment in a Frank network which is exposed to a force per unit length, F , will migrate at a rate

$$v_c = v_D b^2 B_\rho \sqrt{\rho} \left(\exp - \frac{U_{SD}}{kT} \right) 2 \sinh \frac{Fb^2}{kT} \quad (32)$$

where $F = \xi_\rho Gb^2 \sqrt{\rho}$, B_ρ is a constant, U_{SD} is the activation energy for self diffusion, v_D is the Debye frequency and ξ_ρ is a dynamic stress intensity factor, the physical significance of which will be explained in Section 3.5.1.

2. The average subgrain size in a dynamic situation will increase at a rate

$$\frac{d\delta^+}{dt} = \dot{\gamma} \frac{d\delta^+}{d\gamma} = v_D b^2 B_\delta \sqrt{\rho} \left(\exp - \frac{U_{SD}}{kT} \right) 2 \sinh \frac{PV_a}{kT} \quad (33)$$

where $V_a \simeq b^3/\phi$ is an activation volume and the driving pressure is given by $P = 4\gamma_{sb}/\delta = 4\xi_\delta \phi Gb/\delta$. (γ_{sb} the sub-boundary energy, $\gamma_{sb} = (Gb\phi/4\pi(1-\nu)) \ln(e\phi_c/\phi)$ for an infinite equilibrium tilt boundary, ν is Poissons ratio, ϕ the sub-boundary misorientation and ϕ_c is the critical misorientation corresponding to a high angle boundary ($\phi_c \simeq 15^\circ$) and ξ_δ is a stress intensity factor, the value of which will be discussed in Section 3.9.1.

The physical reason for the appearance of the pre-exponential $\sqrt{\rho}$ -term in both equation (32) and equation (33) is because the climb rates under dynamic conditions will be sensitive to the jog density, c_j . In a static situation where climb commonly is analyzed in terms of local vacancy equilibrium and associated diffusion, saturation effects may cause the climb rate to be independent of the density of either thermal or geometric jogs, for a detailed treatment see Hirth and Lothe⁽³¹⁾. In a dynamic situation, however, the climb rate is, in this treatment, assumed to be controlled by thermal activation of vacancies from the jogs along the dislocation core without achieving local equilibria (the jogs simply move too fast), in which case the jog density will appear as a pre-exponential factor in the rate equations. This

jog density will in a dynamic situation be determined by the density of mobile dislocations (as they constantly intersect the stored ones), a density which again is expected to scale with the total density of stored dislocations, i.e. $c_j \propto (1/\sqrt{\rho_m})^{-1} \propto \sqrt{\rho}$. The appropriate values of pre-exponential factors like B_δ and B_ρ will be discussed when incorporating these relationships into the work hardening theory, i.e. in solving equation (3).

Regarding the stability of the subgrain structure, it follows that stress saturation requires the dynamic increase in the subgrain size to balance the generation rate of new boundaries. In the ambient temperature regime such a steady state is obtained at large strains only, strain levels sufficient for generating well defined (sharp) sub-boundaries. equation (33) cannot be applied to a cell structure with more diffuse boundaries, for instance like those schematically illustrated in Fig. 7. Or, in other words, when a loose cell-boundary picture applies, dynamic recovery in this model pertains only to the stability of the cell-interior Frank network.

Some comments are necessary regarding the dislocation structure inside the cell boundaries and the role of dynamic recovery in this context. In order to approach this topic, however, some comments on the cell structure in itself are needed. The physical rationale behind the appearance of a cell structure has not been discussed so far. The statistical approach introduced above says nothing about the mechanism of dislocation storage, it simply recognizes that dislocations are stored in a cell structure, and provides a key to the evolution of such a structure. The reason for the appearance of such a composite microstructure is believed to be the following. Dislocations which are produced and stored during plastic deformation can be stored in the microstructure at two levels: (i) A stable low-energy level within the cell boundaries, where the ultimate low-energy configuration becomes a 2D boundary structure. (ii) A metastable condition, in the form of a 3D dislocation network (Frank network) where the individual segments have the elastic properties associated with 'free' dislocations. equation (32) addresses the thermal stability of the latter structure, and equation (33) gives the stability of the boundary structure.

The dislocation structure inside the cell-boundaries represents a transitional state between these two extremes. Although the cell-boundary structure defined in Sections 3.1 and 3.2 was characterized as being identical to that in the cell interior, except for a difference in scale, this is only true in a very narrow 'pictorial' sense. In physical terms the two structures are of different origin and accordingly their recovery behavior will also be different. The dislocations in the cell boundaries are captured because of the benefit of the screening of long range-stress fields. When a boundary has been formed, it will easily capture other dislocations compatible with the boundary structure. The consequence of this is that the cell boundaries will consist of geometrically necessary dislocations required in order to account for the local misorientations between neighbouring cells. Accordingly, the cell-boundary dislocation structure cannot recover by mutual annihilation like the Frank network in the cell interior. These dislocations can only rearrange themselves into sharper boundaries, a reaction which indeed is observed experimentally⁽⁵²⁾, especially at large strains and at elevated temperatures. However, the flow stress model adopted in this theory (Section 3.1) is not sensitive to such rearrangements of boundary dislocations.

3.5. Work Hardening and Stress Saturation

The elements necessary for an attempt at modelling the stress-strain behaviour of pure fcc-metals have now been assembled. In summary, these elements are:

1. Plastic deformation generates a cell structure which in this context can be adequately characterized by three parameters: (i) the dislocation density inside the cells, (ii) the cell/subgrain size and (iii) the cell/sub-boundary misorientation (or cell wall dislocation density). This microstructural picture can be applied to any strain for any combination of strain-rate and temperature.
2. On the basis of this microstructural description, the flow stress is defined by equation (11).
3. By solving equation (29) a solution to the dislocation storage problem is obtained (i.e., equation (4) is solved).
4. The dynamic recovery behaviour is adequately accounted for by two kinetic equations, one for the stability of the dynamic Frank network (equation (32)) and the other for the subgrain structure (equation (33)).
5. Points 3 and 4 in combination open the way to the solution of equation (3). To solve this fundamental equation is the objective of the following.

It is pointed out that although each of these elements is based on first principles, many assumptions and approximations are involved. The strength, however, of the present approach is that each of these elements are formulated in analytical terms and together provide, for the first time, *the substructure evolution with strain* for any specified combination of strain-rate and temperature. The predictive power of this treatment will, of course, be tested by comparison with experimental results.

3.5.1. Stage II work hardening and Stage II breakdown

3.5.1.1. The zero degree Kelvin solution

In the absence of dynamic recovery, microstructural scaling applies, and integration of equation (29) gives:

$$\gamma = bC\sqrt{\rho} \quad (34)$$

which in combination with equation (11) (assuming $\tau_i \approx 0$) gives:

$$\tau = \theta_{II}\gamma \text{ where } \theta_{II} = \frac{\alpha_1 G}{C} \quad (35)$$

At 0 K, Stage II is characterized by linear work hardening and as long as the principle of scaling applies, the distribution of dislocations into a cell structure has no effect on the work hardening behaviour.

The value of θ_{II} . To quote from the elegant recent contribution by Brown⁽⁵³⁾: “The question is: what sort of theory can produce a dimensionless factor of 1/300? If we were to tackle the problem by numerology, we might suggest $1/\pi^5$! But we are looking for a real answer, compatible with the large amount of information available on the dislocation structure and how it evolves throughout Stage II”. The present theory does not address this question, it simply states that the statistical probability for a mobile dislocation to be stored at any strain rate and temperature is defined by a dimensionless parameter p , the value of which from the 0 K-solution becomes $p = (\theta_{II}/\alpha G)^2 \approx 10^{-4}$. It is pointed out, however, that the first theoretical treatment which accounted for the low value of θ_{II} was the statistical theory of flow stress and work-hardening by Kocks⁽⁸⁾.

3.5.1.2. The general solution

At temperatures $T > 0\text{K}$, dynamic recovery can in principle no longer be ignored, with the consequence that microstructural scaling does not strictly apply, a situation which leads to non-linear hardening. Traditionally, deviation from linearity in the stress–strain response has been taken as the termination of Stage II or the beginning of Stage III. Literally this means, of course, that a true Stage II behaviour is exclusively a 0 K phenomenon. There is, however, nothing special about the τ_{III} -stress level in terms of the mechanisms involved. At any temperature $T > 0\text{K}$ thermally activated processes will cause dynamic recovery, the result of which is first ‘seen’ at τ_{III} (see also Section 3.7). In the present treatment of work hardening, involving two microstructural elements, each of which obey different dynamic recovery characteristics (i.e. equations (32) and (33)), the meaning of the Stage II–III transition needs to be more precisely defined.

To solve the work hardening problem as defined in Section 1 requires that equation (3) can be expressed in terms of the relevant microstructural elements. The recovery contribution to this equation ($d\rho^-/d\gamma$) is obtained from equation (32). The storage term ($d\rho^+/d\gamma$), however, cannot be derived from the 0K solution of equation (29) (i.e. equation (34)) directly. This is because the microstructural evolution contains two elements, ρ_i and δ , which at 0K are coupled through the principles of scaling. At $T > 0\text{K}$, recovery will influence the density of dislocations in the cell interior, and this will again affect the storage rate of new boundaries as determined by equation (29). Accordingly, the principle of scaling is not strictly obeyed. However, as a deviation from a scaling situation becomes significant well into Stage III, a quasi-scaling situation becomes a good approximation in describing the evolution of the cell structure in the Stage II–III transition region for $T > 0\text{K}$. It follows that an increase in the stored dislocation density $d\rho$ will result in corresponding increases in the fractional contributions $d\rho_{nb}$ and $d\rho_{ob}$ (see equations (24) and (25)) given by: $d\rho_{nb} = d\rho_{ob} = -(f(q_c q_b)^2/\delta^3)d\delta$. equation (29) can now be written in the form

$$d\gamma = -bCq_c^2 \frac{\sqrt{(1-f)\rho_i + f(q_c q_b)^2/\delta^2}}{\rho_i \delta^3} d\delta \quad (36)$$

In order to solve this differential equation an expression for the instantaneous dislocation density inside the cells, ρ_i , is needed, which means that the mechanisms controlling the dynamic recovery of the Frank network need to be developed.

3.5.1.3. The effect of dynamic recovery and the definition of τ_{III}^*

The important point, to be focused on here, is the nature of the activation event controlling dynamic recovery, i.e. the growth of the dislocation network inside the cells. Traditionally, the end of Stage II, or the beginning of Stage III, has been defined by the stress level (τ_{III}) where the deviation from a linear stress–strain response is observed. Furthermore, such a deviation has been associated with cross-slip. Although the relevance of cross-slip in this context might seem to have a ‘universal’ acceptance⁽⁷⁾, the reason for singling out cross-slip as the rate controlling mechanism is still open to debate (see Section 2.1 above). The static situation which was discussed in detail in Ref.⁽¹⁾ is of some relevance in this regard. It was argued that network growth was similar to normal grain growth in the sense that an increase in the scale of the network is accomplished by the largest link-lengths growing as a result of the collapse of the smallest. Schematically, the dislocation network will expand owing to recombination or annihilation reactions as illustrated in Fig. 8(a) and (b). Dislocations of opposite signs may react due to glide and climb [Fig. 8(a)]

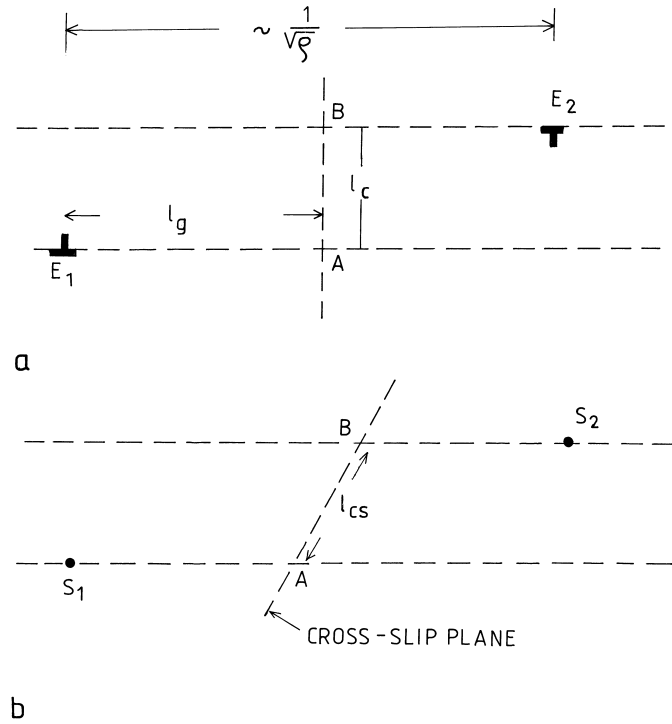


Fig. 8. Annihilation of dislocations of opposite signs due to (a) glide and climb of edge dislocations and (b) glide and cross-slip of screw dislocations (from Ref.⁽¹⁾)

or glide and cross-slip [Fig. 8(b)]. In principle any of these reactions, cross-slip or climb, can be the rate controlling step in the growth of a dislocation network. (In less pure metals and alloys solute drag becomes an alternative rate controlling mechanism).

The principle difference between the static and dynamic situations is that in the dynamic case the stored dislocations are constantly exposed to reactions with the mobile dislocations. Comparing the dynamic and static reactions the probability for a segment in the Frank network to react with a mobile dislocation in the dynamic case is much higher than the probability for recombinations between different network segments in the static case. A model for dynamic recovery of stored dislocations will now be developed on the basis of the assumption that the growth rate of the network is controlled by the collapse of dislocation dipoles where the dipolar configurations are the result of interactions between network dislocations and mobile dislocations. The collapse rate of dipoles in the structure is calculated in terms of a reaction rate equation in the form

$$\frac{dg}{dt} = -gv_g \quad (37)$$

where g is the density of dipoles (or the density of dipole segments of length $1/\sqrt{\rho_i}$ i.e. $g \propto \rho_i^{3/2}$) and v_g is the collapse frequency of such dipoles. During Stage II the principle of similitude applies (ρ_m scales with ρ_i), a consequence of which is that the dipole

density will scale with the density of stored dislocations and it follows that the dynamic recovery rate in the cell interior is given by

$$\frac{d\rho_i^-}{d\gamma} = -\frac{2}{3}\rho_i \frac{v_g}{\dot{\gamma}} \quad (38)$$

The next step becomes to derive expressions for the collapse frequency on the bases of both climb and cross-slip as rate controlling reactions.

The dipoles are the result of interactions which are of a statistical nature, and it follows from the principle of scaling that the distribution of dipole separations will be similar to that of the segments in the stored dislocation network, i.e. the average dipole separation, l_g , will scale with $1/\sqrt{\rho_i}$. Accordingly, in the case of edge dipoles a reasonable assumption becomes that the collapse frequency will scale both with the climb speed and the inverse dipole separation, i.e. $v_g \propto v_c/l_g$ where v_c is the climb speed as given by equation (32). It follows from such a dipole picture that the stress intensity factor ξ_ρ in equation (32) simply reflects the ratio between the average separation of network dislocations and that of the dipoles ($\xi_\rho \propto 1/l_g \sqrt{\rho_i}$). Combination of equations (32) and (38) then gives:

$$\frac{d\rho_i^-}{d\gamma} = -\frac{2}{3}\rho_i^2 b^2 \xi_\rho B_\rho \frac{v_D}{\dot{\gamma}} \left(\exp - \frac{U_{SD}}{kT} \right) 2 \sinh \frac{\xi_\rho G b^4 \rho_i^{1/2}}{kT} \quad (39)$$

Nix *et al.*⁽³³⁾ followed a similar approach in their modelling of the substructure evolution in the cell interior, an analysis which is based on an expression of the same form as equation (38) (although argued in a different way). Nix *et al.*, however, only considered cross-slip as the rate-controlling reaction, applying the theory of thermally activated cross-slip developed by Bonneville and Escaig⁽²⁹⁾. This approach gives the dynamic recovery rate, expressed in terms of the applied stress, τ , in the form:

$$\frac{d}{d\gamma} = -\rho_i B_\rho^{cs} \frac{v_D}{\dot{\gamma}} \left(\exp - \frac{U_{cs}}{kT} \right) 2 \sinh \left[\frac{g(\tau)}{kT} \right] \quad (40)$$

Where B_ρ^{cs} is a constant. According to Bonneville and Escaig $U_{cs} = A_1 (Gb^3)(Gb/\Gamma)(1 - b_o/d_o)^2$ and $g(\tau) = A_1 (Gb/\Gamma)^2 Gb^3 (\tau/G)$, where $A_1 = (\alpha_i/128\pi^2)[0.92(\ln Gb/15\Gamma)^{1/2} - 0.6]$, Γ is the stacking fault energy, α_i is a stress parameter which requires knowledge of the stress state in the cell interior in order to be quantified (was used as an adjustable parameter by Nix *et al.*⁽³³⁾), $d_o = Gb^2/16\pi\Gamma$ and b_o is the conventional width attributed to the recombined perfect screw⁽²⁹⁾, i.e. $b_o = \text{const.} \times b$ where the constant is of the order unity.

If, on the other hand, we apply the pile-up model for thermally activated cross-slip developed by the Stuttgart school⁽²³⁻²⁵⁾, the following alternative expression for the dynamic recovery rate is obtained:

$$\frac{d}{d\gamma} = -\rho_i B_\rho^{cs} \frac{v_D}{\dot{\gamma}} \exp - \frac{g(U_{cs}, \tau)}{kT} \quad (41)$$

According to Wolf⁽²⁵⁾ the obstacle to thermal activation can be written $g(U_{cs}, \tau) = A_2 \ln(\tau_{III}^{OK}/\tau)$ where $A_2 = 0.352Gb^3(1 + w/900)^{-1}(1 + 180\Gamma/Gb)^{-1}$ and $\tau_{III}^{OK} = (2G/w)(0.056 - \Gamma/Gb)$ (w is the number of dislocations in the pile-up).

The Stage III treatment of Kocks and Mecking⁽¹²⁾ is also based on a rate equation of the same form as equation (38). They interpreted this relationship, however, as a first order

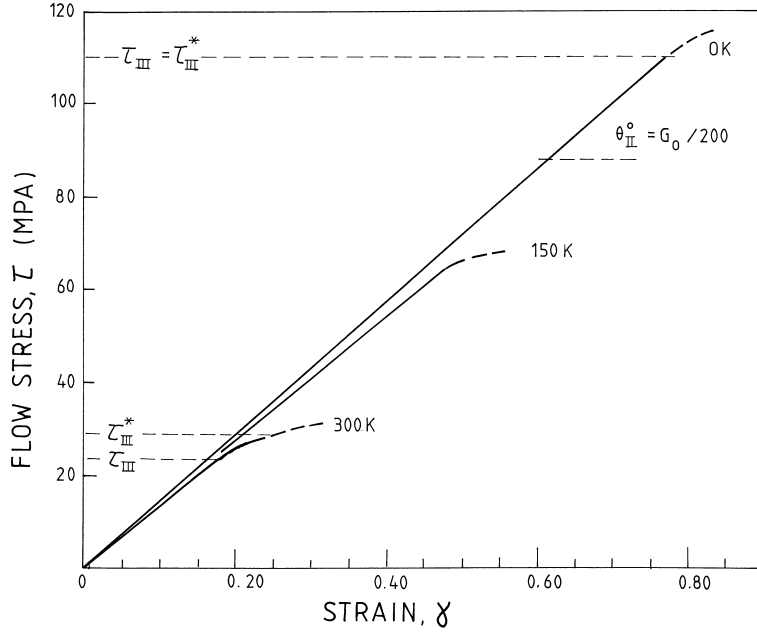


Fig. 9. Stage II and Stage II–III transition stress–strain curves for the temperatures given

reaction⁽¹⁵⁾, i.e. treating v_g as only being temperature dependent (at a constant strain-rate), further comments in Section 3.7.

It was pointed out above that the present work hardening treatment involves two microstructural elements, each of which obeys different recovery characteristics, and it follows that these elements will reach saturation at different stress levels. Furthermore, in Section 2.2, experimental evidence showing that ρ_i had already reached saturation prior to Stage IV was presented. Accordingly, ρ_i appears to saturate during Stage III and in the present treatment this saturation is defined as marking the end of the Stage II–III transition and the associated stress level is labelled τ_{III}^* . The Stage II–III transition ends when the dynamic recovery rate of the Frank network becomes sufficient to balance the dislocation storage rate due to the imposed strain-rate. Accordingly, τ_{III}^* is defined by

$$\left[\frac{d\rho_i^-}{d\gamma} \right]_{\dot{\gamma}, T} + \left[\frac{d\rho_i^+}{d\gamma} \right]_{\dot{\gamma}, T} = 0 \quad (42)$$

The τ_{III}^* -values in Fig. 9 (see below) are defined in this way. The stress level τ_{III}^* corresponds to a dislocation density $\rho_i = \rho_{III}$, which during continued deformation will have to remain approximately* constant. In this theory, strengthening beyond the Stage II–III transition is entirely accomplished by the continuous formation of new cell/subgrain boundaries (as a first order approximation).

The dynamic recovery mechanism responsible for the saturation of ρ_i will require both cross-slip and climb reactions, with climb being the most probable rate-controlling one.

*Since the microstructural evolution in Stage III is far from obeying the principle of scaling the dislocation density inside the cells/subgrains may vary somewhat (because S in equation (29) cannot be treated as a constant). During Stage IV, however, a quasi-scaling situation is established and ρ_i will be constant (see below).

The first sign of a deviation from Stage II hardening may, however, be due to cross-slip controlled events.

3.5.1.4. The stress-strain curve and the temperature dependence of τ_{III} and τ_{III}^*

By solving equations (36) and (39) in combination, stress-strain curves like those shown in Fig. 9 are obtained. These two equations, however, can only be handled separately which means that equation (36) (in combination with equation (39)) must be solved in an interactive way. The procedure for deriving numerical solutions is given in Appendix A (Note that in Stage II and the Stage II–III transition the term $\xi_\rho G b^4 \rho_i^{1/2}$ is expected to be much larger than kT , a situation which implies that the sinh-term can be replaced by an exponential one). The curves in Fig. 9 have been generated by arbitrarily choosing the values of the relevant constants and scaling parameters as follows: $\alpha_1 = 0.3$, $\alpha_2 = 2$ and $q_c = 10$. (The generated curves are based on material constants for copper, i.e. $G_0 = 4.74 \cdot 10^4$ MPa, $G = G_0 \exp - 3.97 \cdot 10^{-4} T$, $b = 2.56 \cdot 10^{-10}$ m and $U_{SD} = 3.4 \cdot 10^{-19}$ J/atom). It follows from equation (39) that the dislocation density at 0 K and $\tau = \tau_{III}^0$ corresponds to $\sqrt{\rho_{III}^0} = U_{SD} / \alpha_3 \xi_\rho G b^4$. By combining this expression for $\sqrt{\rho_{III}^0}$ with equation (35), Stage II will terminate at the stress $\tau_{III}^0 / G_0 = \alpha_3 U_{SD} / \xi_\rho b^3 G_0$ where $\alpha_3 = \alpha_1 + \alpha_2 / q_c$. In Cu this normalized stress is expected to be in the range $2.5 - 3 \cdot 10^{-3}$ (Fig. 2) which gives ξ_ρ in the range 70–90.

The curves in Fig. 9 have been generated on the bases of climb as rate-controlled reaction, which is believed to be a reasonable assumption in predicting τ_{III}^* (saturation in ρ_i), but which may overestimate τ_{III} since this stress level may be controlled by cross-slip. An interesting question now becomes; to which extent can a 3D dislocation network (under dynamic conditions), grow by only allowing cross-slip reactions to occur? The opinion of the present author is; only to a limited extent. This is so because the dynamic network is build up of dislocation segments which are of arbitrary orientations with respect to their Burgers vectors, while cross-slip in its very nature applies to dislocations segments in a singular orientation (pure screws) only. On the other hand, it seems reasonable to assume that thermally activated cross-slip may cause the first detectable deviation from Stage II

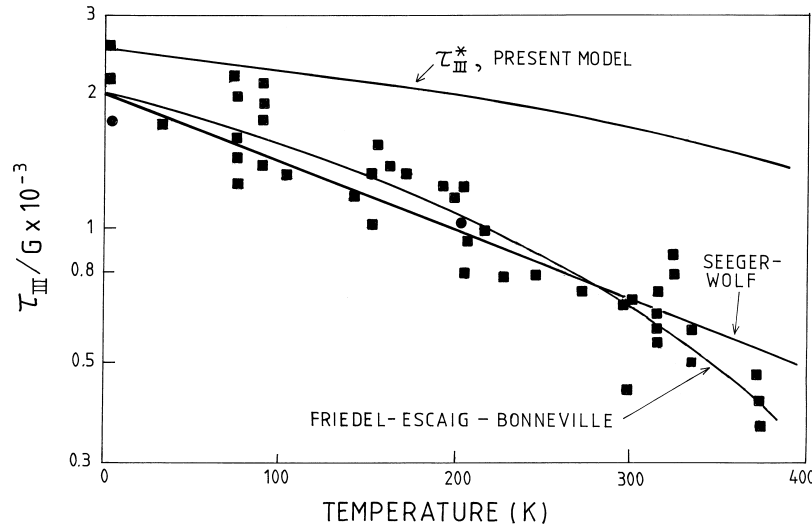


Fig. 10. Temperature dependence of τ_{III} in copper. Experimental data taken from Refs. (54–59)

hardening. However, whatever the magnitude of such a cross-slip contribution, an intrinsic property of the dynamic recovery reactions as they are defined here (either in terms of climb, equation (39), or cross-slip, equations (40) and (41)) is that when the effect of dynamic recovery first becomes detectable (deviation from Stage II) saturation follows almost immediately as shown in Fig. 9. So the idea that Stage II ends at a critical stress level is a reasonable one, at least in terms of a first order approximation, see also Section 3.7. Accordingly, if we assume that the temperature dependence in τ_{III} is due to cross-slip, such a dependence is obtained by solving equation (42) where the $d\rho_i^-/d\dot{\gamma}$ term is defined either by equation (40) or (41) and the athermal storage term $d\rho_i^+/d\dot{\gamma}$ is given by equation (A2). The temperature dependence in τ_{III} as predicted by either the French or the Stuttgart schools then becomes as detailed in the following:

Combination of equations (40), (42) and (A2) (for $g(\tau)/kT > 1$), the French approach gives

$$\frac{\tau_{III}}{G} = \frac{\tau_{III}^{0K}}{G} - \frac{1}{A_1} \left(\frac{\Gamma}{Gb} \right)^2 \left(\frac{kT}{Gb^3} \right) \ln \left(\frac{B_1^{cs}}{\alpha_3} \frac{\nu_D}{\dot{\gamma}} \frac{\tau_{III}}{G} \right) \quad (43)$$

where $B_1^{cs} = CB_\rho^{cs} (1 + f(q_b^2 - 1))/3 \simeq 100B_\rho^{cs}$ (typical values of $C = \alpha_1 G/\theta_{II}$, f and q_b are 100, 0.2 and 5 respectively). The temperature dependence in τ_{III} has been studied most extensively in copper^(54–59) with the result shown in Fig. 10. (The various experiments have been performed using a typical strain rate of 10^{-3} s^{-1}). The prediction due to equation (43) is also included in this diagram, selecting $\tau_{III}^{0K} = 2 \cdot 10^{-3} G_\rho$, the Debye frequency $\nu_D = 10^{13} \text{ s}^{-1}$ and the stacking fault energy equal to 46 mJ/m^2 (see Table 2.2). By this latter selection the apparent activation energy (equation (40)) becomes $g(U, \tau) = (3.12 \cdot 10^{-19} - 158 \tau b^3) \text{ J}$, which implies that the Friedel–Escaig model requires a stress intensity factor for cross-slip of the size 158, a very high value indeed. The curve in Fig. 10, generated on the bases of equation (43), requires $B_1^{cs} \simeq 5 \cdot 10^2$ or $B_\rho^{cs} \simeq 5$. The true value of this pre-exponential factor is difficult to assess from first principles, besides being a factor of the order 1. In the present type of modelling, however, B_1^{cs} needs to be handled as an adjustable (fitting) parameter and considerable deviation in the value of B_ρ^{cs} from unity must be accepted, from which it follows that $B_\rho^{cs} \simeq 5$ becomes a reasonable value indeed.

The Friedel–Escaig model seems to provide a good description of the temperature dependence in τ_{III} . This dependency is nearly linear in temperature, but as seen from equation (43) it is not truly linear because the τ_{III} -stress also appears in the log-term. In the temperature range covered in Fig. 10, however, this log-dependency contribute to a deviation from linearity which is very small. Why Friedel and Escaig claimed a linear relationship was because they obtained their equivalent of equation (43) by applying a kinetics relationship where the stress dependency in the pre-exponential term was overlooked.

The equivalent to equation (43), derived on the bases of the pile-up model for cross-slip (i.e. by combination of equations (41), (42) and (A2), the Stuttgart school) becomes:

$$\ln \frac{\tau_{III}}{G} = \ln \frac{\tau_{III}^{0K}}{G} - \frac{kT}{A_2} \ln \left[\frac{B_1^{cs}}{\alpha_3} \frac{\nu_D}{\dot{\gamma}} \frac{\tau_{III}}{G} \right] \quad (44)$$

The exponential decay in τ_{III} with temperature as predicted by this relationship is illustrated in Fig. 10, using the same values for τ_{III}^{0K} and Γ as used above. Furthermore, in order to calculate the A_2 -parameter (equation (41)) the number of dislocations in the pile-ups are required and here the value recommended by Seeger *et al.*⁽²⁴⁾ has been used ($w = 23$). It is noted, however, that application of equation (44). (with the result illustrated in Fig. 10)

requires a B_1^{cs} factor of the order 10^5 or a B_ρ^{cs} value of the order 10^3 which is a somewhat high value.

In a comment on Fig. 2 above it was mentioned that while τ_{IIIs} , τ_{IV} and τ_s appear to decline with temperature at approximately the same rate, τ_{III} falls off more steeply. This observation has also been made by Gil Sevillano⁽⁷⁾ who suggested that τ_{III} perhaps is controlled only by cross-slip, while τ_{IIIs} may require a process for annihilation of edge components, the latter having a much higher activation energy than the first. In the present multiparameter model the same view pertains to the stability of the intercell dislocation network. While the first sign of deviation from linear hardening may be attributed to cross-slip, the subsequent saturation in ρ_b , which in the present treatment signifies the end of the Stage II–III transition, will require dislocation climb in addition to cross-slip. As pointed out above, climb is expected to be the rate-controlling reaction responsible for this saturation. Accordingly, the end of the Stage II–III transition, as defined by the τ_{III}^* -stress level, is obtained by combining equations (39) and (42) (for $Fb^2/kT = \xi_\rho Gb^4 \rho_i^{1/2}/kT > 1$) and equation A_2 , an exercise which gives:

$$\frac{\tau_{III}^*}{G} = \frac{\tau_{III}^{*0K}}{G} - \left(\frac{\alpha_3}{\xi_\rho} \right) \frac{kT}{Gb^3} \ln \left[\frac{B_1 \xi_\rho v_D}{\alpha_3^3 \dot{\gamma}} \left(\frac{\tau_{III}^*}{G} \right)^3 \right] \quad (45)$$

where $B_1 = CB_\rho(1 + f(q_b^2 - 1))/3 \simeq 100 B_\rho$ (see comments to equation (44)). Note that this relationship for the temperature and strain-rate dependence of the τ_{III}^* -stress is nearly in the same form as that derived above on the basis of the Bonneville and Escaig cross-slip model (equation (43)) for the τ_{III} -stress. The difference pertains to the power to which the stress appears in the logarithmic term, to the first power in equation (43) and to the third in equation (45). This difference is of second order importance for high stresses and/or in the low and ambient temperature range, but becomes a first order difference, indeed, at higher temperatures. This important aspect is discussed in more detail in relation to the temperature and strain-rate dependence of the τ_{IIIs} -stress in the following section below.

The τ_{III}^* -stress level is difficult to determine experimentally. The temperature variation in this stress level as illustrated in Fig. 10 is derived from equation (45) by somewhat arbitrarily selecting $\tau_{III}^{*0K}/G \simeq 2.5 \cdot 10^{-3}$. The activation energy used is that of self diffusion in copper ($3.4 \cdot 10^{-19}$ J/atom), a value which together with the τ_{III}^{*0K} -stress defines the stress intensity factor ξ_ρ , i.e. $\xi_\rho = U_{SD}/b^3 \tau_{III}^{*0K} = 94$. In addition, the slope of the τ_{III}^* -curve in Fig. 10, which is consistent with the temperature variations in τ_{IV} , τ_{IIIs} and τ_s shown in Fig. 2, requires a B_1 constant of the order 10^4 or $B_\delta \simeq 100$, which must be considered to be a reasonable value.

It is pointed out that this climb based treatment of the stability of edge dipoles together with the Friedel–Escaig–Bonneville analyses^(26,28,29) provide an explanation for the differences in the temperature dependence between the stress level signifying the breakdown of Stage II and the saturation stresses associated with the end of the Stage II–III transition (τ_{III}^*), Stage III saturation (τ_{IIIs}) and the final saturation stress (τ_s), see also the following section. It is further interesting to note that these two different approaches predict similar stress dependence in the apparent activation energy involved: $U(\text{apparent}) = U(\text{activation}) - \xi b^3 \tau$, where (in the case of copper) $U(\text{activation}) = 3.1 \cdot 10^{-19}$ J/atom for cross-slip (equation (43)) and $U(\text{activation}) = U_{SD} = 3.4 \cdot 10^{-19}$ J/atom for thermally activated climb. The stress intensity factors involved are $\xi = 158$ and $\xi = 94$ in the cross-slip and climb cases, respectively.

From this comparison of reaction kinetics based on either climb or cross-slip it follows that climb is not only a mechanism to be reserved for high temperature dynamic recovery,

thermally activated climb may compete with cross-slip as rate-controlling reaction at any temperature, and as demonstrated below, is the most probable reaction controlling dynamic recovery during strain hardening beyond a limited stress-range associated with Stage II breakdown. Accordingly, *the important conclusion to be drawn from the present dynamic recovery model is that it modifies the long standing objection to climb as a rate controlling reaction in the low temperature regime of work hardening.* If dynamic recovery is treated by analogy with static recovery, an activation volume of the order $100 b^3$ would be required in order to account for the dynamic recovery rate. In the present treatment, involving interactions between dislocation segments in the Frank network and the mobile dislocations, a consistent model, based on the stability of edge dipoles, has been developed. Comparing now the recovery rates in the dynamic and static cases the present treatment predicts the dynamic case to be speeded up by many orders of magnitude because of: (i) a dipole spacing parameter ξ_ρ which due to the scaling involved also becomes a stress intensity factor ($F = \xi_\rho G b^4 \sqrt{\rho_i}$) of the order 50–100 and (ii) the same parameter also appears as a pre-exponential factor. It is the present opinion that this model makes dynamic recovery based on cross-slip as the rate controlling reaction less realistic beyond the breakdown of Stage II, a view which will be given added support in the treatment of the temperature and strain-rate dependence of τ_{III} in the following section.

3.6. Stage III Work Hardening

This theory defines the Stage II–III transition to end when the dislocation density inside the cells has reached the saturation level ($d\rho_i/d\gamma = 0$). During subsequent deformation, work hardening is due to the strain induced refining of the cell structure, and the flow stress (equation (11)) takes the form

$$\tau(\dot{\gamma}, T) = \alpha_1 G b \sqrt{\rho_{III}(\dot{\gamma}, T)} + \alpha_2 G b \frac{1}{\delta(\dot{\gamma}, T)} \quad (46)$$

where $\delta(\dot{\gamma}, T)$ is obtained by solving equation (29) for the special case $\rho_i = \rho_{III}(\dot{\gamma}, T)$. Before considering how to obtain this solution, some general comments on the Stage III substructure evolution are necessary.

In microstructural terms, Stage II can be described as a reasonably uniform evolution of a cell structure, which refines itself with strain in accordance with the principle of scaling. Stage III introduces a sharp change in this evolutionary picture. In Stage III, deformation becomes inhomogeneous due to localized slip phenomena (microbanding) which transform the structure from that of equiaxed cells into a pan-cake-like subgrain structure which again ‘saturates’ (meaning it remains self-similar during continued straining) at a shear strain of about $\gamma \simeq 1.5$ – 2.5 , a strain which signifies the start of Stage IV. For a detailed TEM description see: on Cu Refs.^(34, 60–63), on α -Fe Refs.^(37, 64), on Ni and Ni–Co alloys Ref.⁽⁶⁵⁾ and on Al an impressive contribution by the Risø-group^(36, 66, 67). As covered in Section 2.2, a characteristic feature of Stage III is a rapid increase in sub-boundary misorientation, from much less than 1 degree at the end of Stage II to about 3 degrees at the end of Stage III, after which the misorientation remains constant (Fig. 6). This increase in misorientation is consistent with microbanding as a controlling slip phenomenon in Stage III. The microbands are associated with severe localized shear^(62, 63), a phenomenon which for compatibility reasons will be associated with the build up of local strain gradients or sub-boundary misorientations.

Accordingly, Stage III will here be treated as a transient or microstructural transition which links the Stages II and IV, an approach which can be labelled a ‘modelling treatment’, in the sense that it requires a certain input from experimental observation. The problem is to solve equation (29) for $\gamma > \gamma_{III}$. As pointed out above, a characteristic aspect of Stage III is a rapid increase in the sub-boundary misorientation, φ , from which it follows that during Stage III the stored dislocation density ρ and the differential term $d\rho_{nb}$ (equations (21) and (25)) needs to be redefined in terms of the misorientation angle φ :

$$\rho = \rho_{III} + \frac{\kappa\varphi}{b\delta} \quad (47)$$

$$d\rho_{nb} = -\frac{\kappa\varphi}{b} \frac{1}{\delta^2} d\delta \quad (48)$$

As long as the effect of dynamic recovery on the subgrain structure can be ignored (i.e. $d\delta^+/d\gamma = 0$), then the sub-structural evolution will be adequately described by a modified version of equation (29) obtained by defining $f = 0$ (i.e. boundaries instead of cell walls) and by introducing the new definitions of ρ and $d\rho_{nb}$ given above, i.e.,

$$\gamma - \gamma_{III} = -\frac{\kappa}{2\rho_{III}C} \int_{\delta_{III}}^{\delta} \frac{\varphi \sqrt{\rho_{III} + \frac{\kappa\varphi}{b\delta}}}{S\delta^2} d\delta \quad (49)$$

In order to solve this integral, functions defining the variation in φ and S with δ are required. An empirical function relating the variation in φ with δ can be extracted from the data contained in Figs 4 and 6, an exercise which indicates that φ scales with δ^{-1} . The storage parameter S , however, which during Stage II was a constant (due to the principle of scaling) can no longer be treated as such during Stage III. As pointed out above, during this stage we have a continuous change in the character of the boundaries from cells of low misorientations towards sharper sub-boundaries. This transition may have effects on both of the storage parameters p_{nb} and f_{ob} ($S_{sc} = 4(1 - f_{ob})p_{nb}$, equation (17)). During Stage IV, however, a state of scaling has been re-established and a reasonable assumption is that S again can be treated as a constant.

On the basis of the assumption that Stage III can be treated as a microstructural transition linking Stage II and IV, the effect of such a transition on Stage III work hardening behaviour will now be explored based on equation (29) (or equation (49)). Let us, for reasons of simplicity, assume that the microstructural transition associated with Stage III is fully characterized by: (i), $\varphi = \varphi_o + K_\varphi b/\delta$ where the constants φ_o and K_φ is defined by the boundary conditions, i.e. $\varphi = \varphi_{III}$ for $\delta = \delta_{III}$ (here $\varphi_{III} = bf(q_c q_b)^2/\kappa\delta_{III}$ is an apparent misorientation*) and $\varphi = \varphi_{IV}$ for $\delta = \delta_{IV}$ and (ii), $S = S_o - K_S b/\delta$ where the constants S_o and K_S is defined by the boundary conditions, i.e. $S = S_{sc}$ for $\delta = \delta_{III}$ and $S = S_{IV}$ for $\delta = \delta_{IV}$. Realizing that $d\tau = -(\alpha_2 G b/\delta^2) d\delta$ (equation (11)), then equation (49) can be reorganized in the following differential form

*As long as microstructural scaling applies the dislocation density due to the cell boundary structures can be written: $f(q_c q_b)^2/\delta^2$ (equation (21)), which can be expressed in terms of an apparent angle of misorientation: $\kappa\varphi/b\delta$, i.e. $\varphi = bf(q_c q_b)^2/\kappa\delta$.

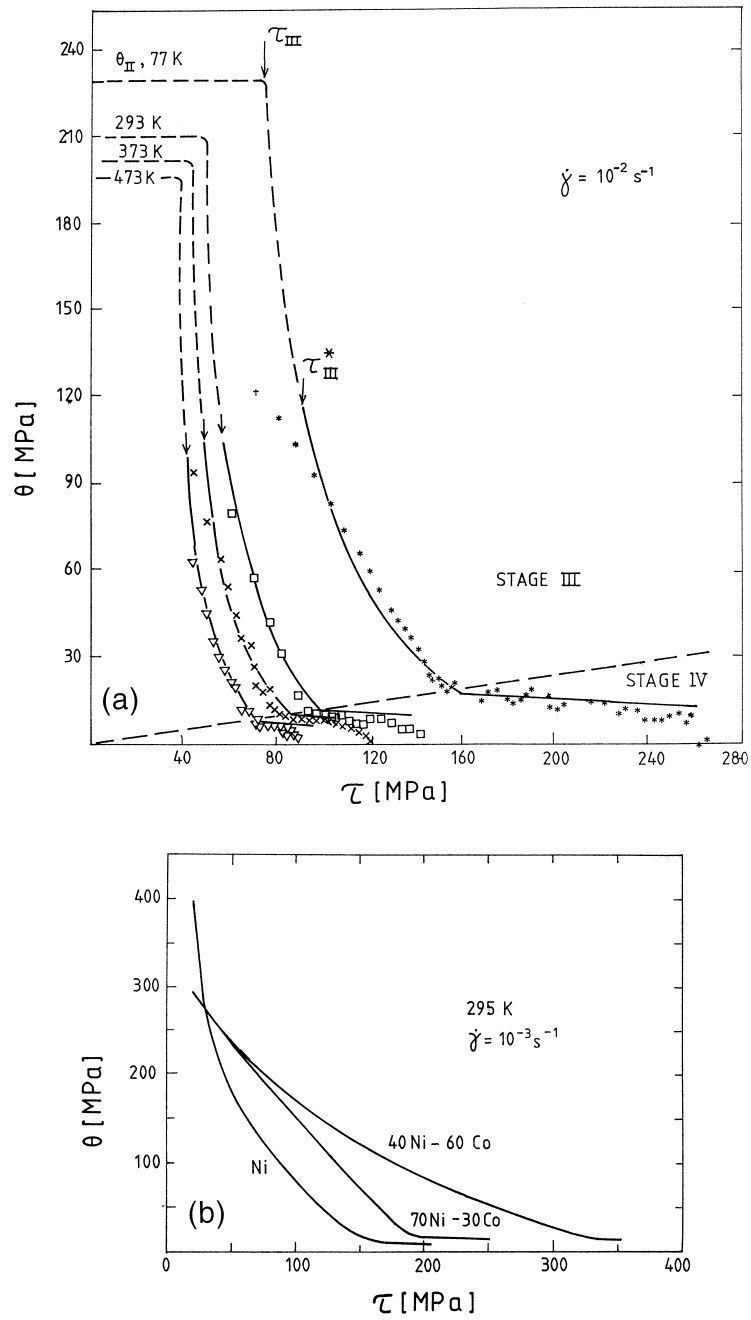


Fig. 11. (a) The predicted variation in strain hardening (θ) vs flow stress (τ) for copper for Stage III equation (49)) and Stage IV equation (61). Experimental data from Ref.⁽²⁰⁾. (b) Experimental θ - τ plots for Ni and Ni-Co alloys⁽³³⁾ illustrating variously shaped curves

$$\frac{d\tau}{d\gamma} = \theta_{III} = \frac{2\alpha_2 q_c C G b S}{\kappa \delta_{III} \phi \sqrt{1 + \frac{\kappa \phi \delta_{III}^2}{b q_c^2 \delta}}} \quad (50)$$

where $\delta_{III}^{-1} = \tau_{III}/\alpha_3 G b q_c$, $\delta^{-1} = (\tau_{III}/\alpha_2 G b)$ ($\tau/\tau_{III} - \alpha_1/\alpha_3$) (from equation (11)). The diagram in Fig. 11(a) illustrates the predictions due to this relationship in comparison to the experimental observations from torsion testing of copper by Zehetbauer and Seumer⁽²⁰⁾. [equation (50) covers only the Stage III parts of the curves in Fig. 11(a), the extension into Stage IV is covered in Section 3.8 below]. In generating these curves reasonable values for the constants involved have been used, i.e. $\alpha_1 = 0.3$, $\alpha_2 = 3$, $f = 0.2$, $q_b = 4$, $q_c = 10$, $\theta_{II} = G/220$, $\phi_{IV} \approx 3^\circ$ and $S_{IV}/S_{sc} \approx 0.5$. Furthermore, the modelling relies on the assumption that the δ_{III}/δ_{IV} -ratio is constant in the temperature range covered by Zehetbauer and Seumer. This is an important assumption which provides a microstructural ‘bridge’ between Stages II and IV. Or in other words, this microstructural modelling treatment assumes a strain-rate and temperature invariant ratio between the cell sizes at the termination of Stage II (δ_{III}) and that at the beginning of Stage IV (δ_{IV}), i.e.

$$\frac{\delta_{III}}{\delta_{IV}} = q_{IV} \quad (51)$$

where $q_{IV} \approx 3$ is used in generating the curves in Fig. 11(a).

In view of the rather simple microstructural transitions used here as a bases for the application of equation (50), the model predictions in Fig. 11(a) are reasonable indeed. It is noted, however, that the modelling curves deviates somewhat from the more straight ‘Voce-law-type’ experimental data lines. In this context it needs to be pointed out that the model pertains to stress levels exceeding τ_{III}^* . The variation in θ between τ_{III} and τ_{III}^* (broken line) has not been modelled. While the value of τ_{III} may be determined by cross-slip, the shape of the θ - τ curve following Stage II break down is difficult to predict from first principles. This is not only so because of the problems associated with the modelling of thermally activated cross-slip. An even more important reason is believed to be that in the early part of Stage III the hardening behaviour is controlled by slip localization (microbanding), while the model is based on an isotropic or global microstructure evolution picture. As the microstructure evolution becomes more uniform, which it does in the latter half of Stage III, the model predictions are much better showing a Voce type behaviour, Fig. 11(a). Furthermore, one should be aware of the fact that there are considerable experimental uncertainties associated with the generation of θ - τ plots, especially for small τ -values. An interesting variety of differently looking θ - τ curves are shown in Fig. 11(b), from torsion testing of Ni and Ni-Co alloys by Nix *et al.*⁽³³⁾. The Kocks–Mecking one-parameter model^{(9), (12)} claims to prove a Voce type hardening during Stage III, discussion of this model is presented in Section 3.7 below.

Additional confidence in the picture developed here for the microstructural evolution during Stage III is obtained from rapid strain rate change experiments, see Section 5.4. The breakdown of microstructural scaling associated with the present description of the Stage II–III transition accounts very well for the deviation from Cottrell–Stokes law in Stage III, see Fig. 22.

3.6.1. The temperature and strain-rate dependence of τ_{III_s}

Stage III terminates at a stress-strain combination identified in Fig. 1 by τ_{IV} and γ_{IV} . As also indicated in Fig. 1, $\tau_{IV} \simeq \tau_{III_s}$ where the term ‘Stage III saturation stress’ (τ_{III_s}) initially was introduced by Kocks⁽⁸⁾. By combining equations (11) and (51), τ_{III_s} can be defined in terms of the saturation dislocation density ρ_{III} , i.e.

$$\tau_{III_s}(\dot{\gamma}, T) = \alpha_4 G b \sqrt{\rho_{III}(\dot{\gamma}, T)} \quad (52)$$

where $\alpha_4 = \alpha_1 + q_{IV} \alpha_2 / q_c$.

As mentioned in Section 2.1, Mecking *et al.*⁽¹⁴⁾ have studied the deformation behaviour of single crystals of Al, Ag, Ni and Cu from which they obtained the temperature dependence of τ_{III_s} as illustrated in Fig. 3. For one of the metals (aluminium) the strain-rate dependence of τ_{III_s} was also studied. These results will in the following be used in order to test the present dynamic recovery model. From their analysis Mecking *et al.* concluded that dynamic recovery in Stage III is influenced most significantly by dislocation rearrangements that *do not* require diffusion in the low-temperature regime, while at higher temperatures (in the case of aluminium) an Arrhenius law behavior with a stress-dependent pre-exponential led to a limiting activation energy compatible with that of self-diffusion (see also comments on Fig. 3 in Section 2.1).

The temperature and strain-rate dependence of ρ_{III} (or $\tau_{III_s}^*$) was in the previous section discussed on the basis of equation (45), i.e. for $Fb^2 = \xi_\rho G b^4 \rho_i^{1/2} > kT$. However, the investigation by Mecking *et al.*⁽¹⁴⁾ covers such extended ranges in both temperatures and strain rates that the quantity Fb^2 will vary from being much larger than kT to much less than kT . Accordingly, equation (42) needs to be solved without making any assumptions regarding the value of the Fb^2/kT -ratio, and such a solution is obtained by using the $d\rho_i^-/d\gamma$ and $d\rho_i^+/d\gamma$ terms as defined by equations (39) and (A2) respectively and subsequently replace $\rho_i^{1/2} = \rho_{III}^{1/2}$ with τ_{III_s} ($\rho_{III}^{1/2} = \tau_{III_s}/(\alpha_4 G b \sqrt{\rho_{III}})$, equation (52)). The result becomes the following analytical expression for the temperature and strain-rate dependence in the Stage III saturation stress:

$$2 \sinh \left[\left(\frac{\xi_\rho}{\alpha_4} \right) \left(\frac{G b^3}{kT} \right) \left(\frac{\tau_{III_s}}{G} \right) \right] = \left(\frac{\alpha_4^3}{B_1 \xi_\rho} \right) \left(\frac{\dot{\gamma}}{v_D} \right) \left(\frac{\tau_{III_s}}{G} \right)^{-3} \exp \frac{U_{SD}}{kT} \quad (53)$$

From this expression the 0K saturation stress becomes: $\tau_{III_s}^0/G = \alpha_4 U_{SD}/\xi_\rho G_0 b^3$. equation (53) cannot be solved explicitly with respect to τ_{III_s}/G , but numerical solutions are easily derived.

Mecking *et al.*⁽¹⁴⁾ investigated the strain-rate and temperature dependence of τ_{III_s} in aluminium single crystals with the results shown in Fig. 12. In order to compare different metals these authors introduced a ‘normalized temperature’, kT/Gb^3 , which is the abscissa in Fig. 12. The present theory (equation (53)) gives the predictions shown in the figure, from which it follows that the observations by Mecking *et al.* are accounted for with great accuracy indeed. The theoretical curves in Fig. 12 have been generated using the following input parameters: $U_{SD} = 120$ kJ/mol, $\tau_{III_s}^0 = 2 \cdot 10^{-3}$ which gives $\xi_\rho = 75$ ($\xi_\rho = \alpha_4 U_{SD}/\tau_{III_s}^0 b^3$) and $B_1 = 2 \cdot 10^3$ s⁻¹. Note that the model predicts an activation energy of 120 kJ/mol over the entire temperature and strain-rate range investigated which is typical of self diffusion in aluminium. In the high temperature range ($Fb^2/kT < 1$) the stress exponent becomes equal to 4, in reasonable agreement with power-law creep. The pre-exponential constant B_1 is defined as $B_1 \equiv B_\rho C(1 + f(q_b^2 - 1))/3 \simeq 100 B_\rho$ (see equation (45)) which gives $B_\rho = 20$, a

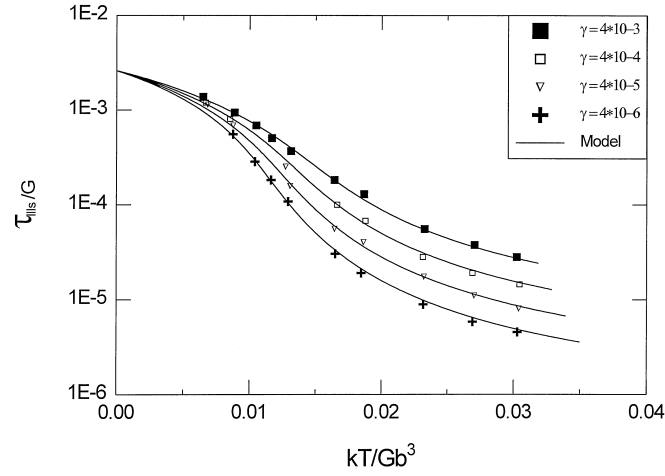


Fig. 12. Stage III saturation stress (τ_{III}/G) in aluminium as a function of kT/Gb^3 , for the strain rates given. (Experimental observations taken from Ref.⁽¹⁴⁾)

reasonable value indeed, considering the uncertainties in quantifying pre-exponential constants of this type. The important conclusion now becomes that the temperature and strain rate dependency of τ_{III} in aluminium can be accounted for by one single activation energy (that of self diffusion), and a constant activation volume (equal to $\xi_\rho b^3$) over the entire temperature range from 0 K to T_m , or in other words, within the framework of one single mechanism.

The next step is to extend this analysis to cover the different metals studied by Mecking *et al.*⁽¹⁴⁾ and presented in Fig. 3. The four metals investigated by them have been assigned the material constants given in Table 1. By this choice the model (equation (53)) predicts the τ_{III}/G vs kT/Gb^3 relationships (at $\dot{\gamma} = 4 \cdot 10^{-4} \text{ s}^{-1}$) for the different metals as illustrated in Fig. 13. Note that in generating these curves only one free modelling parameter has been allowed, namely the activation energy. The ξ_ρ - and B_1 -parameters are kept constant equal to the values used above. This excellent fit is obtained by using activation energies which all are very close to the measured ones for self diffusion, see Table 1.

The conclusion is clear: the temperature and strain-rate dependence of τ_{III} (or τ_{III}^*) in fcc-metals is well accounted for in terms of a model based on one single mechanism, the activation energy of which is that of self diffusion, the most probable reaction is related to the climb stability of edge dipoles. However, since Mecking *et al.*⁽¹⁴⁾ and Kocks⁽¹⁰⁾ in their analyses of the same data reached a rather diametrical conclusion based on cross-slip, some comments are necessary regarding the relevance of the stacking fault energy (SFE) differences involved in the interpretation of the observations in Fig. 13. The relevant stacking

Table 1. The activation energies and selected materials constant used in the modelling for the metals given (the measured U_{SD} values are taken from: Al^(68, 69), Ag^(70, 71), Cu^(72, 73) and Ni⁽⁷⁴⁾)

Metal	$b \cdot 10^{-10} \text{ m}$	$G_0 \cdot 10^4 \text{ MPa}$	$\frac{1}{G} \frac{dG}{dT} \cdot 10^{-4}$	$U_{SD} \text{ kJ/mol}$	
				Model	Measured
Al	2.86	2.99	5.40	120	136–142
Ag	2.89	2.64	4.36	210	179–208
Cu	2.56	4.74	3.97	240	210–236
Ni	2.49	7.87	3.70	270	280

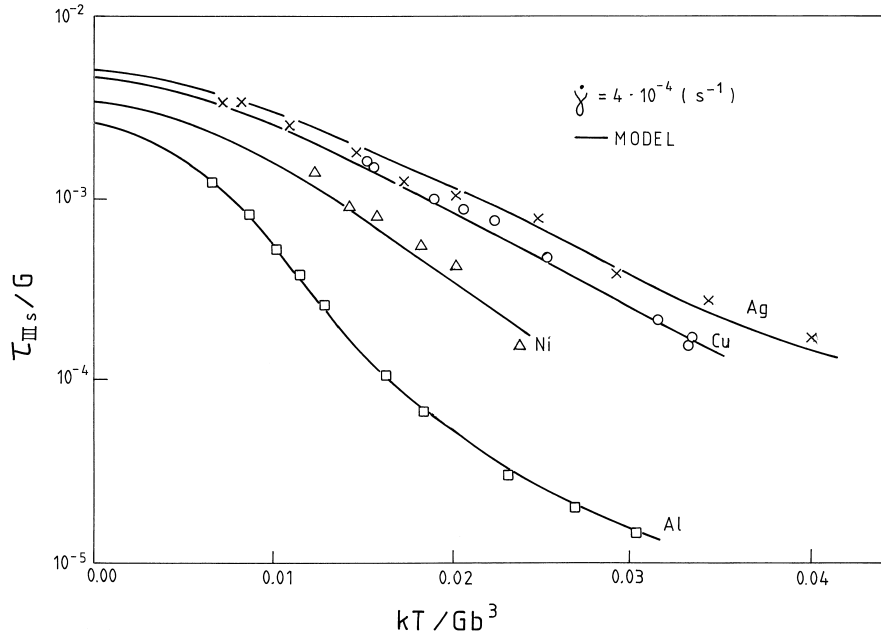


Fig. 13. Stage III saturation stress ($\tau_{III s}/G$) as a function of normalized temperature (kT/Gb^3) for the metals given. Experimental data taken from Ref.⁽¹⁴⁾

fault energies (Γ) are listed in Table 2. As mentioned in Section 2.1 these workers demonstrated that by introducing appropriate scaling factors τ^* and T^* (Table 2), the different curves in Fig. 13 [(or Fig. 3(a))] would fall on the same master line (Fig. 3b). Mecking *et al.*⁽¹⁴⁾ associated these scaling factors with the inverse normalized stacking fault energy, $(\Gamma^{-1})^* = G_0 b / \Gamma$ (in Table 2 normalized with respect to copper) and their conclusion was that the scaling factors correlated well with each other and with the $(\Gamma^{-1})^*$ -parameter, although no comparisons in terms of the real SFE-values involved were included in their paper. Such a comparison is included here and regarding the quality of this correlation with the SFE (data listed in Table 2), the claim of Mecking *et al.*⁽¹⁴⁾ seems to have some merit, at least in a qualitative sense, in relation to the $(\Gamma^{-1})^*$ -parameter. But the point of interest here is to find out how these SFE-values translates into $\tau_{III s}$ predictions according to the theories of both the Stuttgart and the French schools. From the pile-up model by Wolf⁽²⁴⁾ (see also previous section) it follows that $\tau_{III}^{0K} = (2G/w)(0.056 - \Gamma/Gb)$ where w is the number of dislocations in the pile-up. Making the same assumption as that made by

Table 2. The normalized scaling factors τ^* and T^* used in Fig. 3(b) together with: the stacking fault energies, SFE (Ag⁽⁷⁶⁾, Cu⁽⁵⁸⁾, Ni⁽⁷⁷⁾ and Al⁽⁷⁸⁾), the normalized inverse stacking fault energy; $(\Gamma^{-1})^* = (\Gamma/G_0 b)_{Cu}/(\Gamma/G_0 b)$ and the normalized 0K stresses $\tau_{III s}^{0K, SFE}/G$ and $\tau_{III s}^{0K, SD}/G = (U_{SD}/G_0 b^3)/(U_{SD}/G_0 b^3)_{Cu}$. (The G_0 's, U_{SD} 's and b 's used in the calculation of $(\Gamma^{-1})^*$ and $\tau_{III s}^0$ are listed in Table 1)

Metal	τ^*	T^*	Γ m J/m ²	$(\Gamma^{-1})^*$	$\tau_{III s}^{0K, SFE}/G$		$\tau_{III s}^{0K, SD}/G$
					Stuttgart	French	
Ag	1.07	1.15	16	1.81	0.48	0.69	1.07
Cu	1	1	46	1	1	1	1
Ni	0.9	0.8	125	0.58	0.76	0.82	0.82
Al	0.67	0.5	166	0.19	0.3	0.005	0.67

Mecking *et al.*⁽¹³⁾, namely that τ_{III}^{0K} will scale with τ_{III}^{0K} , and by using the w -values recommended by Seeger *et al.*⁽²³⁾ for the metals involved (Ag: $w \simeq 49$, Cu: $w = 23$, Ni: $w = 29$, Al: $w \simeq 41$), then the $\tau_{III}^{0K,SFE}$ predictions become as shown in Table 2. The Bonneville and Escaig model⁽²⁹⁾ predicts $\tau_{III}^{0K} = (\Gamma/G)[1 - 16\pi\Gamma/Gb]^2$ (see also previous section) which gives the $\tau_{III}^{0K,SFE}$ values also listed in Table 2. These predictions do not make much sense for any of the two schools, which in the view of the present author only underscores the fact that severe uncertainties are associated with the application of mechanisms based on cross-slip beyond Stage II breakdown.

In a one-parameter model for Stage III, Mecking and Lücke⁽⁷⁵⁾ suggested that the stacking-fault induced differences in τ_{III}^{0K} (as predicted by the Stuttgart school) is carried over to Stage III in such a way that the prelogarithmic parameter A_2 in the expression for the activation energy used by Wolf⁽²⁵⁾ has the property: $A_2 \propto (\partial \ln \tau / \partial \ln \dot{\gamma})_{T,\theta}$ (for a definition of A_2 see Ref.⁽²⁵⁾ and the discussion following equation (41) above). By inducing such a property into A_2 in combination with the SFE-dependence in this parameter⁽²⁵⁾, they demonstrated that the stacking-fault energies for the metals discussed here became comparable to those given in Table 2. But as demonstrated above, in order to obtain the appropriate τ_{III}^{0K} -scaling from these SFE-values, that requires a different set of pile-up numbers (w) than that recommended by Seeger *et al.*⁽²⁴⁾, which only reinforce the statement made above on the applicability of a cross-slip based model.

Let us, however, assume that the activation energies for cross-slip and the associated τ_{III} -values can be calculated based on available SFE-data, then the present model has been developed to handle dynamic recovery controlled by cross-slip, in principle, and an interesting aspect becomes to find out if the predictions of this model are consistent with the experimental observations discussed here. Applying the Seeger–Wolf and Bonneville and Escaig expressions for the apparent activation energy for cross-slip [i.e. $g(U, \tau) = A_2 \ln \tau_{III}^{0K} / \tau$ where $A_2 = f(w, \Gamma)$ ⁽²⁵⁾ and $g(U, \tau) = (3.12 \cdot 10^{-19} - 158\tau b^3)J$ (see equation (43)) respectively], it follows that this energy for Cu, is less than that for climb ($g(U\tau) = U_{SD} - \xi\tau b^3$) for all temperatures, the implication of which is that cross-slip will be rate-controlling for any combination of strain-rate and temperature. This is in partial agreement with the climb analysis above, at least in the sense that only one rate-controlling mechanism is operative in the temperature and strain rate range investigated. This is also in line with Kocks statement that diffusion is never important for macroscopic plastic flow⁽¹⁰⁾. That high temperature plasticity, however, is controlled by a non-diffusive-mechanism is in conflict with the bulk of evidence from hot deformation and creep. Furthermore, it follows that a model based on the thermal stability of screw dipoles gives a constitutive relationship of the same form as equation (52), the only problem with this relationship in this case is that the stress exponent now becomes equal to 1 in equation (52) or a power law creep exponent of 2, in obvious conflict with the experimental observations in Fig. 12. So, even if reliable SFE's were available, and if such data could be translated into activation energies for cross-slip, such a cross-slip mechanism seems to have problems in accounting for the observations discussed here.

A final point in this context: According to the present analysis we expect the τ^* factor to scale with the normalized 0K saturation stress ($\tau_{III}^{0K,SD}/G \propto U_{SD}/G_o b^3$), and, as can be seen from Table 2, the data do indeed confirm such a correlation. So to reiterate, the conclusion becomes that thermal activation of vacancies is the most likely rate-controlling mechanism important for macroscopic plastic flow in the temperature and strain rate ranges considered in Figs 12 and 13.

3.7. Work Hardening in Terms of a One-Parameter Model

The Kocks–Mecking (K–M) model^(9, 11–13), which dates back 20 years, predicts a Voce type hardening in Stage III on the basis of a one-parameter description. This widely used modelling approach to work hardening is considered here not only for comparing a one-parameter model to the present multi-parameter one, the main reason for the interest in the K–M model is, as pointed out in Section 3.4 above, that it relies on a dynamic recovery concept which is fundamentally different from that applied here. If the problem of dislocation storage in a one-parameter model is analyzed by applying the present approach to dynamic recovery [or that of Nix *et al.*⁽³³⁾], the result does not become a Voce type hardening, to the contrary, Stage II will be followed by only a marginal amount of work hardening prior to stress saturation. This important aspect is covered in more details in the latter part of this section.

The Kocks–Mecking model (for instance as defined in Ref.⁽¹²⁾) rests on the following three expressions, respectively, for the flow stress, athermal hardening and dynamic recovery:

$$\tau = \alpha G b \sqrt{\rho} \quad (54)$$

$$\frac{d\rho^+}{d\gamma} = \frac{2\theta_h}{\alpha G b} \sqrt{\rho} \quad (55)$$

$$\frac{d\rho^-}{d\gamma} = -\rho \frac{v_r}{\dot{\gamma}} \quad (56)$$

where θ_h is the athermal work hardening rate and v_r is the frequency at which the stored dislocations are annihilated. A fundamental assumption in the K–M approach is that this annihilation frequency is only dependent on temperature at a constant strain-rate, i.e. there is no element of static recovery associated with equation (56) in their treatment. Combining equations (55) and (56) gives the following microstructural evolution law:

$$\frac{d\rho}{d\gamma} = \frac{2\theta_h}{\alpha G b} \sqrt{\rho} - \rho \frac{v_r}{\dot{\gamma}} \quad (57)$$

The $d\rho/d\gamma$ -term and the ρ -terms in this equation can be replaced by $d\tau/d\gamma = \theta$ and appropriate expressions for τ (using equation (54)), and equation (57) can be written:

$$\theta = \theta_h \left[1 - \frac{\tau}{\tau_{III}} \right] \quad (58)$$

which is the Voce law in differential form.

Since equations (54)–(56) or their derivatives (equations (57) and (58)), within the framework of a one-parameter description, are assumed to be of general validity for all values of $\tau \leq \tau_{III}$, it follows from this Voce-law type hardening relationship (equation (57)), in combination with the boundary condition imposed by the Stage II–III transition, that the undefined parameters θ_h and v_r must satisfy the following requirements:

1. For $\tau < \tau_{III}$; $\theta = \theta_h$ and $v_r = 0$.
2. For $\tau_{III} \leq \tau \leq \tau_{III}$; $\theta_h = \theta_{III}/(1 - \tau_{III}/\tau_{III})$ and $v_r = 2\theta_h \dot{\gamma}/\tau_{III}$.

And we have a situation where the τ_{III} -stress level becomes a singularity associated with discontinuous changes in both the athermal hardening parameter, θ_h , and the annihilation frequency, ν_r . Some comments are needed regarding these discontinuities. Firstly, in the Voce law regime a necessary assumption becomes that the τ_{III}/τ_{III_s} -ratio is strain-rate and temperature invariant (follows from $\theta_h = \theta_{II}/(1 - \tau_{III}/\tau_{III_s}) = \text{constant}$). As suggested by Gil Sevillano⁽⁷⁾, θ_h can also be interpreted as a common extrapolated work hardening limit for Stage III microstructures. In a one-parameter model, however, there are no microstructural changes associated with a Stage II–III transition*, from which it follows that there is no physical reason for this transition to cause a change in the value of θ_h . This can be formulated in a more stringent way; in a one-parameters description it becomes a *boundary condition* that $\theta_h \equiv \theta_{II}$ for all values of $\tau \leq \tau_{III_s}$. Secondly, on the discontinuous change in ν_r , this may be interpreted in terms of the common idea that Stage III is triggered by a critical stress (see Ref.⁽⁷⁾) followed by a rapid increase in the annihilation frequency to a constant ν_r -level. The idea that Stage III is triggered by a critical stress will be re-examined in the next sub-section below.

The important conclusion, however, to be drawn from this simple inspection is that, in order *not to violate* the boundary condition ($\theta_h \equiv \theta_{II}$ for $\tau \leq \tau_{III_s}$), the annihilation frequency, ν_r , cannot be treated as a constant independent of τ (at a constant temperature and strain rate). This conclusion, namely that $\nu_r = f(\tau)$, is consistent with the analysis of dynamic recovery developed above (equations (39)–(41)). The consequences of having a stress dependent annihilation frequency for the work hardening behavior in Stage III (in a one-parameter analysis) will now be considered in some more details.

3.7.1. A one-parameter model, re-examined

The following analysis will, like the one by Kocks and Mecking above, be developed on the basis of equations (54)–(56). In this treatment, however, θ_h in equation (55) is replaced by θ_{II} and ν_r will be a function of the instantaneous dislocation density in accordance with the dynamic recovery treatment presented in Section 3.5.1.3.

It follows from equation (57) that in order to have $\theta_h = \theta_{II}$ for $\tau \leq \tau_{III}$ requires that the $d\rho^-/d\gamma$ -term in this equation is equal to zero, or $\nu_r = 0$, a situation which is consistent with a critical triggering stress for the initiation of Stage III. It is, however, physically unjustified to assume that $d\rho^-/d\gamma = 0$ in Stage II only because strain hardening appears to be well described by a straight line. The idea of a threshold stress applies strictly only to 0 K. At any other temperature thermally activated dynamic recovery will take place at a rate defined by equation (56). The effect of this recovery, however, during deformation in the ambient temperature range requires large stresses in order *to be seen*. This point is easily illustrated. Let us apply an annihilation term analogous to that given by equations (39)–(41) (Nix *et al.*⁽³³⁾ used a similar expression), but here written in a more generalized form:

$$\frac{d\rho^-}{d\gamma} = -B_\rho b^{(2x-2)} \rho^x \left(\frac{\nu_D}{\dot{\gamma}} \right) \exp\left(-\frac{g(U, \tau)}{kT} \right) \quad (59)$$

Where $g(U, \tau)$ specifies the nature of the ‘obstacle’ to thermal activation, B_ρ is a constant and x can have values of 1 or 2. Let us select both climb and cross-slip as rate-controlling reactions. In the climb case we have $x = 2$ and a linear stress dependence in $g(U, \tau)$, i.e.

*Even in a more realistic cell-boundary-type microstructural description no clear structural changes marks the transition from Stage II to Stage III, as noted by Basinski and Basinski⁽⁵⁰⁾.

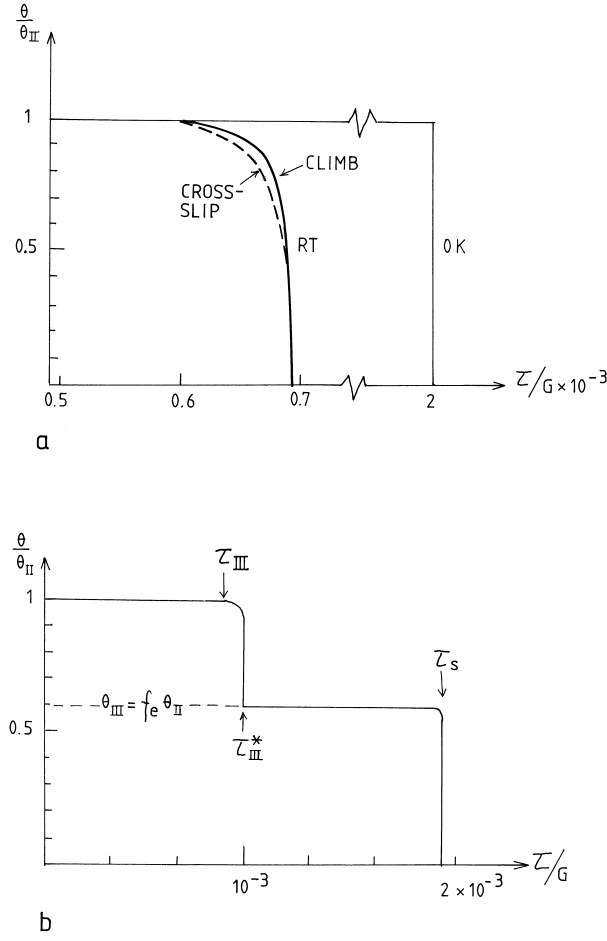


Fig. 14. θ - τ diagrams for copper based on (a) a one-parameter description, and (b) a quasi one-parameter description

$g(U, \tau) = U - \xi \tau b^3$ where ξ is a stress intensity factor to be defined by the 0K saturation stress, i.e. $\xi = U/\tau_s^{0K} b^3$. In the cross-slip case, $x = 1$ and according to Wolf⁽²⁵⁾; $g(U, \tau) = -A_2 \ln(\tau/\tau_{III}^{0K})$ where A_2 is a parameter which is a function of the stacking-fault energy (see comments to equation (41)). Combination of equations (54), (55) and (59) then gives:

$$\theta = \theta_{II} \left[1 - \frac{1}{2} B_\rho C \left(\frac{\tau}{\alpha G} \right)^{2x-1} \left(\frac{\nu_D}{\dot{\gamma}} \right) \exp \left(- \frac{g(U, \tau)}{kT} \right) \right] \quad (60)$$

The θ - τ diagram in Fig. 14(a) illustrates the predictions due to this relationship (in the calculations the parameters used for copper above are also used here). Note the 'blown-up' τ/G -scale in Fig. 14(a), illustrating that at RT the difference between τ_{III} and τ_{IIIs} is a few MPa's only in both of these two cases. (Similarly shaped curve results also from the Bonneville and Escaig model⁽²⁹⁾). At 0K there is of course, no Stage III. This shows that the idea of a Stage III triggering stress is a good approximation indeed.

This work hardening behaviour for the one-parameter model is analogous to that developed above for the dislocation density inside the subgrains (ρ_i) in the present three-parameter description. So the important conclusion becomes that, on the basis of a one-parameter description of work hardening, deviation from linearity in Stage II will immediately be followed by stress saturation, or, *a one-parameter model is only capable of describing Stage II and Stage II breakdown.*

One might argue that this analysis is unrealistic in the sense being developed on the basis of either one of two different and independent rate-controlling mechanisms, each of which was assumed to be capable of causing the dislocation density to saturate, i.e. $d\rho/d\gamma = 0$. In reality deviation from linearity in Stage II might be interpreted as due to cross-slip and the final saturation as being due to climb. However, if such assumptions are incorporated into the model, one no longer has a one-parameter description since the stability of screws and edges will have to be treated independently. That is easily done with the result as schematically outlined in Fig. 14(b). At stress-levels $\tau > \tau_{III}^*$ screw dislocations are no longer stable and continued hardening is due to the storage of edge dislocations only (ignoring the complication such a dislocation storage pattern has in terms of building up realistic 3-D dislocation networks). It follows from equation (60), however, that in this 'refined' one-parameter model, Stage III will also be associated with linear hardening, although at a reduced rate compared to Stage II. This Stage III strain hardening rate becomes; $\theta_{III} = f_e \theta_{II}$, where f_e is the fraction of the mobile dislocations being stored as edges. Eventually these edge dislocations become unstable due to climb and stress saturation occur at a stress τ_s . While this refinement of the one-parameter model above does provide a strain hardening capacity after Stage II, the model prediction in Fig. 14(b) must be rejected as being rather unrealistic indeed. Or in other words, a one-parameter approach to work hardening is not capable of accounting for the stress-strain behaviour of metals beyond Stage II. This is the physical logic of thermally activated reactions (exponential in the applied stress).

A final remark in this context becomes that Mughrabi's^(47, 48) idea of a composite structure was not only important in relation to refining our understanding of the flow-stress in relation to microstructure, it was a necessary refinement in order to understand work hardening beyond Stage II. Having, as demonstrated in this work, two microstructural elements which obey: (i) different principles of athermal storage and (ii) different dynamic recovery kinetics, the inevitable result becomes a multi-stage work hardening behaviour. It should not be necessary to add that such a microstructural evolution picture is experimentally well founded, indeed. A final note becomes that this conclusion (regarding the necessity for (at least) a two-parameter model) is implicit also in the treatments of Nix *et al.*⁽³³⁾ and Prinz and Argon⁽⁴³⁾. Returning now to the present three-parameter model and its predictions in terms of work hardening beyond Stage III.

3.8. Stage IV Work Hardening, a Low Temperature Solution

Following Stage III, a quasi-scaling situation has been re-established in the sense that the cell boundaries are replaced by sub-boundaries of a well defined average misorientation which during Stage IV is strain invariant, i.e. $\varphi = \varphi_{IV} = \text{constant}$ (Section 2.2 and Fig. 6). Accordingly, the variation in the subgrain structure with strain is in this context well described by only one scaling parameter, namely the average subgrain size, δ . The term *quasi-scaling* is applied to describe this situation because the dislocation density within the subgrains does not vary with strain ($\rho_i = \rho_{III} = \text{constant}$). On the assumption that the statistical approach presented in Section 3.2 applies also during Stage IV, it follows that

equation (29) in combination with equation (11) defines the work hardening rate during Stage IV to be:

$$\theta_{IV} = \frac{2\rho_{III}C\alpha_2GbS_{IV}}{\kappa\phi_{IV}\sqrt{\rho_{III} + \frac{\kappa\phi_{IV}}{b\delta}}} \quad (61)$$

where both ρ_{III} and δ can be expressed in terms of τ_{IV} (by combining equations (11) and (51)), i.e. $\rho_{III} = (q_c/\delta_{III})^2 = (q_c/q_{IV}\delta_{IV})^2 = (\tau_{IV}/\alpha_4Gb)^2$ and $\delta^{-1} = (\tau_{IV}/\alpha_2Gb)(\tau/\tau_{IV} - \alpha_1/\alpha_4)$. From the scaling characteristics of the boundary structure a reasonable assumption becomes that S will remain approximately constant throughout Stage IV, i.e. $S = S_{IV}$. The relationship between the strain and microstructural evolution during Stage IV is obtained by solving the following integral (see also equation (49)).

$$\gamma - \gamma_{IV} = -\frac{\kappa}{2\rho_{III}CS_{IV}} \int_{\delta_{IV}}^{\delta} \phi_{IV} \frac{\sqrt{\rho_{III} + \frac{\kappa\phi_{IV}}{b\delta}}}{\delta^2} d\delta \quad (62)$$

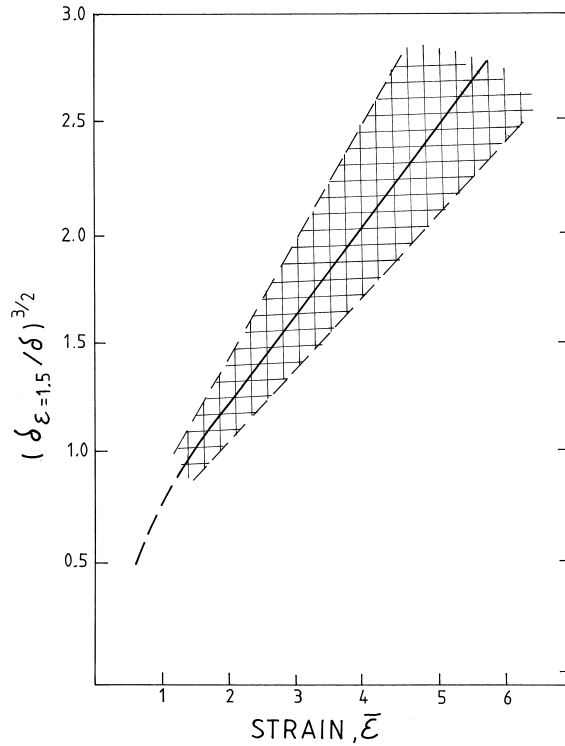


Fig. 15. Subgrain/cell size a function of equivalent strain, a replotting of the data in Fig. 4 as $(\delta_{\epsilon=1.5}/\delta)^{3/2}$ vs $\bar{\epsilon}$. (The shaded area reflects the experimental scatter in Fig. 4)

As ρ_{III} during Stage IV is expected to be much less than $\kappa\phi_{IV}^*/b\delta$, this integral gives

$$\gamma - \gamma_{IV} = \frac{2}{3} K_{IV} b^{-1/2} \phi_{IV}^{3/2} \delta_{IV}^2 \left[\left(\frac{1}{\delta} \right)^{3/2} - \left(\frac{1}{\delta_{IV}} \right)^{3/2} \right] \quad (63)$$

where $K_{IV} = \kappa^{3/2} (q_{IV})^2 / (2q_c^2 C S_{IV})$.

The experimental data presented in Fig. 4 were interpreted as showing a linear variation in $1/\delta$ with strain (γ) while equation (63) predicts a $(1/\delta)^{3/2}$ vs γ relationship. These two relationships, however, are so close that the data in Fig. 4 cannot be used in deciding which relationship is the more accurate, as illustrated by replotting the data in a $(1/\delta)^{3/2}$ vs $\bar{\epsilon}$ diagram, Fig. 15.

The quasi-scaling situation characteristic of Stage IV has a profound effect on the work hardening rate, and the model predictions (equation (61)) are in excellent agreement with observations as illustrated in Fig. 11(a). It is noted that this analysis does not predict a linear hardening in Stage IV, but the deviation from linearity is minor indeed and in accordance with experimental observations as seen from Fig. 11(a). Note also that the models predicts that the initial value of θ_{IV} scales linearly with τ_{IV} . The Stage IV dislocation debris model, developed by Kocks and coworkers^(8,19) will not be discussed besides mentioning that this treatment represents an M extension of the K–M Stage III model, and the present view on that one-parameter analysis is covered in Section 3.7. In contrast to a one-parameter description it is here emphasized that the sequence of different stages which appears beyond Stage II in the present *multi-parameter treatment* is a consequence of the characteristic differences in the associated microstructural evolution, differences which are experimentally well founded and which follows from the differences in both the athermal storage and the dynamic recovery properties of the microstructural elements involved.

A final comment on Stages III and IV. The θ - τ diagram in Fig. 11(a) reveal a strong temperature-dependence especially of Stage III, and a somewhat lower dependence in Stage IV. One should be aware, however, that these stages of work hardening, according to the present model, display no intrinsic temperature-dependence by themselves, the effect of temperature on these stages is entirely inherited from the II–III transition. In this model the temperature-dependencies of Stage III and IV only reflect the temperature-dependence of ρ_{III} , as given by equation (53).

3.9. Work Hardening, the General Solution and the Saturation Stress

At large strains, dynamic recovery of the subgrain structure can no longer be ignored. At sufficiently high stresses the annihilation rate of sub-boundaries due to boundary migration will affect the flow stress, which now requires the solution of the differential equation

$$\frac{d\delta}{d\gamma} = \frac{d\delta^-}{d\gamma} + \frac{d\delta^+}{d\gamma} \quad (64)$$

The first term in this equation is defined by equation (62) which in differential form can be written

$$\frac{d\delta^-}{d\gamma} = - \frac{b^{1/2} \delta^{5/2}}{\phi_{IV}^{3/2} \delta_{IV}^2 K_{IV}} \quad (65)$$

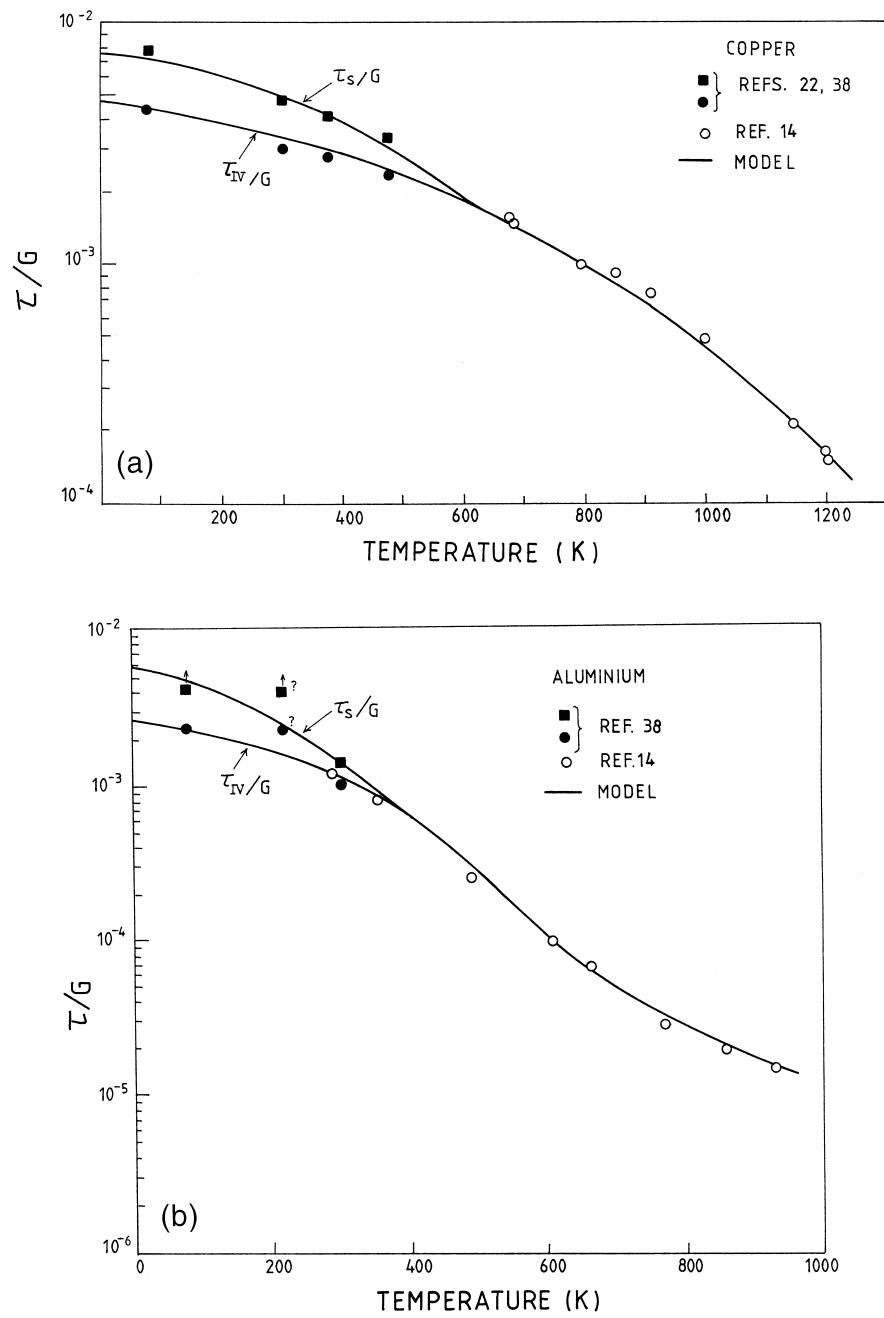


Fig. 16. The theoretically predicted saturation stresses (τ_{IV}/G and τ_s/G) as functions of temperature. (a) aluminium and (b) copper (experimental data from Refs. ^(14, 22, 38))

In a recent paper by the present author⁽⁴⁾, equation (64) was solved using an empirical expression for the work hardening in Stage IV as a basis for deriving $d\delta^-/d\gamma$. That treatment gave $d\delta^-/d\gamma \propto \delta_s^{-1} \delta^2 \propto \delta_{IV}^{-1} \delta^2$, a result which compared to equation (64) is clearly less dependent on δ_{IV} and only slightly less dependent on δ . The $d\delta^+/d\gamma$ -term is given by equation (33). Combination of equations (33), (64) and (65) gives

$$\gamma - \gamma_{IV} = \int_{\delta_{IV}}^{\delta} \left[\frac{2\nu_D b^2 B_\delta \sqrt{\rho}}{\dot{\gamma}} \left(\exp - \frac{U_\delta}{kT} \right) \sinh \frac{PV_a}{kT} - \frac{b^{1/2} \delta^{5/2}}{\phi_{IV}^{3/2} \delta_{IV}^2 K_{IV}} \right]^{-1} d\delta \quad (66)$$

This equation cannot be solved analytically, but numerical solutions can be derived. A more complete treatment of the temperature and strain rate dependence of the entire stress-strain curves will be the topic of a forthcoming paper. However, some general aspects of the stress saturation associated with steady state deformation will be briefly considered in the following.

3.9.1. Stress saturation and the steady-state microstructure

The termination of Stage IV is followed by steady-state deformation characterized by a strain invariant microstructure (dislocation density and subgrain size). The temperature and strain rate dependence of the steady-state subgrain size, δ_s , is obtained by solving equation (64) for the special case $d\delta/d\gamma = 0$, i.e. by combination of equations (33), (64) and (65) which gives

$$2 \sinh \frac{4\xi_\delta G b^4}{\delta_s k T} = \frac{\delta_s^3}{\delta_{IV}^2 b K_s} \left(\frac{\dot{\gamma}}{\nu_D} \right) \exp \frac{U_{SD}}{kT}, \text{ for } \delta_s \leq \delta_{IV} \quad (67)$$

where $K_s = \phi_{IV}^2 B_\delta \kappa^{1/2} K_{IV}$ and ξ_δ is a stress intensity factor, the origin of which will be discussed in a final paragraph at the end of this section. This constitutive law is of a similar form to that regulating the dislocation density inside the subgrains (equation (53) replacing τ_{III_s} with $\tau_{III_s} = \alpha_4 G b \sqrt{\rho_{III}}$). The normalized steady state flow stress (τ_s/G) is obtained by combining equation (67) with equations (46) and (51) (i.e. by replacing δ_s^{-1} with $\delta_s^{-1} = (1/\alpha_2 G b)(\tau_s - (\alpha_1/\alpha_4)\tau_{III_s})$ and $\delta_{IV}^{-1} (\simeq \delta_{III_s}^{-1})$ with $\delta_{IV}^{-1} = (1/\alpha_2 G b)(1 - (\alpha_1/\alpha_4)\tau_{III_s})$). If the temperature of deformation is increased the saturation stress τ_s will decrease and eventually approach the level of τ_{III_s} , and it follows that above a critical temperature $\tau_s \simeq \tau_{III_s}$. Application of equations (53) and (67) to the steady state deformation in copper and aluminium is illustrated in Fig. 16(a) and (b) which show that the experimental observations are well accounted for over the entire temperature range from close to 0K to the melting point (fitting parameters $\xi_\delta(\text{Cu}) = 102$, $B_\delta(\text{Cu}) = 2 \cdot 10^4$, $\xi_\delta(\text{Al}) = 35$ and $B_\delta(\text{Al}) = 511$). The experimental data are taken from Zehetbauer⁽³⁸⁾, Alberdi Garitaonandia⁽²²⁾ and Mecking *et al.*⁽¹⁴⁾ (the latter data are also included in Fig. 13). A similar relationship (to equation (67)) for the steady state subgrain size and flow stress was recently presented by the present author⁽⁴⁾. This latter treatment relied on microstructural scaling, i.e. $\sqrt{\rho} \propto 1/\delta$. The advantage of the present refined treatment is that both the dislocation density and the subgrain size are accounted for independently, which opens up the possibility of testing the idea that steady state-deformation is associated with microstructural scaling (similitude).

Evidence for microstructural scaling during steady-state hot deformation has been provided by Sellars and co-workers^(52,79-81) and Blum^(82,83), the diagram in Fig. 17 has been taken from the latter works. This Blum-diagram shows that the spacing of dislocations inside the subgrains appears to scale with the subgrain size over nearly two

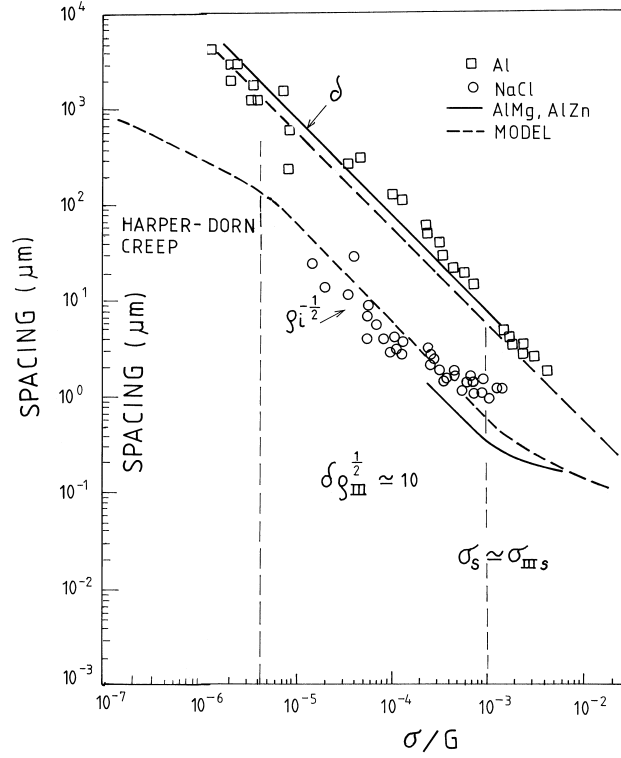


Fig. 17. Steady-state substructure elements (subgrain size δ and dislocation separation $\rho_i^{-1/2}$) as a function of normalized stress, σ/G for aluminium, NaCl, AlMg and AlZn alloys. From Blum^(82,83). Model predictions indicated by broken lines

orders of magnitude in stress. However, the data display considerable scatter with a trend towards a deviation from scaling at high stress levels. The data in Fig. 17 are from pure aluminium, NaCl and solid solution aluminium alloys. The theoretical steady state flow stress predictions given in Fig. 16 can be translated into variations in both $\rho_i = \rho_{III}$ and δ_s and plotted in the Blum-diagram as illustrated by the broken line in Fig. 17. In the high stress region [i.e. when $\tau_s > \tau_{III s}$ in Fig. 16(b)] the saturation subgrain size is defined in terms of the saturation stress $\tau_{III s}$ and τ_s as given by equations (46) and (51), i.e. $\delta_s^{-1} = (1/\alpha_2 G b)(\tau_s - (\alpha_1/\alpha_4)\tau_{III s})$, while the saturation dislocation density is represented by $\rho_{III}^{1/2} = \tau_{III s}/\alpha_4 G b$ (equation (51)). It follows that in this regime these microstructural terms will not be related to each other through a simple scaling law of the form $\delta\sqrt{\rho_{III}} = \text{constant}$, rather, this product will be given by

$$\delta\sqrt{\rho_{III}} = \frac{\alpha_2 \tau_{III s}}{\alpha_4 \tau_s - \alpha_1 \tau_{III s}} \quad (68)$$

As τ_s approaches its limiting value ($\tau_s \approx 2\tau_{III s}$ [see Fig. 16(b)]) the $\delta\sqrt{\rho_{III}}$ -term approaches its minimum value of $\alpha_2/(\alpha_1 + 2q_{IV}\alpha_2/q_c) \approx 2$. The maximum value of this product in the high stress regime is reached when $\tau_s = \tau_{III s}$, in which case $\delta\sqrt{\rho_{III}} = q_c/q_{IV} \approx 5$, see Fig. 17.

It is interesting to note that the experimental data recorded by Blum^(82,83) from Al–Mg/Al–Zn alloys, high purity aluminium and NaCl clearly indicate such a non-scaling trend in the high-stress region, Fig. 17.

The next question becomes; what happens to the $\delta_s \text{ vs. } \sqrt{\rho_{III}}$ relationship with a continued decrease in τ_s ? equation (67) defines the steady-state subgrain size only as long as $\delta_s \leq \delta_{IV}$. By increasing the temperature of deformation or decreasing the strain rate, stress saturation is reached before reaching a Stage IV, and equation (65) needs to be replaced by equation (49) (in differential form). However, if steady-state is reached at progressively lower stresses the generation rate of new boundaries will eventually be defined by that typical of Stage II, in which case $d\delta^-/d\gamma$ is obtained by rewriting equation (A2) in the form

$$\frac{d\delta^-}{d\gamma} = - \frac{\delta^2}{bq_c^2 C \sqrt{1 + f(q_b^2 - 1)}} \quad (69)$$

The steady state subgrain size during high temperature deformation (i.e. when $PV_a/kT = 4\xi_\delta Gb^4/\delta_s kT \ll 1$) is then obtained by solving equation (64) for $d\delta/d\gamma = 0$, which gives

$$\delta_s^{-1} = \left[K_\delta \left(\frac{kT}{Gb^3} \right) \left(\frac{\dot{\gamma}}{v_D} \right) \exp \frac{U_{SD}}{kT} \right]^{1/4} \quad (70)$$

where $K_\delta = [8b^{-4} \xi_\delta q_c^2 C B_\delta (1 + f(q_b^2 - 1))]^{-1}$. The corresponding relationship for the intercell dislocation density, ρ_{III} , is obtained by rewriting equation (53) in the form

$$\sqrt{\rho_{III}} = \left[K_\rho \left(\frac{kT}{Gb^3} \right) \left(\frac{\dot{\gamma}}{v_D} \right) \exp \frac{U_{SD}}{kT} \right]^{1/4} \quad (71)$$

where $K_\rho = [b^4 \xi_\rho^2 C B_\rho \sqrt{1 + f(q_b^2 - 1)}]^{-1}$. And it follows that *high temperature creep is associated with microstructural scaling*, with the scaling constant defined as follows:

$$\delta_s \sqrt{\rho_{III}} = (K_\rho / K_\delta)^{1/4} \quad (72)$$

For the special case of aluminium and aluminium alloys illustrated in Figs 16 and 17, $\delta_s \sqrt{\rho_{III}} \simeq 10$, in reasonable agreement with the observations by Blum. In generating the broken line in Fig. 17 the values of the constants involved are the same as used in obtaining Figs 12 and 16(b), and it follows that the product $\delta \sqrt{\rho_{III}}$ changes with decreasing saturation stress from having a value of about 2 at 0K to a value of about 5 at intermediate stress levels (when the τ_s and τ_{III_s} values merge together) reaching a value of about 10 when τ_s ($= \tau_{III_s}$) approaches the stress level of τ_{III}^* . As the change over from 5 to 10 is rather marginal when generating a diagram like Fig. 17, the conclusion becomes that a quasi-scaling situation is established already when the two stresses τ_s and τ_{III_s} merges together. Or in other words, a clear non-scaling situation becomes apparent only under conditions where the stress–strain curve is associated with a well defined Stage IV.

If one continues to conduct experiments with progressively lower saturation stresses a deviation from scaling is reached also in the low stress region as indicated in the diagram in Fig. 17. This new situation corresponds to a phenomenon referred to as Harper–Dorn creep^(84,85), the mechanism of which will be considered in the following section on creep.

Firstly, however, some comments are needed regarding the physical meaning of the stress intensity ξ -parameters (ξ_ρ and ξ_δ are of the order 100 and 50 respectively). The origin of ξ_ρ

is dealt with in Section 3.5.1 and related to the stability of ‘dynamic’ edge dipoles. A similar argument may apply also to the migration of sub-boundaries, i.e. that they in a dynamic situation are ‘speeded up’ by the mobile dislocations. We must remember that during hot deformation the sub-boundaries are quite mobile indeed. This follows from the common observation that the dynamic subgrain structure remains equiaxed during hot deformation, which implies that any boundary will have to migrate at an average speed defined by the product: $\delta \cdot \dot{\gamma}$. In order to maintain such an equiaxed boundary structure it also follows that during hot deformation sub-boundaries have to be flexible. A sub-boundary can be flexible, however, only if it is able to continuously change the composition of the dislocation configuration forming the boundary network, for a detailed discussion of this point, see Ref.⁽²⁾. In a dynamic situations boundaries are exposed to moving dislocations of a multiplicity of Burgers vectors, which will provide flexibility and stimulate mobility. It is suggested that such an interaction between mobile dislocations and migrating boundaries is responsible for the enhanced boundary mobility in dynamic situations compared to static ones.

4. CONSTITUTIVE LAWS FOR STEADY-STATE CREEP

This section has been included for two reasons, firstly to demonstrate that the present approach provides constitutive relationships for steady state creep which are consistent with what are to be expected and secondly to show that this approach also opens for an alternative interpretation of Harper–Dorn creep. If the objective is to derive physically based constitutive equations for the steady-state deformation at a constant stress it is the opinion of the present author that this can be achieved *only* by balancing the quasi-athermal dislocation storage reaction with that due to the annihilation caused by dynamic recovery, i.e. one has to properly define and solve equation (3) (for the condition of steady state, $d\rho/d\gamma = 0$) or its variants equations (42) and (64) (which has been tried above). In the creep literature such an approach has, so far, not been followed in a stringent way, an important reason for which is that a satisfactory treatment of athermal storage of dislocations in a multi-parameter situation has not been available. Instead, state equations are derived on the basis of phenomenological descriptions, for a recent review see Ref.⁽⁸²⁾. Such state equations can be classified as being of two kinds: a high temperature/low stress variant of the form.

$$\dot{\gamma} = \text{const.} \cdot v_D \frac{kT}{Gb^3} \left(\frac{\sigma}{G} \right)^n \exp - \frac{U_{SD}}{kT} \quad (73)$$

where $3 \leq n \leq 5$, or a high stress variant

$$\dot{\gamma} = \text{const.} \cdot v_D \left(\frac{\sigma}{G} \right)^{n-1} \left(\exp - \frac{U_{SD}}{kT} \right) \sinh \frac{V_a \sigma}{kT} \quad (74)$$

The problem is that the physics behind these laws will remain somewhat in the dark as long as their appearance does not explicitly relate to equation (3).

The present treatment opens for a more complete analysis which results in different creep laws depending on whether a Stage IV is present prior to steady-state or not. In the high stress regime ($\tau_s > \tau_{IV}$) the steady-state creep rate is given by equation (67) which can be brought on the same forms as equation (74) with $n = 4$. On the other hand, during high

temperature creep (i.e. when τ_{III} is only marginally larger than τ_{III}^*) the creep rate is given by equation (53) which is well represented by equation (73) with $n = 4$ also in this case.

It is interesting to note that by deriving the creep laws from first principles, i.e. by solving equation (3) (for $d\rho/d\gamma = 0$, as done here), high temperature creep is found to be associated with a power-law-exponent $n = 4$, and not $n = 3$, which is the exponent in the natural creep law referred to in the literature⁽⁸²⁾. It is well established, however, that in high temperature creep of pure metals this exponent may be as high as 5. The reason why the present approach fails to predict $n = 5$ instead of the obtained $n = 4$ may be due to an inadequate solution of the dynamic recovery problem. More specifically, the collapse rate of dipoles, v_g , is derived using a model based on dipoles of average separation l_g (see equation (39)). The physical reality, however, is that l_g is the mean separation in a distribution of dipole separations. A more rigorous treatment, incorporating such a distribution in the model, may result in a collapse frequency which will scale with l_g to a higher power than one. (See Li's static recovery model based on the collapse of dipoles⁽⁸⁶⁾). An alternative explanation for having a stress exponent of 5 instead of 4 relates to a possible change in the stress dependence in the slip length, L , at the low saturation stresses typical of high temperature creep. At such low stress levels the dislocation density in the cell interior is of the order 10^9 m^{-2} , δ is of the order $500 \text{ } \mu\text{m}$ (see Fig. 17) and the slip length L (equation (14)) take values in the mm-range. It follows that at such stress levels the slip length may no longer be determined by statistical interactions between mobile and stored dislocations as described in Section 3.2, i.e. equation (14) may no longer be valid. In polycrystalline metals the slip length is then expected to be determined by the grain size and even in single crystals the slip length may no longer be determined by intrinsic dislocation interactions. The physical implication of this is that the slip length L in equation (4) becomes a constant, the consequence of which is that the dislocation storage term, $d\rho_i^+/d\gamma$ in equation (42), also becomes independent of ρ_i , with the result that the stress exponent in equation (52) becomes equal to 4 instead of 3, i.e. signifying a power law creep behaviour with $n = 5$ in the high temperature regime.

4.1. Harper–Dorn (*H–D*) Creep

The slope change in the low stress region in Fig. 17 was associated above with H–D creep. It needs to be pointed out, however, that Fig. 17 indicates that during H–D creep the dislocation separation ($1/\sqrt{\rho}$) varies with the inverse square root of the saturation stress. This is in accordance with the prediction of the present H–D treatment, to be presented below, but *not* in accordance with H–D creep as described in the literature. From the creep literature (for reviews see Refs.^(87–89)), the primary characteristics of H–D creep can be summarized as follows: (i) H–D creep is a steady state phenomenon, (ii) The stress exponent is equal to 1; (iii) The activation energy is that of lattice self-diffusion. (iv) The dislocation density is very low (of the order $5 \cdot 10^7 \text{ m}^{-2}$), and *independent* of stress, and (v) there is a random and reasonably uniform distribution of dislocations mainly of edge character.* Although no convincing theory of H–D creep has been developed the phenomenon is commonly interpreted in terms of a dislocation mechanism^(87–89).

*It needs to be mentioned that whether H–D creep is a real phenomenon or not has been questioned, instead of a true steady state phenomenon it has been classified both as an artifact and as a transient^(82,89). The continued discussion assumes, however, that what has been summarized above (in points (i)–(iii)) is representative for this phenomenon.

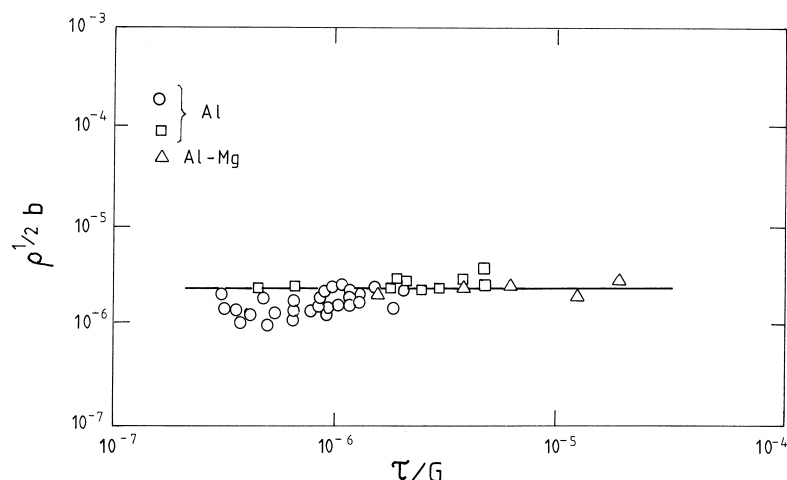


Fig. 18. Square root of the normalized dislocation density vs normalized shear stress for Al and Al-Mg tested at stress levels within the Harper–Dorn region, from Refs.^(91, 92)

However, if a constant dislocation density is a characteristic of H–D creep, an interpretation in terms of a dislocation mechanism becomes difficult indeed, the reason for which simply being that if $\rho = \text{constant}$ (in the sense of being stress and strain rate invariant) then the fundamental steady-state-equation (corresponding to $d\rho/d\dot{\gamma} = 0$ in equation (3)) will have *no physically meaningful solution*.* The only exception here is a situation where the frictional stress in equation (11) (see comment to equation (2) and Section 3.7 below) becomes comparable to the internal stresses due to the stored dislocations. In such a case the solution of equation (3) may no longer be relevant and an alternative mechanism needs to be developed. The Peierls stress model developed by Nabarro⁽⁹⁰⁾ is an example of such a mechanism. It is, however, in the following assumed that H–D creep can be interpreted in terms of a mechanisms which is similar to the ones considered above in the sense that the flow stress is controlled by long range interactions between the mobile dislocations and the stored steady-state dislocation structure. Before proposing such a model, the experimental evidence in support of a constant dislocation density will be briefly reviewed.

The diagram in Fig. 18, taken from Yavari *et al.*⁽⁹¹⁾ (using data collected by Mohamed *et al.*⁽⁹²⁾), rather convincingly demonstrates a constant dislocation density over almost two orders of magnitude in stress. The vast majority of the data points in Fig. 18 refers to an X-ray topography investigation of aluminium single crystals by Nøst and Nes⁽⁹³⁾. However, for unknown reasons, their data have been misrepresented by Mohamed *et al.* The Nøst and Nes data do not reflect a stress invariant dislocation density at all during H–D creep as clearly revealed in Fig. 19. It is interesting to note, however, that the measured dislocation densities do indicate a reduced stress dependency under the conditions investigated, i.e. a

*Steady-state is defined by equation (3): $d\gamma^+/d\dot{\gamma} + d\rho^-/d\dot{\gamma} = 0$. Regarding the dislocation density and strain-rate dependencies of these terms we can write $d\rho^+/d\dot{\gamma} \propto \rho^x$ (this term is independent of $\dot{\gamma}$ and during Stage II $x = 0.5$ (equation (55))) and $d\rho^-/d\dot{\gamma} \propto -\rho^y/\dot{\gamma}$, (y is typical of size 2, see equations (39) and (59)). In the general case $x \neq y$, and it follows that the steady-state dislocation density during H–D creep necessarily becomes a function of $\dot{\gamma}$. In the singular case $x = y$ no steady-state dislocation density can be defined, the physical meaning of such a case is less clear.

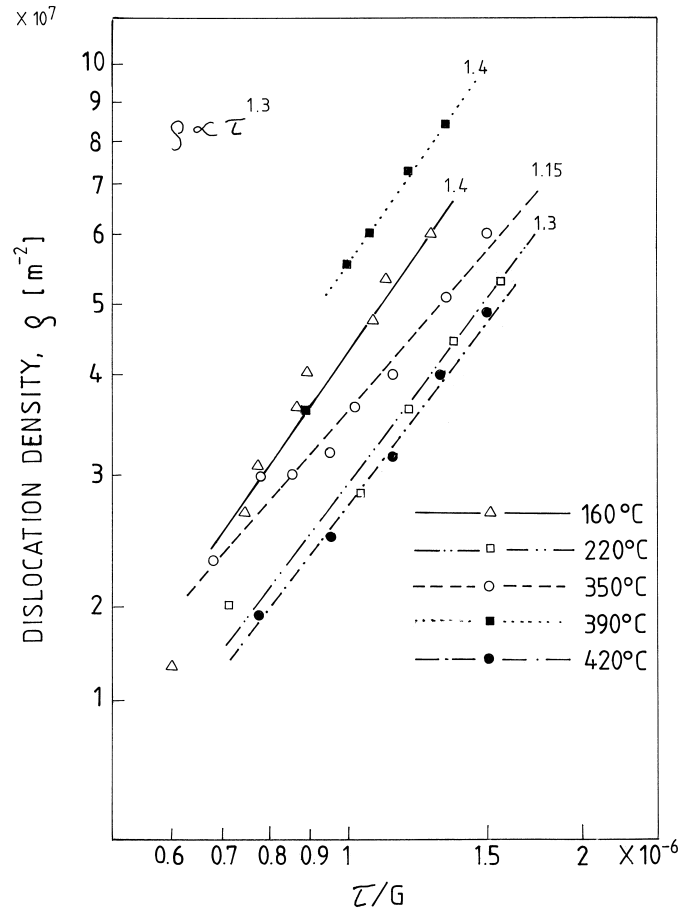


Fig. 19. Dislocation density vs normalized shear stress for aluminium tested at stress levels within the Harper–Dorn region, data from an *in situ* X-ray topography investigation by Nøst and Nes⁽⁹³⁾. Note that the dislocation density on average scales with $\sigma^{1.3}$. The numbers of the top of each line refer to the stress exponent typical of each experiment, the temperature of which is given

reduction from the commonly observed second power in stress to close to the first power, Fig. 19 shows $\rho \propto \sigma^{1.3}$. As also mentioned above, another important observation relating to H–D creep is a uniform distribution of dislocations, mainly of edge orientation, as revealed by a TEM investigation by Yavari *et al.*⁽⁹¹⁾. It is difficult to understand why these workers, after having quantified the dislocation density of H–D creep (etch pits measurements) to be around $5 \cdot 10^7 \text{ m}^{-2}$, tried to characterize these dislocations in TEM. An elementary calculation shows that more than 100 foils would be required in order to find only one ‘H–D-dislocation’. Still, a rather detailed TEM investigation was carried out by Yavari *et al.* on foils from H–D crept specimens containing dislocation densities around 10^{13} m^{-2} or 5 to 6 orders of magnitude higher than that typical of H–D creep.

As a conclusion on experimental evidence; regarding the nature of the dislocation structures generated during H–D creep, the most reliable data by Nøst and Nes⁽⁹³⁾ indicate that

the dislocation density follows a ρ vs $\sigma^{1.3}$ -relationship, and further, their *in situ* X-ray topography examination revealed a non-uniform dislocation distribution displaying some tendency to pile-up formation.

4.1.1. Modelling Harper–Dorn creep

Returning for a moment to the construction of the Blum diagram, Fig. 17. It was pointed out that as the saturation stress, σ_s , approached the value of σ_{III}^* a situation of microstructural scaling will prevail. However, if experiments are conducted with ever decreasing steady-state stresses, a situation will eventually be reached where the substructure generation no longer will be determined by that typical of Stage II but rather by a quasi Stage I situation, i.e. a condition characterized by single slip. (This will apply even to polycrystals since in H–D creep steady state is reached at strains less than 1%, in which case a ‘Sachs-stress analysis’ predicts the activation of only one slip system in each grain). With only one slip system active, microstructural scaling as defined in Section 3.2 no longer has any meaning, and accordingly the athermal storage rate of dislocations can no longer be based on equation (29). An alternative expression, valid for Stage I, will now be derived, based on Seeger’s Stage I treatment^(94–96).

On the generation of dislocations during H–D creep the fundamental assumptions in this model are: (i) A stress independent density of dislocation sources, each characterized by having generated N loops covering an average slip length L , which again are forming slip lines separated by a distance y (in Stage I, L and y are stress/strain invariant, i.e. the source density $= (y/L^2)^{-1}$ and the macroscopic dislocation density becomes $\rho \simeq 2\pi N/L^2 y = 2\pi N/Ly$). (ii) The dislocation storage rate can be written⁽⁹⁵⁾.

$$\frac{dN^+}{d\gamma} = \frac{y}{b} \quad \text{or} \quad \frac{d\rho^+}{d\gamma} = \frac{2\pi}{bL} \quad (75)$$

and, (iii) according to Seeger⁽⁹⁵⁾ the flow stress can be written:

$$\tau_s = \frac{NGb}{2\pi} (yL^3)^{-1/4} = \frac{(Ly^3)^{1/4}}{4\pi^2} Gb\rho \quad (76)$$

The steady-state value of ρ which corresponds to the conditions during H–D creep is obtained by solving equation (3). Note that in this ‘Stage I-approach’ to the modelling of H–D creep the steady-state flow stress scales with the dislocation density to the first power, which is a consequence of having a constant source density. It is interesting to note, however, that this linear relationship is in reasonable agreement with the observations by Nøst and Nes⁽⁹³⁾, Fig. 19. In order to solve equation (3) (for $d\rho/d\gamma=0$) the $d\bar{\rho}/d\gamma^-$ term needs to be identified, and like in the steady-state situations treated above, the dynamic recovery rate during H–D creep is expected to be controlled by the collapse of edge dipoles, which means that equation (38) now can be written in the form

$$\frac{dN^-}{d\gamma} = -N \frac{v_g}{\dot{\gamma}} \quad (77)$$

In calculating v_g for H–D creep the only differences, compared to the treatment above, is that l_g in this case is related to the slip line separation y , i.e. $l_g = y/\xi_\rho$ where ξ_ρ in this case becomes a constant of order unity. Note that in the case of H–D creep the slip line spacing parameter y is a constant, the consequence of which is that the collapse frequency becomes independent of the applied stress (as opposed to the treatment above where l_g scales with

$1/\sqrt{\rho}$ which again scales with the inverse applied stress). Another important difference is that the climb rate (equation (32)) in this case does not contain $\rho^{1/2}$ as a pre-exponential factor. During H–D creep the dislocation core is assumed to be saturated with vacancies in which case the climb rate is independent of the jog density, see Hirth and Lothe⁽³¹⁾. By combining equation (3) with equations (75) and (77) the following expression for the steady-state dislocation density, ρ , is obtained

$$\rho = \frac{K_{HD}}{b^2} \left(\frac{\dot{\gamma}}{v_D} \right) \left(\frac{kT}{Gb^3} \right) \exp \frac{U_{SD}}{kT} \quad (78)$$

where $K_{HD} = \pi y^2 / (bLB_\rho \xi_\rho^2)$. Combining this expression for the steady-state dislocation density with equation (76) gives the following steady-state creep rate:

$$\dot{\epsilon} = K_{HD}^* v_D \left(\frac{\sigma b^3}{kT} \right) \exp - \frac{U_{SD}}{kT} \quad (79)$$

where $K_{HD}^* = 4\pi^2 b / (K_{HD} (Ly^3)^{1/4} M^2) \simeq b^2 L / (y^2 (Ly^3)^{1/4})$. Since the dislocation density during H–D creep is about 10^7 m^{-2} , reasonable estimates for the slip length, L , and slip line separation, y , become in the ranges 10^{-2} – 10^{-3} m and 10^{-4} – 10^{-5} m , respectively, which corresponds to a K_{HD} -value somewhere in the range 10^{-8} – 10^{-10} . This value compares well with the experimentally determined one of $4 \cdot 10^{-9}$. The most important aspect to be pointed out, however, is that this analysis correctly predicts the well confirmed characteristic of H–D creep, namely a stress exponent $n = 1$.

The Peierls stress model for H–D creep proposed by Nabarro⁽⁹⁰⁾ assumes that the internal stresses (of size $\alpha Gb\sqrt{\rho}$) is equal to the Peierls stress which again is comparable to applied stress. In his analysis Nabarro estimated the Peierls stress for aluminium to be about $4 \cdot 10^{-6} \text{ G}$. A comment to this becomes that in the investigation by Nøst and Nes⁽⁹³⁾ the applied stresses (τ/G in the range $(0.5\text{--}2)10^{-6}$) correlate very well with the corresponding separations of dislocations within slip lines. Furthermore, these results which refer to direct *in situ* X-ray topography observations (see Fig. 19) indicate that the Peierls stress for aluminium is less than $5 \cdot 10^{-7} \text{ G}$, and it follows that such a low stress level makes a Peierls model for H–D creep somewhat uncertain.

5. THE FRICTIONAL STRESS AND COTTRELL–STOKES LAW

5.1. The Dynamic Contribution to the Flow-Stress

If the stress applied to a piece of metal results in plastic deformation, the instantaneous flow stress can be divided into two parts: (i) a *quasi-athermal* contribution, τ_{AT} , and (ii) a *dynamic* (instantaneous) or thermal contribution* τ_T , i.e.

$$\tau = \tau_T + \tau_{AT} \quad (80)$$

The second term, τ_{AT} , has been labelled *quasi-athermal* because it is athermal only in an ‘instantaneous’ sense, that is, in situations where the material is subjected to sudden changes in temperature or strain rate. This stress term is defined by equation (46) and is not truly athermal because it reflects the effects of strain rate and temperature on the micro-

On nomenclature, these quantities have been assigned different symbols, e.g. in a recent review on work hardening by Gil Sevillano⁽⁷⁾ τ_{AT} and τ_T are referred to as τ_c^{strong} and τ_c^{weak} respectively, while in a review on creep by Blum⁽⁸²⁾ the respective symbols are called τ_G and τ_ .

structure, i.e., the storage and annihilation of dislocations due to work hardening and dynamic recovery (see Section 3.4 above). The dynamic term (τ_T , also referred to as the effective stress) can be interpreted as a flow stress contribution due to the short range obstacle, or alternatively as a dynamic contribution to the flow stress due to obstacle limited glide i.e., $\tau_T \equiv \tau_i$, where the frictional stress τ_i is defined in relation to equation (1). This τ_T -term, which has been ignored in the present treatment until now, was derived for the first time by Seeger⁽⁹⁷⁾.

Consider a single crystal which has been deformed to a strain γ , the result of which is a microstructure defined by ρ_i and δ . Further deformation requires the applied stress to *exceed*

$$\tau_{AT} = \alpha_1 Gb\sqrt{\rho_i} + \alpha_2 Gb \frac{1}{\delta} \quad (81)$$

Let us assume that this τ_{AT} -stress is increased by an increment τ_T . This stress increment will cause a number of dislocations (ρ_m) to migrate at a rate

$$v = bB_I v_D \left(\exp - \frac{U_I}{kT} \right) 2 \sinh \frac{\tau_T V_a}{kT} \quad (82)$$

where V_a is the activation volume associated with the *rate-controlling* obstacle to dislocation glide, U_I is the activation energy and B_I is a constant. The macroscopic effect of this stress increment τ_T is that the crystal will start to deform at a rate determined by the Orowan equation

$$\dot{\gamma} = v \rho_m b \quad (83)$$

By combining equations (82) and (83) the dynamic contribution to the flow stress is obtained by solving

$$2 \sinh \frac{\tau_T V_a}{kT} = [b^2 \rho_m B_I]^{-1} \left(\frac{\dot{\gamma}}{v_D} \right) \exp \frac{U_I}{kT} \quad (84)$$

This equation cannot be solved analytically with respect to τ_T (or the frictional stress $\tau_i \equiv \tau_T$), but for $T < 1/3 T_m$ the ratio $\tau_i V_a / kT$ is expected to be larger than one in which case a good approximate solution of equation (84) becomes

$$\tau_i = \frac{U_I}{V_a} + \frac{kT}{V_a} \ln \frac{\dot{\gamma}}{b^2 v_D \rho_m B_I} \quad (85)$$

This equation may be applied to a spectrum of short range interactions which can be overcome by thermal activation*. Two important cases will be briefly considered in the following, firstly the pure metal case and secondly the effect of solute atoms on τ_i .

*The constitutive laws resulting from the K-M model reflects the combined use of a microstructural evolution law (equation (3)), and a kinetics law where the one proposed by Kocks *et al.*⁽⁹⁸⁾ is used: $\dot{\gamma} = \dot{\gamma}_0 \exp \left[- (\Delta U / kT) (\tau / \hat{\tau})^{p_1} p_2 \right]$, $\hat{\tau} \propto \sqrt{\rho}$. In this empirical relationship p_1 and p_2 are fitting parameters and ΔU an activation energy. A kinetic law of this type is not required in the present treatment because dynamic recovery is accounted for in terms of a physical model and accordingly the kinetics aspect of dynamic recovery is contained within the microstructure evolution law. The Orowan equation (equation (83)) is, in this treatment, the appropriate kinetics law taking care of the strain-rate/temperature sensitivity due to short range interactions.

In a pure metal the rate-controlling reaction is assumed to be thermal activations associated with the short range interactions between mobile dislocations and the stored ones, i.e. ‘cutting of trees’ or dragging of jogs, in which case $U_I = \beta_1 G b^3$ where β_1 is in the range 0.2–0.4. The activation volume can be written $V_a \simeq l_\rho b^2$, where l_ρ scales with the spacing of the stored dislocations, i.e. $l_\rho = \omega / \sqrt{\rho}$, where ω is a geometric constant of order unity. Furthermore, in a situation where microstructural scaling applies it follows, by combining equations (80), (81) and (85), that the flow stress takes the form

$$\tau = \frac{\sqrt{\rho}}{\omega b^2} \left[\beta_1 G b^3 - kT \ln \frac{b^2 \rho_m v_D B_I}{\dot{\gamma}} \right] + \alpha G b \sqrt{\rho} \quad (86)$$

The τ_i -contribution to this equation is relatively weak with a maximum at 0K given by:

$$\left(\frac{\tau_i}{\tau} \right)_{0K} = \left(1 + \frac{\alpha \omega}{\beta_1} \right)^{-1} \quad (87)$$

A reasonable assumption is that ω is in the range 2–4 and since $\alpha \simeq 0.3$ the τ_i/τ_o ratio is expected to be in the range 15–30%.

By differentiation of equation (82) the strain-rate sensitivity of fcc metals in the ambient temperature regime becomes as follows

$$\frac{d\tau}{d\dot{\gamma}} = \frac{kT}{\alpha \omega G b^3} \frac{\tau}{\dot{\gamma}} \quad (88)$$

Commonly, the strain-rate sensitivity of metals is accounted for by an empirical relationship of the form⁽⁹⁹⁾

$$\tau = [\text{const.} \cdot \dot{\gamma}^m]_{\gamma, T} \quad (89)$$

where the exponent m is referred to as the ‘strain-rate sensitivity’. It is interesting to note that differentiation of this empirical relationship with respect to $\dot{\gamma}$ gives an expression on the same form as equation (88) provided $m \equiv kT/\alpha \omega G b^3$. This definition of the strain rate sensitivity gives (around room temperature) $m \approx 10^{-2}$, a value of the right magnitude. It is further emphasized that the empirical equation (89) is of limited validity. The constitutive equation for the dynamic flow stress contribution which determines the true strain rate sensitivity is given by equation (84).

Solute pinning. In less pure metals and in solid solution alloys short range interactions between the mobile dislocations and solute atoms may determine the frictional stress. A large number of theories have been created to explain solid solution hardening and excellent recent reviews have been published by Haasen⁽¹⁰⁰⁾ and Neuhäuser and Schwink⁽¹⁰¹⁾. The effect of solute atoms will be only briefly considered here, the objective being to explore the potential of applying equation (84) (or 85) in the interpretation of rate sensitivity effects, especially in the transition from pure metal to alloy behaviour. When applying equation (85) in order to quantify the frictional stress in solute alloys the idea is that the activation energy U_I now reflects the interaction between solute atoms and the dislocation core ($U_I = U_{sa} = \beta_2 G b^3$ where β_2 is expected to be smaller or much smaller than β_1 , typically of the order $10^{-1} \beta_1$), and that the activation volume is determined by the solute concentration. On the assumption that the mobile dislocations migrate at such a high rate that no solute atmosphere is established (i.e. short range interactions involving only solute atoms engaged in the glide plane, a breakaway situation), then an estimate for the activation volume

becomes: $V_a = l_{sa} b^2$ where l_{sa} is the average separation of core solute atoms, i.e. $l_{sa} = \omega_{sa} b c^{-e}$ where c is the solute concentration and ω_{sa} a constant. To identify the value of the exponent e is a key objective of the various theories of solid solution strengthening^(100,101), providing values in the range $0.3 < e < 1$. Experimentally, however, it appears to be well confirmed that the activation volume V_a scales with $b^3 c^{-2/3}$ in both hcp⁽¹⁰²⁾ and fcc⁽¹⁰³⁾ alloys.

In the case of solute pinning the strain rate sensitivity becomes

$$\frac{d\tau}{d\dot{\gamma}} = \frac{kTc^e}{\omega_{sa}b^3} \left[\frac{1}{\dot{\gamma}} - \frac{1}{\rho_m} \frac{d\rho_m}{d\dot{\gamma}} \right] \quad (90)$$

In the pure metal case a reasonable assumption is that $d\rho_m/d\dot{\gamma} \approx 0$ (the assumption made in deriving equation (88)). This is necessarily not the case when solute pinning controls the thermally activated glide reaction. Then, instantaneous changes in strain rate may cause the release of temporarily locked dislocations, the result becoming an avalanche effect where the second term within the brackets in equation (86) dominates over the first term. The result is the well known negative strain-rate sensitivity effect associated with some solid solution alloys (e.g. Al-Mg alloys).

If a solute containing metal is subjected to increasing plastic deformation, a transition from solute controlled to cutting of trees controlled thermal activation is expected, for the simple reason that as the forest dislocation density increases l_ρ may (in dilute alloys) eventually become smaller than l_{sa} . It follows from equation (85), which can be defined in terms of both solute and 'forest' pinning that a transition from the first to the second mechanism will occur, at 0K, when the deformation causes the flow stress to exceed a critical value, τ_{tr} , given by:

$$\tau_{tr} \simeq \frac{\alpha\omega G \beta_2}{\omega_a \beta_1} c^e \simeq 0.1 G c^e \quad (91)$$

This is expected to be a good estimate for the transition stress also in the ambient temperature range where an atomic fraction c in the range 10^{-3} – 10^{-2} will correspond to a τ_{tr} value typical of Stage II. The relevance of this transition in relation to strain rate changes performed around room temperature will be further considered when discussing the effect of solute atoms on the Cottrell–Stokes law.

5.2. Cottrell–Stokes (C–S) Law

In a situation where microstructural scaling applies equation (86) provides a straight forward interpretation of the C–S law⁽¹⁰⁴⁾:

Due to strain rate variation $\Delta\dot{\gamma}$:

$$\frac{\Delta\tau}{\tau} = \frac{kT}{\alpha\omega G b^3} \Delta \ln \dot{\gamma} \quad (92)$$

Due to temperature variation ΔT :

$$\frac{\Delta\tau}{\tau} = \frac{k\Delta T}{\alpha\omega G b^3} \ln \frac{b^2 \rho_m v_D B_I}{\dot{\gamma}} \quad (93)$$

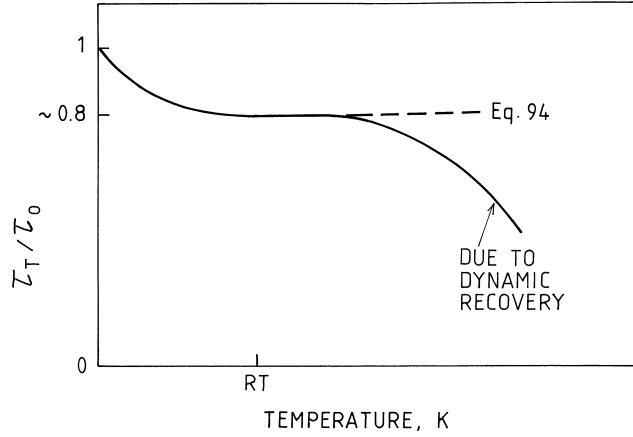


Fig. 20. The flow stress ratio (quench ratio) vs temperature as predicted by equation (94) (broken line)

or expressed as the ratio between the flow stress at a reference temperature, τ_o (here 0K has been selected as a reference temperature) and the flow stress τ_T , which is the flow stress measured at T after an instantaneous change in temperature from 0K to T :

$$\frac{\tau_T}{\tau_0} = \frac{\beta_1 - \frac{kT}{Gb^3} \ln \frac{b^2 \rho_m v_D B_I}{\dot{\gamma}} + \alpha \omega}{\beta_1 + \alpha \omega} \quad (94)$$

where τ_o is the flow stress as given by equation (86) at 0K. The response to sudden changes in strain rate or temperature will be of a pseudo-elastic nature. Rate changes have to be accommodated by the available mobile dislocations which must respond by corresponding changes in the migration rate (equation (82)). With the definition of α given by equation (30) these $\Delta\tau/\tau$ -ratios are not constant but will increase slightly by increasing the forest dislocation density (or increasing τ), as originally pointed out by Basinski⁽¹⁰⁵⁾.

Selecting $\alpha \simeq 0.3$, these relationships are all in accordance with Cottrell–Stokes observations⁽¹⁰⁵⁾, provided $\omega \simeq 2-3$, a value which is very reasonable indeed. This value of ω gives $(\tau_i/\tau)_{0K} \simeq 0.2$ (equation (87)) and it follows from equation (94) that τ_T/τ_0 will vary with temperature as qualitatively illustrated in Fig. 20. Moving out of Stage II where scaling is no longer a good approximation, deviations from the predictions of equations (92)–(94) are expected, see special section below. While rate sensitivity effects due to rapid changes in strain-rate and temperature were not intended as a topic of major concern in this approach to work hardening, this brief coverage does illustrate that the τ_f -contribution to the flow stress is only about 20% or less of the 0K-strength and rapidly drops off with increasing temperature, and it follows that the assumption made above (see equation (11)) that the τ_f -contribution to the flow stress is of second order is justified.

5.3. Effect of Solute Atoms on Cottrell–Stokes Law

It follows from equation (85) (with $V_a = \omega_{sa} c^{-e} b^3$) that in solute containing metals and alloys the result of rapid strain rate changes, provided $\tau < \tau_{ir}$ (equation (91)), becomes

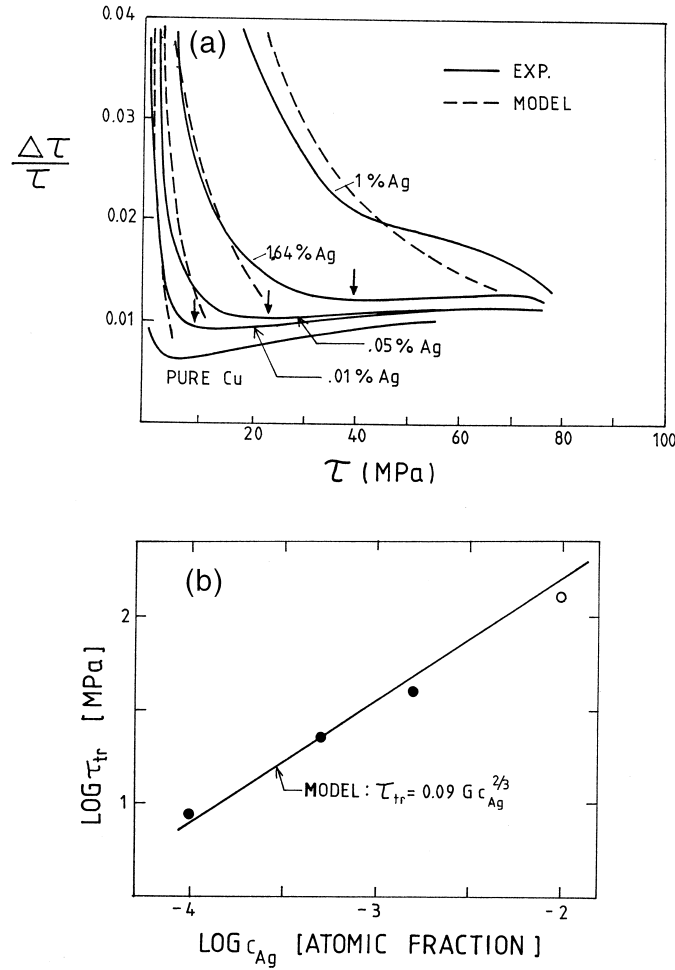


Fig. 21. (a) Strain rate sensitivity as a function of total flow-stress, experimental results on copper and dilute Cu–Ag alloys taken from⁽⁵⁰⁾ (strain rate changes by a factor of 10). (b) The transition stress τ_{tr} vs concentration of Ag. (The transition stresses used are indicated by arrows in (a), open symbol extrapolated value)

$$\Delta\tau = \frac{kTc^e}{\omega_{sa}b^3} \Delta \ln \dot{\gamma} \quad (95)$$

The transition from solute to ‘forest’ controlled strain-rate change behaviour is nicely illustrated by the experiments by Basinski and Basinski⁽⁵⁰⁾ who studied pure copper and dilute Cu–Ag alloys, Fig. 21(a). For stress levels $\tau < \tau_{tr}$, where τ_{tr} is indicated by arrows in Fig. 21(a), the $\Delta\tau/\tau$ -ratio declines with increasing stress according to equation (95). Selecting e equal to $2/3$ the composition dependency is accounted for by the present analysis with very good precision, given the uncertainties associated with this type of experiment. It is interesting to note that similar effects of solid solution atoms on strain rate sensitivity in Mg–Cd alloys have recently been reported by Lukac⁽¹⁰²⁾. It is further noted that the log–log, τ_{tr} vs c_{Ag} , plot in Fig. 21(b) nicely confirms the predictions of equation (91). For stress-

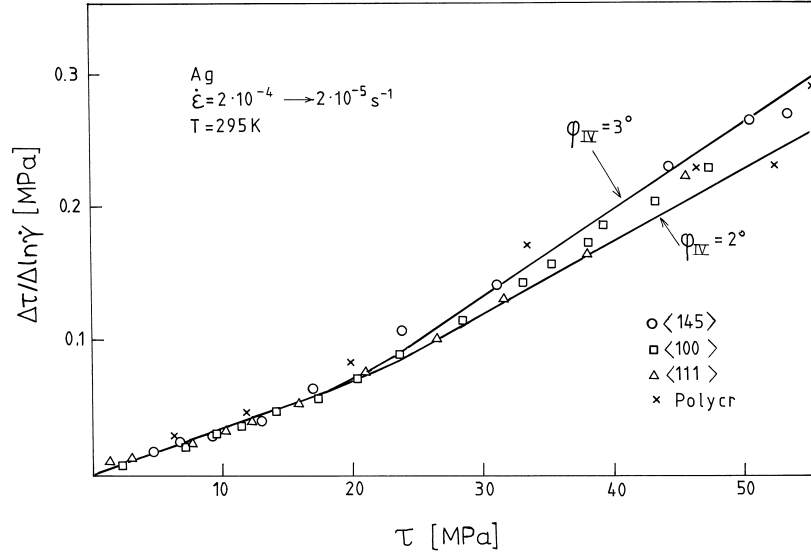


Fig. 22. Haasen plot of rate sensitivity vs shear stress. Experimental data on silver single and polycrystals from Mecking and Kocks⁽¹²⁾

ses $\tau > \tau_{tr}$ normal 'pure metal behavior' is observed, also in accordance with model predictions.

As pointed out by Basinski and Basinski⁽⁵⁰⁾, it seems to be generally accepted that the initial enhanced values of $\Delta\tau$ are a result of impurities, the question, however, of whether this is a direct or indirect cause is still disputed. It is in this context interesting to note that the present simple model, developed on the bases of *direct* short range interactions between the mobile dislocations and the solute atoms, provides a satisfactory interpretation of the experimental observations.

5.4. Deviation from Cottrell–Stokes Law at Large Strain

By performing rapid strain rate change experiments over a wide stress range results as shown in Fig. 22 (Haasen plot) are found, taken from a study of silver by Mecking and Kocks⁽¹²⁾. This figure clearly reveals a deviation from C–S law above a critical stress level. Such a deviation from C–S law has commonly been interpreted in qualitative terms as resulting from a 'similitude breakdown' associated with the Stage II–III transition, for a detailed discussion see Ref.⁽⁷⁾. The present model opens for a more direct, quantitative, test of the results of such a breakdown.

Rapid strain rate variations will, according to the present model (equation (84)) result in flow stress variations as given by:

$$\Delta\tau = \frac{kT}{V_a} \Delta \ln \dot{\gamma} = \frac{kT}{\omega b^2} \sqrt{\rho} \Delta \ln \dot{\gamma} \quad (96)$$

As long as microstructural scaling applies this equation is identical to equation (92). Beyond Stage III, however, (i.e. for stresses $\tau > \tau_{III}^*$) we no longer have any simple scaling law between $\rho_I (= \rho_{III})$ and δ . If, under such conditions, ρ is expressed in terms of the flow stress, τ , which is achieved by combining equations (11) and (47), equation (96) takes the

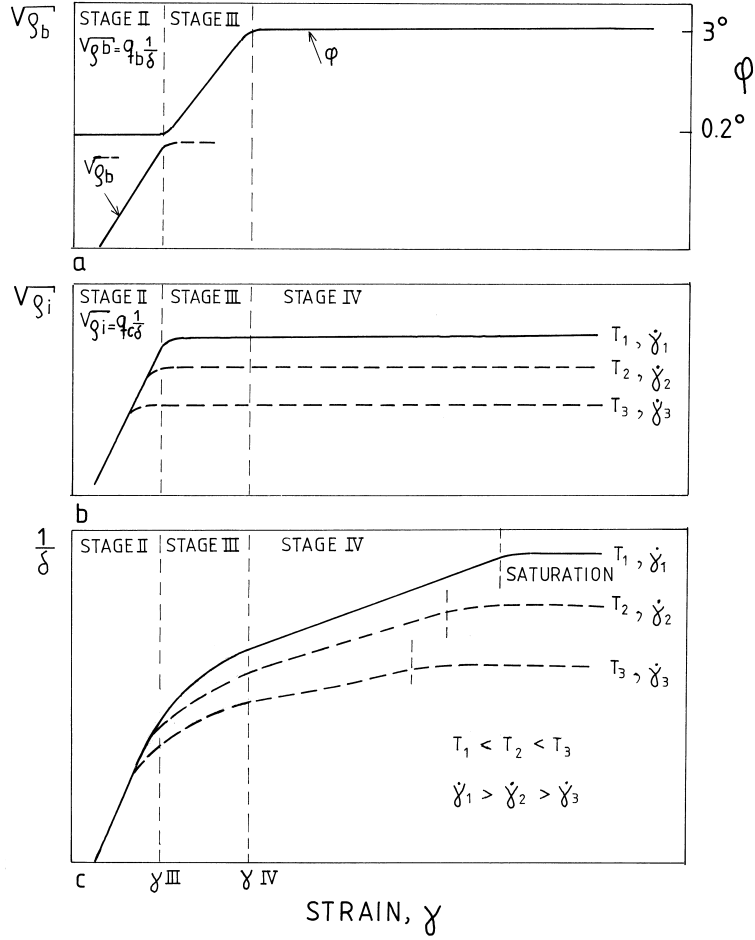


Fig. 23. Schematic representation of the microstructural evolution as a function of strain

following from:

$$\Delta\tau = \frac{\tau_{III}^*}{\alpha_3\omega} \left(\frac{kT}{Gb^3} \right) \left[1 + \kappa\phi \left(\frac{\alpha_3}{\alpha_2} \right) \left(\frac{G}{\tau_{III}^*} \right) \left(\alpha_3 \frac{\tau}{\tau_{III}^*} - \alpha_1 \right) \right]^{1/2} \Delta \ln \dot{\gamma} \quad (97)$$

where ϕ gives the variation in sub-boundary misorientation during Stage III deformation (on the variation in ϕ with δ (or τ), see equation (50)). Application of equations (92) and (97) during Stages II and III, respectively, in the interpretation of the result for silver given in Fig. 22, requires $\omega \simeq 7$ for the selection of ϕ_{IV} -values indicated in the figure. It is interesting to see that the model correctly accounts for the experimentally determined slope change in the $\Delta\tau$ vs τ relationship going from Stage II to Stage III.

6. SUMMARY AND CONCLUSIONS

A work hardening theory has been developed based on a microstructural concept comprising three elements; the cell/subgrain size, δ , the dislocation density inside the cells, ρ_i ,

and the cell boundary dislocation density or the sub-boundary-misorientation, ρ_b or φ . The theory is based on a statistical approach to the storage of dislocations. This approach predicts that the slip length, L , scales with the inverse square root of the stored dislocation density, $\rho_s^{-1/2}$, and also, predicts a substructure evolution which is consistent with the concept of microstructural scaling (similitude) at zero degree Kelvin, at stress $\tau < \tau_{III}$. The model provides a solution to the basic 'dislocation-book-keeping-problem' by defining a differential equation which regulates the storage of dislocations into (i) a cell interior dislocation network, (ii) increases in boundary misorientation and (iii) the creation of new cell boundaries.

The predictions of the model in terms of the microstructural evolution are illustrated schematically in Fig. 23(a)–(c). As shown in Fig. 23(b) the theory predicts that the square root of the dislocation density inside the cells, $\sqrt{\rho_i}$, increases linearly with strain until reaching saturation, which occurs during the Stage II–III transition. During continued deformation ρ_i remains constant. The prediction in terms of the inverse cell/subgrain size, $1/\delta$, is a linear increase with strain in Stage II, Fig. 23(c), followed by rapidly falling rate-increase entering Stage III, the rate-change decreases and stagnates at the end of Stage III, resulting in an almost linear increase in $1/\delta$ in Stage IV. This close to linear $1/\delta$ vs γ relationship is, in microstructural terms, the most characteristic aspect of Stage IV, and aspect which is experimentally well documented. The dislocation density within the cell boundaries scales with the dislocation density in the cell interior during Stage II, Fig. 23(a). At the start of Stage III the sub-boundary misorientation starts to increase rapidly and the model now treats the boundary structure in terms of boundary misorientation. The evolution of the boundary misorientation has not been modelled, but the broken line in Fig. 23(c) gives a well established experimental picture. The rapid increase in boundary misorientation is, in microstructural terms, one of the most characteristic aspects of Stage III. How this microstructural evolution is affected by changes in strain-rate or temperature are indicated in a qualitative way in the figure.

By combining this microstructural evolution picture with the flow stress model ($\tau = \tau_i + \alpha_1 Gb\sqrt{\rho_i} + \alpha_2 Gb/\delta$) stress–strain curves can be generated. In the present treatment the shape of the stress–strain curves in Stage III and Stage IV are analyzed in terms of θ - τ plots. It follows from such plots that the model predicts a close to Voce type hardening in the latter part of Stage III and a nearly linear hardening in Stage IV. These predictions are shown to correlate well with experimental observations in copper.

Dynamic recovery has been modelled in terms of dynamic dislocation network growth for the dislocation population inside the subgrains and as dynamic subgrain growth for the subgrain structure. A detailed analysis of the temperature dependence of the Stage II saturation stress, τ_{IIIs} , has been performed and applied to results from hot deformation of Ag, Cu, Ni and Al. The constitutive law developed accounts for the experimental observations with remarkable precision and the activation energy of the rate-controlling reaction was found to be that of self diffusion.

It has been demonstrated that, on the basis of a one-parameter approach to work hardening, it is not possible to account for the shape of the stress–strain curve beyond Stage II.

General steady-state creep laws are presented, covering the entire range from low temperature creep to Harper–Dorn creep.

A model for the frictional stress (effective stress), due to short range interactions with forest dislocations and/or elements in solid solution, has been presented. The constitutive laws which have been developed account for the characteristic strain-rate and temperature sensitivity observed experimentally, especially the effect of solute atoms on the strain rate sensi-

tivity is well accounted for by the model, and so is also the deviation from C–S law in Stage III.

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REFERENCES

1. E. Nes, *Acta Metall. Mater.* **43**, 2189 (1995).
2. T. Furu, Ørsund R. and E. Nes, *Acta Metall. Mater.* **43**, 2209 (1995).
3. E. Nes and J. A. Sæter, *Proc. 16th Risø Int. Symp.* (edited by N. Hansen et al.) pp. 169–192, Risø National Laboratory, Roskilde (1995).
4. E. Nes, *Scripta Metall. Mater.* **33**, 225 (1995).
5. E. Nes and T. Furu, *Scripta Metall. Mater.* **33**, 87 (1995).
6. E. Nes, T. Pettersen and K. Marthinsen, *Proc. Int. Conf. on Thermo-Mechanical Processing, Theory, Modelling, Practice*. The Swedish Society for Materials Technology, Stockholm (1996).
7. J. Gil Sevillano, *Material Science and Technology. A Comprehensive Treatment*, vol. 6 (edited by R. W. Cahn et al.), pp. 19–88, VCH, Weinheim (1993).
8. U. F. Kocks, *Phil. Mag.*, **13** (1966).
9. U. F. Kocks, *J. Eng. Mater. Tech. (ASME-H)* **98**, 76 (1976).
10. U. F. Kocks, in *Dislocations and Properties of Real Materials*, pp. 125–143, Institute of Metals, London (1985).
11. U. F. Kocks and H. Mecking, in *Strength of Metals and Alloys* (edited by P. Haasen, V. Gerold and G. Kostorz), Vol I, p. 345, Pergamon Press, Oxford (1979).
12. H. Mecking and U. F. Kocks, *Acta Met.* **29**, 1865 (1981).
13. H. Mecking, in *Dislocation Modelling of Physical Systems* (edited by M. F. Ashby et al.), p. 197. Pergamon Press, Oxford (1981).
14. H. Mecking, B. Nicklas, N. Zarubova and U. F. Kocks, *Acta Met.* **34**, 527 (1986).
15. Y. Estrin and H. Mecking, *Acta Met.* **32**, 57 (1984).
16. H. Mecking and Y. Estrin, in *Constitutive Relations and their Physical Basis* (edited by S. I. Andersen et al.), pp. 123–145, Risø National Laboratory, Roskilde, Denmark (1987).
17. Y. Estrin, in *Unified Constitutive laws of Plastic Deformation* (edited by A. S. Krausz and K. Krausz) Academic Press (1995).
18. A. D. Rollett and U. F. Kocks, *Solid State Phenomena* **1**, 35–36 (1994).
19. A. D. Rollet, U. F. Kocks and R. D. Doherty, in *Formability and Metallurgical Structure*, p. 211, TMS-AIME, Philadelphia, USA (1987).
20. M. Zehetbauer and V. Seumer, *Acta Metall. Mater.* **41**, 577 (1993).
21. T. E. Mitchell, in *Progress in Applied Materials Research*, (edited by E. G. Stanford et al.), pp. 117–237, Temple Press Books, London (1964).
22. J. M. Alberdi Garitaonandia, Ph.D. Thesis, University of Navarra, San Sebastian (1984).
23. G. Schoeck and A. Seeger, *Report of the Bristol Conference on Defects in Crystalline Solids*, p. 440. Physical Society, London (1955).
24. A. Seeger, R. Berner and H. Wolf, *Z. Physik* **155**, 247 (1959).
25. H. Wolf, *Z. Naturf. (A)* **15**, 180 (1960).
26. J. Friedel, *Dislocations and Mechanical Properties of Crystals*, Wiley, New York (1957).
27. F. R. N. Nabarro, *Acta Met.* **37**, 1521 (1989).

28. B. Escaig, *Thèse de Doctorat d'Etat*, Université de Paris, Centre d'Orsay No. A. 382 (1968).
29. J. Bonneville and B. Escaig, *Acta Met.* **27**, 1477 (1979).
30. M. S. Duesbery, N. P. Louat and K. Sadananda, *Acta metall. mater.* **40**, 149 (1992).
31. J. P. Hirth and J. Lothe, *Theory of Dislocations*, McGraw-Hill, New York (1968).
32. C. M. Sellars and W. T. McG. Tegart, *Mem. Sci. Rev. Met.* **63**, 731 (1966).
33. W. D. Nix, J. C. Gibeling and D. A. Hughes, *Met. Trans. A* **16A**, 2215 (1985).
34. J. Gil Sevillano, P. Van Houtte and E. Aernoudt, *Prog. Material Sci.* **25**, 69 (1980).
35. M. Hatherly, *ICSMA 6*, (edited by R. C. Gifkins), p. 1181, Pergamon Press, Oxford (1983).
36. N. Hansen, *Mat. Sci. Tech.* **6**, 1039 (1990).
37. G. Langford and M. Cohen, *Trans. ASM* **62**, 623 (1969).
38. M. Zehetbauer, *Acta Metall Mater.* **41**, 589 (1993).
39. F. Schuh and M. von Heimendahl, *Z. Metallkunde* **65**, 346 (1987).
40. E. Nes, A. L. Dons and N. Ryum, *Proc. ICSMA6* (Ed. Gifkins, R. C.), pp. 425 Pergamon Press, London 425 (1982).
41. F. J. Torrealdea Folgado, PhD thesis, Universidad de Navarra (1981).
42. T. Furu, Ørsund R. and E. Nes, *Mat. Sc. Eng.* **A214**, 122 (1996).
43. F. B. Prinz and A. S. Argon, *Acta Met.* **32**, 1021 (1984).
44. P. Haasen, *J. Phys. France* **50**, 2445 (1989).
45. D. Kuhlmann-Wilsdorf, *Met. Trans.* **1**, 3173 (1970).
46. D. Kuhlmann-Wilsdorf, in *Work Hardening in Tension and Fatigue* (edited by A. W. Thompson), The Metallurgical Society of AIME, New York (1977).
47. H. Mughrabi, in *Strength of Metals and Alloys (ICSMA5)* (edited by P. Haasen et al.) p. 1615, Pergamon Press, Oxford (1988).
48. H. Mughrabi, *Mater. Sci. Eng.* **85**, 15 (1987).
49. O. B. Pedersen, L. M. Brown and W. M. Stobbs, *Acta Metall.* **24**, 1843 (1981).
50. S. J. Basinski and Z. S. Basinski, *Dislocations in Solids*, vol. 4, pp. 261–362, (Ed. F. R. N. Nabarro) North Holland, Amsterdam (1979).
51. H. Mecking, private communication.
52. F. R. Castro-Fernandez, C. M. Sellars and J. A. Whiteman, *Mat. Sci. and Tech.* **6**, 453 (1990).
53. L. M. Brown, *Met. Trans. A* **22A**, 1693 (1991).
54. T. H. Blewitt, R. R. Coltman and J. K. Redman, *Defects in Crystalline Solids*, pp. 369, The Physical Society, London (1955).
55. J. Diehl and R. Z. Berner, *Z. Metallk.* **51**, 522 (1960).
56. J. W. Schu, O. Buck and E. Ko, *Z. Metallk.* **53**, 172 (1962).
57. P. Haasen and A. King, *Z. Metallk.* **51**, 722 (1960).
58. P. R. Thornton, T. E. Mitchell and P. B. Hirsch, *Phil. Mag.* **7**, 1349 (1962).
59. A. Beneteau and P. Coulomb, *Scripta Met.* **12**, 741 (1978).
60. J. Washburn and G. Murty, *Can. J. Phys.* **45**, 523 (1967).
61. T. Kosel and J. Washburn, *Proceedings of the 2nd Int. Conf. of Metals and Alloys*, pp. 459, Am. Soc. Metals (1970).
62. A. S. Malin and M. Hatherly, *Mat. Sci.* **13**, 463 (1979).
63. A. S. Malin, J. Huber and M. Hatherly, *Z. Metallk.* **72**, 310 (1981).
64. H. J. Roven and E. Nes, *Metal Sci.* **18**, 515 (1984).
65. D. A. Hughes and W. D. Nix, *Mat. Sci. Eng.* **A122**, 153 (1989).
66. B. Bay, N. Hansen and D. Kuhlmann-Wilsdorf, *Mat. Sci. Eng.* **A113**, 385 (1989).
67. B. Bay and N. Hansen, in *Deformation of polycrystals: Mechanisms and microstructure*. Risø National Laboratory, pp. 137–144, Risø (1989).
68. T. S. Lundy and J. F. Murdoch, *J. Appl. Phys.* **53**, 1671 (1962).
69. T. E. Volin, K. H. Lie and R. W. Balluffi, *Acta metall.* **19**, 263 (1971).
70. N. Q. Lam, S. J. Rothman, H. Mehrer and L. J. Nowicki, *Phys. Status Solidi (b)* **57**, 225 (1973).
71. S. J. Rothman, N. L. Peterson and J. T. Robinson, *Phys. Status Solidi* **39**, 635 (1970).
72. S. J. Rothman and N. L. Peterson, *Phys. Status Solidi* **35**, 305 (1969).
73. K. Maier, *Phys. Status Solidi (a)* **44**, 567 (1977).
74. M. Badia and A. Vignes, *Acta Met.* **17**, 177 (1969).
75. H. Mecking and K. Lücke, *Proceedings of the Second International Conference on the Strength of Metals and Alloys*. Amer. Soc. Met., pp. 470–474 (1970).
76. H. Saka, T. Iwata and T. Imura, *Phil. Mag.* **37A**, 273 (1978).
77. C. B. Carter and S. M. Holmes, *Phil. Mag.* **35**, 1161 (1977).
78. L. E. Murr, *Interfacial Phenomena in Metals and Alloys*, Addison-Wesley, Reading, Mass. (1975).
79. C. M. Sellars, *Material Science Forum* **29**, 113–115 (1993).
80. J. J. Urcola and C. M. Sellars, *Acta Met. et Mater.* **35**, 2637 (1987).
81. J. J. Urcola and C. M. Sellars, *Acta Met. et Mater.* **35**, 2649 (1987).
82. W. Blum, In *Materials Science and Technology* Volume 6 (edited by R. W. Cahn et al.), pp. 359–405, VCH, Weinheim (1993).

83. W. Blum, in *Hot Deformation of Aluminium Alloys* (edited by T. Langdon et al.), pp. 181–200, Warrendale: TMS (1991).
84. J. G. Harper and J. E. Dorn, *Acta Met.* **5**, 654 (1957).
85. J. G. Harper, L. A. Shepard and J. E. Dorn, *Acta Met.* **6**, 509 (1958).
86. J. C. M. Li, *Recrystallization, Grain Growth and Textures*, pp. 49–94, AMS, Ohio (1966).
87. J. Cadec, *Creep in Metallic Materials*, pp. 150–155, Elsevier, Amsterdam (1988).
88. H. Oikawa and T. G. Langdon, in *Creep Behaviour of Crystalline Solids* (edited by B. Wilshire and R. W. Evans), pp. 33–82, Pineridge Press, Swansea (1985).
89. F. A. Mohamed and J. Wolfenstine, in *Hot Deformation of Aluminium Alloys* (edited by T. G. Langdon et al.), pp. 223–237, TMS, Warrendale (1991).
90. F. R. N. Nabarro, *Acta Metall.* **37**, 2217 (1989).
91. P. Yavari, D. A. Miller and T. G. Langdon, *Acta Metall.* **30**, 871 (1982).
92. F. A. Mohamed, K. L. Murty and J. W. Morris, *Metal. Trans.* **4**, 935 (1973).
93. ØstB. N and E. Nes, *Acta Metall.* **17**, 13 (1969).
94. A. Seeger, *Kristallplastizität Hdb.d. Phys.* VII/2, Springer-Verlag, Berlin (1958).
95. A. Seeger, H. Kronmüller, S. Mader and H. Traub, *Phil. Mag.* **6**, 639 (1961).
96. A. Seeger, J. Diehl, S. Mader and H. Rebstock, *Phil. Mag.* **2**, 1 (1957).
97. A. Seeger, *Dislocation and Mechanical Properties of Crystals*, p. 243, Wiley, New York (1957).
98. U. F. Kocks, A. S. Argon and M. F. Ashby, *Thermodynamics and Kinetics of Slip*, Progress in Materials Science, Vol. 19 Pergamon Press, Oxford (1975).
99. G. E. Dieter, *Mechanical Metallurgy*, pp. 295–301, McGraw-Hill, New York (1986).
100. P. Haasen, Mechanical Properties of Solid Solutions, in: *Physical Metallurgy* (Ed. R. W. Cahn and P. Haasen) pp. 2009–2073, North-Holland Publ. Co. Amsterdam (1996).
101. H. Neuhäuser and C. Schwink, *Solid Solution Strengthening in Material Science and Technology. A Comprehensive Treatment*, vol. 6 (edited by R. W. Cahn et al.) pp. 191–250, VCH, Weinheim (1993).
102. P. Lukac, *phys. stat. sol. (a)* **131**, 377 (1992).
103. R. A. Kloske and M. E. Fine, *Trans. AIME* **245**, 217 (1969).
104. A. H. Cottrell and R. J. Stokes, *Proc. Roy. Soc. A* **233**, 17 (1955).
105. Z. S. Basinski, *Scripta Met.* **8**, 1301 (1974).

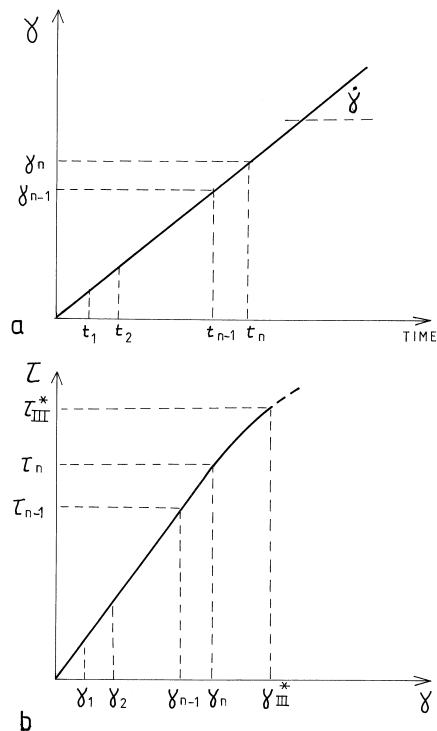


Fig. 24. Schematic representation of the step-by-step procedure used in calculating the stress-strain curve

7. APPENDIX A

The objective is to calculate $\tau = f(\gamma\dot{\gamma}, T)$ for $\tau < \tau_{III}^*$. This will be achieved by analyzing the deformation in terms of a succession of strain increments $\Delta\gamma$, each applied at time intervals $\Delta t = \Delta\gamma/\dot{\gamma}$ as shown in Fig. 24(a). The result is a stress-strain curve as given in Fig. 24(b). After the n -th strain increment the corresponding stress/strain values become

$$\tau_n = \alpha_1 Gb\sqrt{\rho_i^n} + \alpha_2 Gb \frac{1}{\delta_n} \text{ and } \gamma_n = n\Delta\gamma \quad (\text{A1})$$

The dislocation density ρ_i^n is calculated by solving equation (3), which here has been written in the form

$$\frac{d\rho_i}{dt} = \frac{d\rho_i^+}{dt} + \frac{d\rho_i^-}{dt}$$

where the second term is given by equation (39) and the first term is obtained from the OK-solution, i.e. by differentiation of equation (34).

$$\frac{d\rho_i^+}{dt} = \frac{2\dot{\gamma}\sqrt{\rho_i}}{bC\sqrt{1+f(q_b^2-1)}} \quad (\text{A2})$$

and ρ_i^n is obtained from the integral

$$\int_{\rho_i^{n-1}}^{\rho_i^n} \frac{bC\sqrt{1+f(q_b^2-1)}d\rho_i}{2\dot{\gamma}\sqrt{\rho_i} - \rho_i^2 \xi_\rho b^3 v_D B_1 \exp - \frac{U_{SD} - \xi_\rho Gb^4 \sqrt{\rho_i}}{kT}} = \frac{\Delta\gamma}{\dot{\gamma}} \quad (\text{A3})$$

where $B_1 = CB_\rho (1 + f(q_b^2 - 1))/3$. The cell size, δ_n , after the n -th strain increment is obtained by integration of equation (36), assuming ρ_i to be a constant within each time interval Δt_n .

$$\Delta\gamma = -\frac{bCq_c^2}{\rho_i^{n-1}} \int_{\delta_{n-1}}^{\delta_n} \sqrt{(1-f)\rho_i^{n-1} + f(q_c q_b)^2/\delta^2} \cdot \frac{1}{\delta^3} d\delta \quad (\text{A4})$$

And by combining equations (A3) and (A4) with equation (A1), stress-strain curves as illustrated in Fig. 9 are obtained.